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[Intervention Review]

Methods of induction of labour: a network meta-analysis

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ABSTRACT

Rationale

Induction of labour (IoL) aims to initiate labour when the risks of continuing pregnancy outweigh the benefits. Over 10 methods are currently available, yet the most effective and safest method remains unclear.

Objectives

To compare the benefits and harms of various cervical ripening and IoL methods at or beyond term labour and to rank the methods.

Search methods

We searched Cochrane Pregnancy and Childbirth's Trials Register, ClinicalTrials.gov, and the WHO ICTRP until 1 February 2023.

Eligibility criteria

We included randomised controlled trials (RCTs) evaluating IoL methods among women with a live fetus at or beyond term labour (gestational age ≥ 37 weeks). We focused on the methods currently recommended by international guidelines and those proposed by previous Cochrane reviews.

Outcomes

Our critical outcomes included failure to achieve vaginal delivery within 24 hours, caesarean section due to any causes, caesarean section due to non-reassuring fetal status, uterine hyperstimulation with changes in the heartbeat of the baby before birth, perinatal death, and severe neonatal morbidity.

Risk of bias

We assessed bias and trustworthiness using the Cochrane RoB 1 tool and the Cochrane Pregnancy and Childbirth Trustworthiness Tool (CPS-TST), respectively. The final analyses only included trials that met the CPS-TST.

Synthesis methods

We performed pair-wise meta-analyses using a random-effects model and network meta-analyses (NMA) with a frequentist approach. We reported analyses and results separately for two populations: 1) women without a previous caesarean section and a mix of women with or without a previous caesarean section (where more than 50% of participants had no previous caesarean section), and 2) women with a previous caesarean section. We used risk ratio (RR) and mean difference (MD) to present treatment effects with 95% confidence intervals (CIs). We assessed the certainty of evidence for critical outcomes using the GRADE approach for NMA. We used the surface under the cumulative ranking curve (SUCRA) to estimate treatment ranking.

Included studies

The analysis included 106 RCTs assessing 13 IoL methods among 30,348 women. Most trials (84.9%) were conducted in inpatient settings. Only two RCTs recruited women with previous caesarean section, and seven RCTs recruited a mix of women with or without a previous caesarean section.

Synthesis of results

We evaluated the effects of the following IoL methods: vaginal misoprostol ($\leq 50 \mu\text{g}$), oral misoprostol ($\leq 50 \mu\text{g}$), sublingual or buccal misoprostol ($\leq 50 \mu\text{g}$), controlled-release misoprostol vaginal pessary, vaginal dinoprostone (tablet or gel), controlled-release dinoprostone vaginal pessary, oxytocin (alone), nitric oxide donors, balloon catheters, osmotic cervical dilators, oxytocin plus amniotomy (Oxytocin + Amniotomy), balloon catheters plus oxytocin (Balloon + Oxytocin), balloon catheters plus vaginal/oral misoprostol ($\leq 50 \mu\text{g}$) (Balloon + Misoprostol $\leq 50 \mu\text{g}$), and inactive methods (placebo, no intervention, and expectant management).

The following results show the relative effects of NMA for four of the six critical outcomes among women without previous caesarean section and a mix of women with or without previous caesarean section. The findings of women with previous caesarean section are not presented here due to limited evidence. We used vaginal misoprostol ($\leq 50 \mu\text{g}$) as the reference method for presenting results, where data for this method were available in the network.

1) Failure to achieve vaginal delivery within 24 hours

There was no clear evidence that any of the methods were more effective than vaginal misoprostol ($\leq 50 \mu\text{g}$) in reducing the risk of this outcome. Oxytocin + Amniotomy (RR 0.41, 95% CI 0.14 to 1.24, 22.4% fewer, moderate-certainty evidence), Balloon + Misoprostol $\leq 50 \mu\text{g}$ (RR 0.85, 95% CI 0.60 to 1.19, 5.7% fewer, high-certainty evidence), and Balloon + Oxytocin (RR 0.94, 95% CI 0.71 to 1.26, 2.3% fewer, low-certainty evidence) likely resulted in little to no difference in this outcome compared with vaginal misoprostol ($\leq 50 \mu\text{g}$). Despite the comparable relative treatment effects between some methods and vaginal misoprostol ($\leq 50 \mu\text{g}$), these three methods ranked highest.

2) Caesarean section due to non-reassuring fetal status

There was no clear evidence that any of the methods were more effective than vaginal misoprostol ($\leq 50 \mu\text{g}$) in reducing the risk of this outcome. Balloon + Oxytocin (RR 0.77, 95% CI 0.47 to 1.25, 2.0% fewer, moderate-certainty evidence), controlled-release dinoprostone vaginal pessary (RR 0.86, 0.62 to 1.18, 1.2% fewer, low-certainty evidence), and balloon catheters (RR 0.88, 95% CI 0.70 to 1.11, 1.0% fewer, low-certainty evidence) may result in little to no difference in this outcome when compared with vaginal misoprostol ($\leq 50 \mu\text{g}$). Despite the comparable relative treatment effects between all methods and vaginal misoprostol ($\leq 50 \mu\text{g}$), these three methods ranked highest.

3) Uterine hyperstimulation with changes in the heartbeat of the baby before birth

Nitric oxide donors (RR 0.05, 95% CI 0.01 to 0.47, 3.9% fewer, moderate-certainty evidence), osmotic cervical dilators (RR 0.07, 95% CI 0.01 to 0.42, 3.8% fewer, moderate-certainty evidence), balloon catheters (RR 0.38, 95% CI 0.21 to 0.69, 2.5% fewer, moderate-certainty evidence), and oral misoprostol ($\leq 50 \mu\text{g}$) (RR 0.62, 95% CI 0.39 to 0.99, 1.6% fewer, moderate-certainty evidence) probably reduce this outcome compared with vaginal misoprostol ($\leq 50 \mu\text{g}$). When compared with inactive interventions (placebo, no intervention, or expectant management), vaginal misoprostol ($\leq 50 \mu\text{g}$) (RR 3.47, 95% CI 1.16 to 10.35) probably increases the risk of this outcome. The three highest-ranked methods were nitric oxide donors, osmotic cervical dilators, and inactive methods (placebo, no intervention, and expectant management).

4) Perinatal death

There was no clear evidence for this outcome, as it was rare (10 cases reported across trials).

Authors' conclusions

For women without previous caesarean section and a mix of women with or without previous caesarean section, there was no clear evidence that any of the IoL methods were more effective than vaginal misoprostol ($\leq 50 \mu\text{g}$) for the outcomes of failure to achieve

vaginal delivery within 24 hours, caesarean section due to non-reassuring fetal status, and perinatal death. Nitric oxide donors, osmotic cervical dilators, balloon catheters, and oral misoprostol ($\leq 50 \mu\text{g}$) probably reduce the risk of uterine hyperstimulation with changes in the heartbeat of the baby before birth.

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Registration

Protocol (2023): <https://doi.org/10.1002/14651858.CD015234>

PLAIN LANGUAGE SUMMARY

Which method is best to induce labour for pregnant women at or after 37 weeks?

Key messages

Findings from studies that included women without a previous caesarean section and studies that included a mix of women with or without a previous caesarean section (where most had no previous caesarean section):

- It is unclear whether any of the induction of labour methods worked better than inserting a low-dose ($\leq 50 \mu\text{g}$) of misoprostol into the vagina in achieving vaginal delivery within 24 hours, reducing the risk of caesarean section due to concerns about the baby's well-being, and preventing the death of the baby.
- Nitric oxide donors, osmotic cervical dilators, balloon catheters, and taking a low-dose ($\leq 50 \mu\text{g}$) of misoprostol by mouth probably help reduce the risk of excessive uterine contractions that can distress the baby compared to low-dose ($\leq 50 \mu\text{g}$) vaginal insertion of misoprostol.
- Future research is needed to improve our confidence in the evidence.

What is induction of labour?

Induction of labour is the process of starting labour using medical methods, which is often needed if a pregnancy goes beyond the due date or if there are concerns about the health of the mother or baby. There are five types of labour induction methods, described below.

- Medication-based methods (hormonal medications like misoprostol, dinoprostone, and oxytocin are given vaginally, orally (through the mouth), or intravenously (into a vein))
- Mechanical methods (typically a balloon catheter is inserted vaginally to dilate (open) the cervix)
- Surgical methods (known as amniotomy, this involves intentionally rupturing the membrane ('breaking the water'))
- Combination methods (using both medication and mechanical or surgical techniques to improve effectiveness)
- Alternative methods (such as acupuncture and herbal supplements).

What did we want to find out?

We wanted to know the most effective and safest labour induction method for pregnant women at or after 37 weeks with a live baby.

What did we do?

We searched for studies comparing labour induction methods in women at or beyond 37 weeks of pregnancy. We looked at methods recommended in international labour induction guidelines and in previous Cochrane reviews. We compared and summarised the results of the studies and rated our confidence in the evidence based on factors such as study methods and sizes.

What did we find?

We reviewed 106 studies involving 30,348 women from 35 countries, assessing 13 labour induction methods and one control intervention (no routine labour induction method or a policy of waiting (expectant management)).

The studies included the following groups of women.

- Most studies focused on women without a previous caesarean section.
- Two studies specifically included only women with a previous caesarean section.
- Seven studies included a mix of women (with and without a previous caesarean section, where most had no previous caesarean section)
 - Three studies clearly reported that they included a mix of women.
 - Four studies did not report this information; however, because more than half of women were first-time mothers, we assumed that these studies also included a mix of women with most having no previous caesarean section.

We assessed the following methods.

- Eight medication-based methods: low-dose (≤ 50 μ g) misoprostol given via oral, vaginal, or sublingual/buccal (inside the mouth) routes; controlled-release misoprostol vaginal pessary (tampon-like device); dinoprostone (vaginal gel/tablet or controlled-release vaginal pessary); oxytocin; and nitric oxide donors
- Two mechanical methods: balloon catheters and osmotic cervical dilators
- Three combination methods: oxytocin with amniotomy; balloon catheter with oxytocin; and balloon catheter with low-dose (≤ 50 μ g) misoprostol

The following findings summarise the results for our four key outcomes among women without a previous caesarean section and a mix of women with or without a previous caesarean section: failure to give birth vaginally within 24 hours, caesarean section due to concerns about the baby's well-being, uterine contractions that can distress the baby, and death of the baby.

- There was no clear evidence that any method is more effective than low-dose (≤ 50 μ g) vaginal insertion of misoprostol for such key outcomes as failure to give birth vaginally within 24 hours, caesarean section due to concerns about the baby's well-being, and death of the baby.
- When compared with low-dose (≤ 50 μ g) vaginal insertion of misoprostol, nitric oxide donors, osmotic dilators, balloon catheters, and low-dose (≤ 50 μ g) oral ingestion of misoprostol probably reduce the risk of excessive uterine contractions that can distress the baby.

There is not enough evidence to indicate the safest and most effective labour induction method for women with previous caesarean section.

What are the limitations of the evidence?

We have mostly moderate to little confidence in the evidence. Our confidence was reduced due to problems with the study methods and because there were not enough studies to be certain about the results. Many studies did not collect or report serious unwanted effects for women and babies such as death of the baby and excessive uterine contractions that can distress the baby. Lastly, the studies did not look at all methods of inducing labour that we were interested in for our planned outcomes.

How up-to-date is this evidence?

The evidence is current to 1 February 2023.

SUMMARY OF FINDINGS

Summary of findings 1. Failure to achieve vaginal delivery within 24 hours (women without previous caesarean section and a mix of women with or without previous caesarean section)

Failure to achieve vaginal delivery within 24 hours

Patient or population: Pregnant women at or beyond term labour (gestational age ≥ 37 weeks) without previous caesarean section and a mix of women with or without previous caesarean section

Setting: Inpatient or outpatient settings

Interventions for which direct evidence available: Oral misoprostol (≤ 50 μg), sublingual or buccal misoprostol (≤ 50 μg), controlled-release misoprostol vaginal pessary, vaginal dinoprostone (tablet or gel), controlled-release dinoprostone vaginal pessary, oxytocin (alone), nitric oxide donors, balloon catheters, osmotic dilators, oxytocin plus amniotomy, balloon catheters plus oxytocin, balloon catheters plus misoprostol (≤ 50 μg ; oral or vaginal routes)

Comparison: Vaginal misoprostol (≤ 50 μg)

IoL methods	Direct evidence		Indirect evidence		Network evidence		Anticipated absolute effects for network estimate*		
	RR (95% CI)	Certainty	RR (95% CI)	Certainty	RR (95% CI)	Certainty	Risk with vaginal misoprostol (≤ 50 μg)	Risk with IoL methods	Risk difference with IoL methods
Oral misoprostol (≤ 50 μg)	1.32 (1.00,1.74)	⊕⊕⊕⊕ Moderate a	1.25 (1.00,1.57)	⊕⊕⊕⊕ Moderate b	1.26 (1.08,1.48)	⊕⊕⊕⊕ High ^c	379 per 1000	478 per 1000 (409 to 561)	99 more per 1000 (30 more to 182 more)
SL or BC misoprostol (≤ 50 μg)	1.11 (0.93,1.33)	⊕⊕⊕⊕ High	0.68 (0.40,1.16)	⊕⊕⊕⊕ Very low ^d	1.04 (0.85,1.26)	⊕⊕⊕⊕ High ^e	379 per 1000	394 per 1000 (322 to 478)	15 more per 1000 (57 fewer to 99 more)
CTR misoprostol	Not estimable	Not applicable	1.13 (0.70,1.84)	⊕⊕⊕⊕ Very low ^d	1.13 (0.70,1.84)	⊕⊕⊕⊕ Very low ^f	379 per 1000	428 per 1000 (265 to 697)	49 more per 1000 (114 fewer to 318 more)
Vaginal dinoprostone	1.19 (0.99,1.44)	⊕⊕⊕⊕ Low ^g	1.21 (0.90,1.62)	⊕⊕⊕⊕ Very low ^d	1.20 (1.02,1.41)	⊕⊕⊕⊕ Low ^h	379 per 1000	455 per 1000 (387 to 534)	76 more per 1000 (8 more to 155 more)
CTR dinoprostone	1.21 (1.01,1.44)	⊕⊕⊕⊕	1.09 (0.74,1.59)	⊕⊕⊕⊕	1.19 (0.96,1.47)	⊕⊕⊕⊕	379 per 1000	451 per 1000 (364 to 557)	72 more per 1000 (15 fewer to 178 more)

	Moderate ⁱ		Low ^j		Low ^k				
Oxytocin (alone)	2.07 (1.27,3.37)	⊕⊕⊕⊕ Low ^l	2.01 (1.27,3.17)	⊕⊕⊕⊕ Low ^m	2.03 (1.41,2.91)	⊕⊕⊕⊕ Low ⁿ	379 per 1000	769 per 1000 (534 to 1000)	390 more per 1000 (155 more to 724 more)
Nitric oxide donors	1.81 (1.29,2.52)	⊕⊕⊕⊕ Moderate ^a	2.53 (0.00,∞)	⊕⊕⊕⊕ Low ^o	1.81 (1.11,2.93)	⊕⊕⊕⊕ Moderate ^k	379 per 1000	686 per 1000 (421 to 1000)	307 more per 1000 (42 more to 731 more)
Balloon catheters	0.92 (0.68,1.24)	⊕⊕⊕⊕ Moderate ^a	1.57 (1.28,1.94)	⊕⊕⊕⊕ Very low ^d	1.43 (1.17,1.75)	⊕⊕⊕⊕ Low ^p	379 per 1000	542 per 1000 (443 to 663)	163 more per 1000 (64 more to 284 more)
Osmotic dilators	Not estimable	Not applicable	1.34 (0.98,1.84)	⊕⊕⊕⊕ Very low ^d	1.34 (0.98,1.84)	⊕⊕⊕⊕ Very low ^f	379 per 1000	508 per 1000 (371 to 697)	129 more per 1000 (8 fewer to 318 more)
Oxytocin plus amniotomy	Not estimable	Not applicable	0.41 (0.14,1.24)	⊕⊕⊕⊕ Moderate ^b	0.41 (0.14,1.24)	⊕⊕⊕⊕ Moderate ^q	379 per 1000	155 per 1000 (53 to 470)	224 fewer per 1000 (326 fewer to 91 more)
Balloon plus oxytocin	0.94 (0.68,1.30)	⊕⊕⊕⊕ Low ^l	0.96 (0.53,1.74)	⊕⊕⊕⊕ Low ^r	0.94 (0.71,1.26)	⊕⊕⊕⊕ Low ^s	379 per 1000	356 per 1000 (269 to 478)	23 fewer per 1000 (110 fewer to 99 more)
Balloon plus misoprostol (≤ 50 µg)	0.77 (0.55,1.06)	⊕⊕⊕⊕ High	0.95 (0.59,1.54)	⊕⊕⊕⊕ Very low ^d	0.85 (0.60,1.19)	⊕⊕⊕⊕ High ^e	379 per 1000	322 per 1000 (227 to 451)	57 fewer per 1000 (152 fewer to 72 more)

*The assumed risks in the vaginal misoprostol (≤ 50 µg) group are based on weighted means of baseline risks from the studies with vaginal misoprostol (≤ 50 µg) arm in the network meta-analysis. The corresponding risks for each method of induction of labour (and their 95% CIs) are based on the assumed risk in the vaginal misoprostol (≤ 50 µg) group and the relative effect of the individual treatment intervention, when compared with the vaginal misoprostol (≤ 50 µg) group (and its 95% CI) as derived from the network meta-analysis.

CI: confidence interval; **RR:** risk ratio; **IoL:** induction of labour; **SL or BC misoprostol (≤ 50 µg):** sublingual or buccal misoprostol (≤ 50 µg); **CTR misoprostol:** controlled-release misoprostol vaginal pessary; **CTR dinoprostone:** controlled-release dinoprostone vaginal pessary; **Balloon catheters:** Foley single balloon catheter or double balloon devices (Atad/Cook balloon or other types); **Balloon plus oxytocin:** balloon catheters plus oxytocin; **Balloon plus misoprostol (≤ 50 µg):** balloon catheters plus misoprostol (≤ 50 µg; oral or vaginal routes).

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate; the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited; the true effect may be substantially different from the estimate of the effect.
Very low certainty: we have very little confidence in the effect estimate; the true effect is likely to be substantially different from the estimate of effect.

- ^aDirect evidence downgraded –1 due to serious imprecision.
- ^bIndirect evidence downgraded –1 due to serious imprecision.
- ^cNetwork evidence downgraded –1 due to moderate-certainty direct and indirect evidence; however, upgraded +1 due to precision of network estimate (no intransitivity or incoherence).
- ^dIndirect evidence downgraded –3 due to multiple limitations in trial design, unexplained statistical heterogeneity, and serious imprecision.
- ^eNetwork evidence not initially downgraded given high-certainty direct evidence (no intransitivity or incoherence).
- ^fNetwork evidence downgraded –3 due to very low-certainty indirect evidence (no intransitivity or incoherence).
- ^gDirect evidence downgraded –2 due to multiple limitations in trial design and unexplained statistical heterogeneity.
- ^hNetwork evidence downgraded –2 due to low-certainty direct evidence (no intransitivity or incoherence).
- ⁱDirect evidence downgraded –1 due to multiple limitations in trial design.
- ^jIndirect evidence downgraded –2 due to unexplained statistical heterogeneity and serious imprecision.
- ^kNetwork evidence downgraded –1 due to moderate direct evidence (no intransitivity or incoherence).
- ^lDirect evidence downgraded –2 due to multiple limitations in trial design and serious imprecision.
- ^mIndirect evidence downgraded –2 due to multiple limitations in trial design and serious imprecision.
- ⁿNetwork evidence downgraded –2 due to low-certainty direct and indirect evidence (no intransitivity or incoherence).
- ^oIndirect evidence downgraded –2 due to very serious imprecision.
- ^pNetwork evidence downgraded –1 due to moderate direct evidence and downgraded further –1 due to incoherence between the direct and indirect estimates (no intransitivity).
- ^qNetwork evidence downgraded –1 due to moderate indirect evidence (no intransitivity or incoherence).
- ^rIndirect evidence downgraded –2 due to multiple limitations in trial design and serious imprecision.
- ^sNetwork evidence downgraded –1 due to low-certainty direct and indirect evidence (no intransitivity or incoherence).

Summary of findings 2. Caesarean section due to non-reassuring fetal status (women without previous caesarean section and a mix of women with or without previous caesarean section)

Caesarean section due to non-reassuring fetal status

Patient or population: Pregnant women at or beyond term labour (gestational age ≥ 37 weeks) without previous caesarean section and a mix of women with or without previous caesarean section

Setting: Inpatient or outpatient settings

Interventions for which direct evidence available: Oral misoprostol (≤ 50 μ g), sublingual or buccal misoprostol (≤ 50 μ g), controlled-release misoprostol vaginal pessary, vaginal dinoprostone (tablet or gel), controlled-release dinoprostone vaginal pessary, oxytocin (alone), nitric oxide donors, balloon catheters, osmotic dilators, oxytocin plus amniotomy, balloon catheters plus oxytocin, balloon catheters plus misoprostol (≤ 50 μ g; oral or vaginal routes)

Comparison: Vaginal misoprostol (≤ 50 μ g)

IoL methods	Direct evidence	Indirect evidence	Network evidence	Anticipated absolute effects for network estimate*
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	RR (95% CI)	Certainty	RR (95% CI)	Certainty	RR (95% CI)	Certainty	Risk with vaginal misoprostol ($\leq 50 \mu\text{g}$)	Risk with IoL methods	Risk difference with IoL methods
Oral misoprostol ($\leq 50 \mu\text{g}$)	0.50 (0.24,1.03)	⊕⊕⊕⊕ Low ^a	1.05 (0.77,1.43)	⊕⊕⊕⊕ Low ^b	0.89 (0.68,1.15)	⊕⊕⊕⊕ Low ^c	87 per 1000	77 per 1000 (59 to 100)	10 fewer per 1000 (28 fewer to 13 more)
SL or BC misoprostol ($\leq 50 \mu\text{g}$)	1.35 (0.87,2.10)	⊕⊕⊕⊕ Low ^a	0.88 (0.18,4.24)	⊕⊕⊕⊕ Very low ^d	1.30 (0.92,1.83)	⊕⊕⊕⊕ Low ^e	87 per 1000	113 per 1000 (80 to 159)	26 more per 1000 (7 fewer to 72 more)
CTR misoprostol	Not estimable	Not applicable	1.16 (0.57,2.36)	⊕⊕⊕⊕ Low ^b	1.16 (0.57,2.36)	⊕⊕⊕⊕ Low ^f	87 per 1000	101 per 1000 (50 to 205)	14 more per 1000 (37 fewer to 118 more)
Vaginal dinoprostone	1.17 (0.84,1.63)	⊕⊕⊕⊕ Low ^a	1.09 (0.78,1.53)	⊕⊕⊕⊕ Low ^b	1.13 (0.89,1.44)	⊕⊕⊕⊕ Low ^c	87 per 1000	98 per 1000 (77 to 125)	11 more per 1000 (10 fewer to 38 more)
CTR dinoprostone	0.78 (0.46,1.32)	⊕⊕⊕⊕ Low ^a	0.91 (0.61,1.36)	⊕⊕⊕⊕ Low ^b	0.86 (0.62,1.18)	⊕⊕⊕⊕ Low ^c	87 per 1000	75 per 1000 (54 to 103)	12 fewer per 1000 (33 fewer to 16 more)
Oxytocin (alone)	1.25 (0.35,4.52)	⊕⊕⊕⊕ Very low ^g	1.77 (0.66,4.80)	⊕⊕⊕⊕ Very low ^d	1.56 (0.71,3.42)	⊕⊕⊕⊕ Very low ^h	87 per 1000	136 per 1000 (62 to 298)	49 more per 1000 (25 fewer to 211 more)
Nitric oxide donors	0.18 (0.04,0.79)	⊕⊕⊕⊕ Moderate ⁱ	1.49 (0.92,2.42)	⊕⊕⊕⊕ Low ^b	1.22 (0.77,1.92)	⊕⊕⊕⊕ Low ^j	87 per 1000	106 per 1000 (67 to 167)	19 more per 1000 (20 fewer to 80 more)
Balloon catheters	0.97 (0.68,1.40)	⊕⊕⊕⊕ Low ^a	0.81 (0.60,1.10)	⊕⊕⊕⊕ Low ^b	0.88 (0.70,1.11)	⊕⊕⊕⊕ Low ^c	87 per 1000	77 per 1000 (61 to 97)	10 fewer per 1000 (26 fewer to 10 more)
Osmotic dilators	Not estimable	Not applicable	1.09 (0.52,2.29)	⊕⊕⊕⊕ Low ^b	1.09 (0.52,2.29)	⊕⊕⊕⊕ Low ^f	87 per 1000	95 per 1000 (45 to 199)	8 more per 1000 (42 fewer to 112 more)
Oxytocin plus amniotomy	Not estimable	Not applicable	3.90 (0.16,93.18)	⊕⊕⊕⊕ Very low ^d	3.90 (0.16,93.18)	⊕⊕⊕⊕ Very low ^k	87 per 1000	339 per 1000 (14 to 1000)	252 more per 1000 (73 fewer to 8020 more)

Balloon plus oxytocin	0.97 (0.55,1.72)	⊕⊕⊕⊖ Moderate ⁱ	0.40 (0.15,1.05)	⊕⊕⊕⊖ Low ^b	0.77 (0.47,1.25)	⊕⊕⊕⊖ Moderate ^l	87 per 1000 (41 to 109)	67 per 1000 (41 to 109)	20 fewer per 1000 (46 fewer to 22 more)
Balloon plus misoprostol (≤ 50 µg)	1.04 (0.70,1.53)	⊕⊕⊕⊖ Low ^a	0.41 (0.18,0.92)	⊕⊕⊕⊖ Low ^b	0.87 (0.61,1.23)	⊕⊕⊕⊖ Low ^c	87 per 1000 (53 to 107)	76 per 1000 (53 to 107)	11 fewer per 1000 (34 fewer to 20 more)

*The assumed risks in the vaginal misoprostol (≤ 50 µg) group are based on weighted means of baseline risks from the studies with vaginal misoprostol (≤ 50 µg) arm in the network meta-analysis. The corresponding risks for each method of induction of labour (and their 95% CIs) are based on the assumed risk in the vaginal misoprostol (≤ 50 µg) group and the relative effect of the individual treatment intervention, when compared with the vaginal misoprostol (≤ 50 µg) group (and its 95% CI) as derived from the network meta-analysis.

CI: confidence interval; **RR:** risk ratio; **IoL:** induction of labour; **SL or BC misoprostol (≤ 50 µg):** sublingual or buccal misoprostol (≤ 50 µg); **CTR misoprostol:** controlled-release misoprostol vaginal pessary; **CTR dinoprostone:** controlled-release dinoprostone vaginal pessary; **Balloon catheters:** Foley single balloon catheter or double balloon devices (Atad/Cook balloon or other types); **Balloon plus oxytocin:** balloon catheters plus oxytocin; **Balloon plus misoprostol (≤ 50 µg):** balloon catheters plus misoprostol (≤ 50 µg; oral or vaginal routes).

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate; the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited; the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate; the true effect is likely to be substantially different from the estimate of effect.

^aDirect evidence downgraded –2 due to multiple limitations in trial design and serious imprecision.

^bIndirect evidence downgraded –2 due to multiple limitations in trial design and serious imprecision.

^cNetwork evidence downgraded –2 due to low-certainty direct and indirect evidence (no intransitivity or incoherence).

^dIndirect evidence downgraded –3 due to multiple limitations in trial design and very serious imprecision.

^eNetwork evidence downgraded –2 due to low-certainty direct evidence (no intransitivity or incoherence).

^fNetwork evidence downgraded –2 due to low-certainty indirect evidence (no intransitivity or incoherence).

^gDirect evidence downgraded –3 due to multiple limitations in trial design and very serious imprecision.

^hNetwork evidence downgraded –3 due to very low-certainty direct and indirect evidence (no intransitivity or incoherence).

ⁱDirect evidence downgraded –1 due to serious imprecision.

^jNetwork evidence downgraded –1 due to moderate-certainty direct evidence and downgraded further –1 due to incoherence between the direct and indirect estimates (no intransitivity).

^kNetwork evidence downgraded –3 due to very low-certainty indirect evidence (no intransitivity or incoherence).

^lNetwork evidence downgraded –1 due to moderate-certainty direct evidence (no intransitivity or incoherence).

Summary of findings 3. Uterine hyperstimulation with changes in the heartbeat of the baby before birth (women without previous caesarean section and a mix of women with or without previous caesarean section)

Uterine hyperstimulation with changes in the heartbeat of the baby before birth

Patient or population: Pregnant women at or beyond term labour (gestational age ≥ 37 weeks) without previous caesarean section and a mix of women with or without previous caesarean section

Setting: Inpatient or outpatient settings

Interventions for which direct evidence available: Oral misoprostol (≤ 50 μ g), sublingual or buccal misoprostol (≤ 50 μ g), controlled-release misoprostol vaginal pessary, vaginal dinoprostone (tablet or gel), controlled-release dinoprostone vaginal pessary, oxytocin (alone), nitric oxide donors, balloon catheters, osmotic dilators, oxytocin plus amniotomy, balloon catheters plus oxytocin (≤ 50 μ g)

Comparison: Vaginal misoprostol (≤ 50 μ g)

IoL methods	Direct evidence		Indirect evidence		Network evidence		Anticipated absolute effects for network estimate*		
	RR (95% CI)	Certainty	RR (95% CI)	Certainty	RR (95% CI)	Certainty	Risk with vaginal misoprostol (≤ 50 μ g)	Risk with IoL methods	Risk difference with IoL methods
Oral misoprostol (≤ 50 μ g)	0.56 (0.30,1.05)	⊕⊕⊕⊕ Low ^a	0.69 (0.35,1.35)	⊕⊕⊕⊕ Low ^b	0.62 (0.39,0.99)	⊕⊕⊕⊕ Moderate ^c	41 per 1000	25 per 1000 (16 to 41)	16 fewer per 1000 (25 fewer to 0 fewer)
SL or BC misoprostol (≤ 50 μ g)	1.43 (0.85,2.41)	⊕⊕⊕⊕ Low ^a	0.22 (0.06,0.76)	⊕⊕⊕⊕ Low ^b	1.10 (0.64,1.88)	⊕⊕⊕⊕ Very low ^d	41 per 1000	45 per 1000 (26 to 77)	4 more per 1000 (15 fewer to 36 more)
CTR misoprostol	Not estimable	Not applicable	1.39 (0.46,4.21)	⊕⊕⊕⊕ Very low ^e	1.39 (0.46,4.21)	⊕⊕⊕⊕ Very low ^f	41 per 1000	57 per 1000 (19 to 173)	16 more per 1000 (22 fewer to 132 more)
Vaginal dinoprostone	0.92 (0.60,1.40)	⊕⊕⊕⊕ Low ^a	1.33 (0.56,3.16)	⊕⊕⊕⊕ Low ^b	1.00 (0.64,1.56)	⊕⊕⊕⊕ Low ^g	41 per 1000	41 per 1000 (26 to 64)	0 fewer per 1000 (15 fewer to 23 more)
CTR dinoprostone	0.80 (0.37,1.75)	⊕⊕⊕⊕ Low ^a	1.84 (0.70,4.79)	⊕⊕⊕⊕ Very low ^e	1.11 (0.60,2.06)	⊕⊕⊕⊕ Low ^h	41 per 1000	46 per 1000 (25 to 84)	5 more per 1000 (16 fewer to 43 more)

Oxytocin (alone)	Not estimable	Not applicable	0.64 (0.32,1.29)	⊕⊕⊕⊕ Low ^b	0.64 (0.32,1.29)	⊕⊕⊕⊕ Low ⁱ	41 per 1000	26 per 1000 (13 to 53)	15 fewer per 1000 (28 fewer to 12 more)
Nitric oxide donors	0.05 (0.00,0.89)	⊕⊕⊕⊕ Low ^j	0.06 (0.00,1.51)	⊕⊕⊕⊕ Very low ^e	0.05 (0.01,0.47)	⊕⊕⊕⊕ Moderate ^k	41 per 1000	2 per 1000 (0 to 19)	39 fewer per 1000 (41 fewer to 22 fewer)
Balloon catheters	0.33 (0.01,7.89)	⊕⊕⊕⊕ Low ^a	0.38 (0.21,0.70)	⊕⊕⊕⊕ Moderate ^l	0.38 (0.21,0.69)	⊕⊕⊕⊕ Moderate ^m	41 per 1000	16 per 1000 (9 to 28)	25 fewer per 1000 (32 fewer to 13 fewer)
Osmotic dilators	Not estimable	Not applicable	0.07 (0.01,0.42)	⊕⊕⊕⊕ Moderate ^k	0.07 (0.01,0.42)	⊕⊕⊕⊕ Moderate ⁿ	41 per 1000	3 per 1000 (0 to 17)	38 fewer per 1000 (41 fewer to 24 fewer)
Oxytocin plus amniotomy	Not estimable	Not applicable	3.30 (0.13,84.05)	⊕⊕⊕⊕ Very low ^e	3.30 (0.13,84.05)	⊕⊕⊕⊕ Very low ^f	41 per 1000	135 per 1000 (5 to 1000)	94 more per 1000 (36 fewer to 3405 more)
Balloon plus oxytocin	0.66 (0.11,3.85)	⊕⊕⊕⊕ Low ^k	2.34 (0.20,28.07)	⊕⊕⊕⊕ Low ^o	1.03 (0.23,4.51)	⊕⊕⊕⊕ Low ^g	41 per 1000	42 per 1000 (9 to 185)	1 more per 1000 (32 fewer to 144 more)
Balloon plus misoprostol (≤ 50 µg)	Not reported by included studies	Not applicable	Not reported by included studies	Not applicable	Not reported by included studies	Not applicable	Not applicable	Not applicable	Not applicable

*The assumed risks in the vaginal misoprostol (≤ 50 µg) group are based on weighted means of baseline risks from the studies with vaginal misoprostol (≤ 50 µg) arm in the network meta-analysis. The corresponding risks for each method of induction of labour (and their 95% CIs) are based on the assumed risk in the vaginal misoprostol (≤ 50 µg) group and the relative effect of the individual treatment intervention, when compared with the vaginal misoprostol (≤ 50 µg) group (and its 95% CI) as derived from the network meta-analysis.

CI: confidence interval; **RR:** risk ratio; **IoL:** induction of labour; **SL or BC misoprostol (≤ 50 µg):** sublingual or buccal misoprostol (≤ 50 µg); **CTR misoprostol:** controlled-release misoprostol vaginal pessary; **CTR dinoprostone:** controlled-release dinoprostone vaginal pessary; **Balloon catheters:** Foley single balloon catheter or double balloon devices (Atad/Cook balloon or other types); **Balloon plus oxytocin:** balloon catheters plus oxytocin; **Balloon plus misoprostol (≤ 50 µg):** balloon catheters plus misoprostol (≤ 50 µg; oral or vaginal routes).

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate; the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited; the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate; the true effect is likely to be substantially different from the estimate of effect.

^aDirect evidence downgraded –2 due to multiple limitations in trial design and serious imprecision.

^bIndirect evidence downgraded –2 due to multiple limitations in trial design and serious imprecision.

^cNetwork evidence downgraded –2 due to low-certainty direct and indirect evidence; however, upgraded +1 due to precision of network estimate (no intransitivity or incoherence).

^dNetwork evidence downgraded –2 due to low-certainty direct and indirect evidence and downgraded further –1 due to incoherence between the direct and indirect estimates (no intransitivity).

^eIndirect evidence downgraded –3 due to multiple limitations in trial design and very serious imprecision.

^fNetwork evidence downgraded –3 due to very low-certainty indirect evidence (no intransitivity or incoherence).

^gNetwork evidence downgraded –2 due to low-certainty direct and indirect evidence (no intransitivity or incoherence).

^hNetwork evidence downgraded –1 due to moderate-certainty direct evidence (no intransitivity or incoherence).

ⁱNetwork evidence downgraded –2 due to low-certainty indirect evidence (no intransitivity or incoherence).

^jDirect evidence downgraded –2 due to very serious imprecision.

^kNetwork evidence downgraded –2 due to low-certainty direct evidence; however, upgraded +1 due to precision of network estimate (no intransitivity or incoherence).

^lIndirect evidence downgraded –1 due to multiple limitations in trial design.

^mNetwork evidence downgraded –1 due to moderate-certainty indirect evidence (no intransitivity or incoherence).

ⁿNetwork evidence downgraded –1 due to moderate-certainty indirect evidence.

^oIndirect evidence downgraded –2 due to very serious imprecision.

Summary of findings 4. Perinatal death (women without previous caesarean section and a mix of women with or without previous caesarean section)

Perinatal death

Patient or population: Pregnant women at or beyond term labour (gestational age ≥ 37 weeks) without previous caesarean section and a mix of women with or without previous caesarean section

Setting: Inpatient or outpatient settings

Interventions for which direct evidence available: Oral misoprostol (≤ 50 μ g), sublingual or buccal misoprostol (≤ 50 μ g), vaginal dinoprostone (tablet or gel), nitric oxide donors, balloon catheters

Comparison: Vaginal misoprostol (≤ 50 μ g)

IoL methods	Direct evidence	Indirect evidence	Network evidence	Anticipated absolute effects for network estimate*
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	RR (95% CI)	Certainty	RR (95% CI)	Certainty	RR (95% CI)	Certainty	Risk with vaginal misoprostol ($\leq 50 \mu\text{g}$)	Risk with loL methods	Risk difference with loL methods
Oral misoprostol ($\leq 50 \mu\text{g}$)	0.33 (0.01,7.97)	⊕⊕⊕⊕ Very low ^a	1.15 (0.00,∞)	⊕⊕⊕⊕ Very low ^b	0.33 (0.01,7.97)	⊕⊕⊕⊕ Very low ^c	12 per 1000	4 per 1000 (0 to 96)	8 fewer per 1000 (12 fewer to 84 more)
SL or BC misoprostol ($\leq 50 \mu\text{g}$)	3.20 (0.13,77.04)	⊕⊕⊕⊕ Very low ^a	Not estimable	Not applicable	3.20 (0.13,77.10)	⊕⊕⊕⊕ Very low ^c	12 per 1000	38 per 1000 (2 to 925)	26 more per 1000 (10 fewer to 953 more)
CTR misoprostol	Not reported by included studies	Not applicable	Not reported by included studies	Not applicable	Not reported by included studies	Not applicable	See comment ¹	See comment ²	See comment ³
Vaginal dinoprostone	1.60 (0.20,12.90)	⊕⊕⊕⊕ Very low ^a	Not estimable	Not applicable	1.60 (0.20,12.90)	⊕⊕⊕⊕ Very low ^c	12 per 1000	19 per 1000 (2 to 155)	7 more per 1000 (10 fewer to 143 more)
CTR dinoprostone	Not reported by included studies	Not applicable	Not reported by included studies	Not applicable	Not reported by included studies	Not applicable	See comment ¹	See comment ²	See comment ³
Oxytocin (alone)	Not reported by included studies	Not applicable	Not reported by included studies	Not applicable	Not reported by included studies	Not applicable	See comment ¹	See comment ²	See comment ³
Nitric oxide donors	Disconnected	Disconnected	Disconnected	Disconnected	Disconnected	Disconnected	See comment ⁴	See comment ⁴	See comment ⁴
Balloon catheters	Not estimable	Not applicable	0.33 (0.01,16.36)	⊕⊕⊕⊕ Very low ^d	0.33 (0.01,16.36)	⊕⊕⊕⊕ Very low ^e	12 per 1000	4 per 1000 (0 to 196)	8 fewer per 1000

									(12 fewer to 184 more)
Osmotic dilators	Not reported by included studies	Not applicable	Not reported by included studies	Not applicable	Not reported by included studies	Not applicable	See comment 1	See comment 2	See comment 3
Oxytocin plus amniotomy	Not reported by included studies	Not applicable	Not reported by included studies	Not applicable	Not reported by included studies	Not applicable	See comment 1	See comment 2	See comment 3
Balloon plus oxytocin	Not reported by included studies	Not applicable	Not reported by included studies	Not applicable	Not reported by included studies	Not applicable	See comment 1	See comment 2	See comment 3
Balloon plus misoprostol (≤ 50 µg)	Not reported by included studies	Not applicable	Not reported by included studies	Not applicable	Not reported by included studies	Not applicable	See comment 1	See comment 2	See comment 3

*The assumed risks in the vaginal misoprostol (≤ 50 µg) group are based on weighted means of baseline risks from the studies with vaginal misoprostol (≤ 50 µg) arm in the network meta-analysis. The corresponding risks for each method of induction of labour (and their 95% CIs) are based on the assumed risk in the vaginal misoprostol (≤ 50 µg) group and the relative effect of the individual treatment intervention, when compared with the vaginal misoprostol (≤ 50 µg) group (and its 95% CI) as derived from the network meta-analysis.

¹No included studies, or there was no event in the included studies to estimate the baseline risk.

²Absolute risk with methods of induction of labour cannot be estimated in the absence of absolute risk with vaginal misoprostol (≤ 50 µg).

³Risk difference cannot be estimated in the absence of absolute risks with intervention and vaginal misoprostol (≤ 50 µg).

⁴The direct comparison between nitric oxide donors and placebo, no treatment, or expectant management disconnected to the network. The RR for this comparison is 7.06 (95% CI 0.37 to 136.45). The certainty of this direct evidence is very low (⊕⊕⊕⊕). The direct evidence was downgraded –3 due to limitation in trial design and very serious imprecision.

CI: confidence interval; **RR:** risk ratio; **IoL:** induction of labour; **SL or BC misoprostol (≤ 50 µg):** sublingual or buccal misoprostol (≤ 50 µg); **CTR misoprostol:** controlled-release misoprostol vaginal pessary; **CTR dinoprostone:** controlled-release dinoprostone vaginal pessary; **Balloon catheters:** Foley single balloon catheter or double balloon devices (Atad/Cook balloon or other types); **Balloon plus oxytocin:** balloon catheters plus oxytocin; **Balloon plus misoprostol (≤ 50 µg):** balloon catheters plus misoprostol (≤ 50 µg; oral or vaginal routes).

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate; the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited; the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate; the true effect is likely to be substantially different from the estimate of effect.

^aDirect evidence downgraded –3 due to multiple limitations in trial design and very serious imprecision.

^bIndirect evidence downgraded –3 due to multiple limitations in trial design and very serious imprecision.

^cNetwork evidence downgraded –3 due to very low-certainty direct evidence (no intransitivity or incoherence).

^dIndirect evidence downgraded –3 due to multiple limitations in trial design and very serious imprecision.

^eNetwork evidence downgraded –3 due to very low-certainty indirect evidence (no intransitivity or incoherence).

BACKGROUND

Please see [Supplementary material 8](#) for a glossary of terms.

Description of the condition

Labour induction is the use of medical procedures or medications to induce labour by ripening the cervix or stimulating uterine contractions, or both. The goal of inducing labour is to safely achieve vaginal birth within a specified time frame. It is typically recommended when the risk of continued pregnancy is considered to outweigh the benefits [1]. There are various maternal and fetal indications for induction of labour [2, 3]. Common indications from a maternal perspective are pre-eclampsia with severe features and pre-labour rupture of membranes. Fetal indications are intrauterine growth restriction, risk of stillbirth in prolonged gestation, and other concerns regarding fetal well-being [1, 4]. The incidence of labour induction has increased steadily during the last few decades. In high-income countries, approximately one-quarter of term labour (at or beyond 37 weeks of gestation) is preceded by labour induction, while in the UK large maternity hospitals report rates well over 30% [5, 6]. The rates of labour induction in low- and middle-income countries, however, are generally lower than that of high-income countries [7, 8].

There are numerous clinical components that must be examined prior to induction to assess the chances of success of induction and decrease the likelihood of complications including induction failure, leading to potentially avoidable caesarean section and adverse neonatal outcomes. Factors associated with the success rates of labour induction include gestational age, estimated fetal birthweight, presence of a favourable cervix, maternal age, maternal parity status, and body mass index (BMI) [1].

Although induction of labour may lower the number of caesarean sections, stillbirths, and admissions to the neonatal unit, it may also be potentially associated with excessive uterine contractions (uterine hyperstimulation) compared with no treatment [9, 10]. Excessive uterine contractions may result in uterine rupture and uterine atony [11], as well as place the fetus at risk of fetal brain oxygenation deficiency resulting in neonatal seizures [12]. Furthermore, it is reported that induced labour may be more painful than spontaneous labour [2]. The likelihood of complications is likely to differ between the different methods of induction of labour used as well as depend on the characteristics of the pregnant woman and fetus, including a history of caesarean section and preterm labour [13].

There appear to be significant differences in the effectiveness and safety of cervical ripening and labour induction methods used in 1) women with a history of caesarean delivery and 2) women without a history of caesarean delivery. For women with a previous caesarean section who require delivery before the onset of spontaneous labour, the evidence remains unclear whether labour should be induced or a repeat elective caesarean section should be performed [14]. Most of the available evidence evaluating the benefits and harms of induced labour in women with a previous caesarean section has been drawn from non-randomised studies. One systematic review of eight observational studies concluded that induction of labour is likely to increase the risk of uterine rupture/dehiscence and postpartum haemorrhage in women attempting labour after a previous caesarean section when compared to spontaneous labour [15]. Methods that are

recommended to induce labour for women with a previous caesarean section from evidence-based guidelines, such as the National Institute for Health and Care Excellence (NICE), American College of Obstetricians and Gynecologists (ACOG), and Society of Obstetricians and Gynaecologists of Canada (SOGC), vary [4, 16, 17]. There is a limited amount of evidence from randomised controlled trials on this question [18].

The indications for inducing labour differ depending on whether induction is for preterm labour (before 37⁺⁰ completed weeks of gestation) or at or beyond term labour (gestational age at or beyond 37⁺⁰ weeks). This is because these two groups differ in terms of the clinical characteristics of the mother or fetus, as well as the severity of adverse effects that may occur when each induction of labour method is used in each population [2]. For women at first pregnancy with no contraindications, physicians might attempt to inhibit labour for preterm pregnancy in order to accelerate fetal lung maturation. In women who experienced preterm premature rupture of membranes (PPROM) prior to 37⁺⁰ weeks of gestation, waiting for labour to begin naturally with careful monitoring results in better outcomes for both mother and baby when compared to early planned birth [19]. Our review thus focuses on induction of labour at or beyond term labour only.

Description of the intervention and how it might work

Methods of induction of labour are designed to induce the onset of labour by softening and maturing the cervix or stimulating uterine contractions, or both [4]. The methods currently used to induce labour can be categorised into five different groups [1, 10]:

- mechanical methods;
- surgical methods;
- pharmacological methods;
- combined methods; and
- alternative methods.

Each category has different mechanisms of action and a variety of effects. The choices of methods depend on maternal characteristics, national clinical guidelines, policies, clinical experiences, and the resources available in each healthcare setting [10].

Mechanical methods

Mechanical induction methods aim to ripen the cervix. This can be accomplished directly by cervical canal dilatation or indirectly through released prostaglandin or oxytocin secretions. The mechanical methods used for inducing labour include various types of catheters and osmotic cervical dilators [1].

The conventional Foley urinary catheter, double-balloon catheter, or extra-amniotic saline infusion can be utilised to ripen the cervix. The catheter is inserted into the cervical canal and inflated with 30 to 60 cm³ of water to dilate the cervix [20]. The balloon(s) at the end of the catheter exert(s) gentle pressure on the cervix, which can stimulate cervical ripening.

Osmotic cervical dilators are made from sterile seaweed (laminaria) or synthetic materials (Dilapan-S). This method dilates the cervical canal in the same way as Foley catheters. Once the devices are inserted into the cervical canal, they absorb water from

the surrounding cervical tissue and swell. This swelling creates pressure that gradually opens the cervix [21].

Surgical methods

Surgical methods are performed by a physician to create a tiny hole in the amniotic sac or detach the inferior pole of the membranes from the lower uterine segment. These procedures could promote the onset of labour by releasing prostaglandins and oxytocin from intrauterine tissues, which can increase uterine contractions and accelerate cervical maturation [22].

Artificial (iatrogenic) rupture of the membranes, often known as amniotomy, is the intentional rupture of the amniotic sac performed by a physician during labour. With the use of a special tool, a physician punctures the amniotic sac through the vagina. Prostaglandin levels in the uterine cavity are expected to rise as a result of this operation, which is expected to shorten the uterine muscle and allow the uterus to contract more often and more strongly [23]. However, if this technique is employed alone for labour induction, the onset of labour is uncertain [1].

Membrane sweeping is a method in which a healthcare provider inserts one or two fingers into a woman's cervix and uses a circular motion to detach the membranes from a lower uterine segment. This approach may reduce the need for formal induction of labour methods (pharmacological methods or amniotomy) [24]. The aim of membrane sweeping is to soften and ripen the cervix by releasing prostaglandins from the intrauterine tissues and increase uterine activity by releasing oxytocin [25].

Pharmacological methods

In recent decades, pharmacological methods have been the preferred approach for inducing labour due to their considered effectiveness. There are various pharmacological methods available for inducing labour. These include prostaglandin E1, prostaglandin E2, oxytocin, and nitric oxide donors. The main pharmacological effects of these medications work by stimulating uterine contractions and maturing the cervix. Some of these medications can be administered via multiple routes, such as orally, vaginally, intravenously, or intracervically [26]. The success of inducing labour and the chances of adverse outcomes may depend on the drug class and route of administration. Pharmacological methods should be used with caution for women with a history of caesarean section delivery, since they might increase the chance of uterine rupture compared to women who delivered with spontaneous labour [15].

- Prostaglandin E1: misoprostol is a synthetic prostaglandin E1 (PGE1) analogue. This medication induces labour by maturing the cervix and stimulating uterine contraction [27]. Misoprostol can be administered in multiple routes (oral, buccal, sublingual, vaginal). Misoprostol tablets are available in 25, 100, and 200 µg. These tablets can be cut into smaller pieces to administer lower doses. The optimal dose is still unknown, but for obstetric use, low doses of 25 and 50 µg are recommended to minimise adverse effects such as tachysystole [4, 26]. Each route of administration has a different mechanism and onset of action. Oral misoprostol is rapidly absorbed and has a rapid onset of action, whereas vaginal misoprostol is slowly absorbed but has a longer duration of action [28, 29]. Sublingual and buccal misoprostol appear to have a quicker onset of action than oral and vaginal misoprostol because they avoid

the first-pass metabolism by the liver [30]. Controlled-release misoprostol vaginal pessary has been developed and evaluated in clinical trials over the last decade to minimise the number of applications and vaginal examinations required during labour induction.

- Prostaglandin E2: dinoprostone is a synthetic prostaglandin E2 (PGE2) analogue. It is used to ripen the cervix and stimulate the myometrium of the gravid uterus to contract more intensely. Dinoprostone is available in various forms (gels, tablets, suppositories, or pessaries) and routes (oral, intravenous, vaginal, or intracervical) [4]. Oral and intravenous dinoprostone have been found to be associated with more gastrointestinal side effects than other routes and methods, therefore their use is not currently recommended [31, 32]. Vaginal dinoprostone and controlled-release dinoprostone vaginal pessary are two forms that are recommended in current practice.
- Prostaglandin F2 alpha: dinoprost is a synthetic prostaglandin F2 alpha (PGF2alpha) analogue. Prostaglandin F2 alpha acts via the PGF2 alpha receptor to stimulate myometrial contractility [33]. Routes of administration include oral and intracervical. Nowadays, dinoprost is partly substituted by dinoprostone.
- Oxytocin: oxytocin is a hormone made in a part of the brain that causes the uterus to contract [4]. A synthetic form of oxytocin is usually used to induce labour via the intravenous route when the cervix is favourable (Bishop score > 6). There are two regimens for inducing labour: lower dose and higher dose. A lower-dose regimen typically involves an initial dose of 0.5 to 2.0 milliunits (mU)/minute, followed by a maintenance dose of 1.0 to 2.0 mU/min every 15 to 60 minutes. A higher-dose regimen involves an initial dose of 6.0 mU/min, followed by a maintenance dose of 2.0 to 6.0 mU/min every 15 to 40 minutes [26]. A higher-dose regimen likely leads to a shorter time to delivery, but might increase the risk of adverse effects such as hyperstimulation [26].
- Progesterone antagonists (also known as progesterone blockers): progesterone is a hormone that decreases uterine contractility. Mifepristone is one progesterone antagonist agent that can inhibit progesterone activities, thus increasing uterine contractions and sensitivity of the uterus to the actions of prostaglandins [34].
- Nitric oxide: nitric oxide donors, including isosorbide mononitrate (ISMN), isosorbide dinitrate (ISDN), nitroglycerin, and sodium nitroprusside, are considered to be potential mediators for cervical ripening while having no impact on uterine contraction. The relaxation of cervical smooth muscle is the main mechanism of this drug [35].

Combined methods

Combination methods for labour induction involve the simultaneous or sequential use of multiple techniques to enhance the effectiveness of the induction process. These approaches often integrate both mechanical and pharmacological methods. Two combined methods are recommended for use in clinical practices: 1) oxytocin plus amniotomy and 2) balloon catheter plus oxytocin. The NICE guideline recommends the use of oxytocin plus amniotomy for women with a favourable cervix (Bishop score > 6) [2], while the World Health Organization (WHO) recommends balloon catheter plus oxytocin as an alternative when misoprostol or dinoprostone are unavailable or contraindicated [3].

Three commonly studied combinations include: 1) oxytocin plus amniotomy, 2) balloon catheter plus misoprostol, and 3) balloon catheter plus dinoprostone [26]. Evidence from randomised controlled trials and systematic reviews suggests that combined methods, particularly balloon catheter combined with misoprostol, may reduce the time to delivery compared to single induction methods [26]. However, the safety profile of these combination approaches remains uncertain, with limited data on potential adverse outcomes such as uterine hyperstimulation with fetal heart rate changes, pain related to induction of labour, and perinatal mortality [36].

Alternative methods

Acupuncture, castor oil, herbal supplements, homoeopathy, and sexual intercourse are considered to be alternative methods for inducing labour. The international and national evidence-based guidelines state that there is still insufficient information regarding their efficacy and safety, and therefore do not recommend using them to induce labour [2, 3].

Why it is important to do this review

There is one non-Cochrane review assessing the effectiveness and safety of various methods of induction of labour using network meta-analysis (NMA) [9, 10]. That review included 611 trials (900 reports), and the effectiveness and safety of 31 active interventions for third-trimester labour induction were assessed. The NMA findings suggested that the most effective interventions in achieving vaginal births were oxytocin with amniotomy and vaginal misoprostol. Insufficient data were available or there was too much uncertainty to determine which treatments were the safest. However, the recommendations of that review were not made in conjunction with the GRADE approach [37, 38].

As further trials evaluating the methods of induction of labour have been published in recent years, we proposed to conduct a systematic review and NMA to update the evidence regarding the effectiveness and safety of the methods for induction labour for at or beyond term labour with or without previous caesarean section, and provide the evidence summaries using the GRADE approach. In addition, we only included trials with trusted data, screened using the Cochrane Pregnancy and Childbirth Trustworthiness Screening Tool.

OBJECTIVES

To compare the benefits and harms of various cervical ripening and induction of labour methods at or beyond term labour and to rank the methods.

METHODS

The protocol of this review was published in 2023 [39]. Some methods in this protocol have changed (see [Supplementary material 9](#)). This review was reported in accordance with the reporting guidance provided in the PRISMA Extension Statement for Reporting of Systematic Reviews Incorporating Network Meta-analyses of Health Care Interventions (PRISMA-NMA) [40].

Criteria for considering studies for this review

Types of studies

We included randomised controlled trials (RCTs) and cluster-RCTs. Studies published in abstract form were eligible if sufficient information was provided. We excluded cross-over trials, quasi-randomised trials (trials in which the investigator randomly allocates participants to intervention or control groups using an allocation method that falls short of genuine randomisation, including date of birth), non-randomised trials/quasi-experimental studies (studies in which the investigator assigns participants to intervention or control group without randomisation techniques), and observational studies.

Types of participants

All pregnant women, singleton or multiple, with a live fetus requiring cervical ripening and induction of labour at or beyond term labour (gestational age $\geq 37^{+0}$ weeks). We included studies that recruited women with or without health conditions or complications in this review. We included induction of labour, either in outpatient or inpatient settings, in this review. We did not include studies involving premature or mixed-term labour where the term and preterm labour data cannot be separated. We included trials that recruited only women with previous caesarean section and trials that recruited only women without previous caesarean section. For trials that recruited a mix of women with or without previous caesarean section, we included such trials if over 50% of women did not have previous caesarean section.

During study screening, we identified two types of studies that had different research objectives and different characteristics compared to the majority of studies and thus decided to exclude them from our review. Firstly, we did not include trials that compared the effects of labour induction methods in women who did not respond to a first labour induction method. The reason for this exclusion was that it could introduce heterogeneity and bias to network meta-analysis due to different prognostic profiles. Secondly, we also excluded trials that compared labour induction methods with expectant management initiated among women with different gestational ages. Such studies typically aim to determine the optimal time point for inducing labour, particularly among late-term pregnancies, rather than comparing the effects of labour induction methods.

Types of interventions

We included all cervical ripening and induction of labour methods recommended in any of the five guidelines of international reach ([Supplementary material 10](#)). We used the following guidelines of international reach to define our interventions of direct interest: ACOG guidelines [41, 42], NICE guideline [2], Royal College of Obstetricians and Gynaecologists (RCOG), International Federation of Gynecology and Obstetrics (FIGO) [43], and WHO guidelines [3, 11, 44].

This review also included any other cervical ripening and labour induction methods where Cochrane reviews have recommended further research to determine their true effects on important labour processes and delivery outcomes ([Supplementary material 11](#)). However, we did not include methods that are not currently used in clinical practice (such as extra-amniotic prostaglandins, intravenous prostaglandin, and relaxin) or that are considered to

be alternative methods (such as castor oil or enemas, hypnosis, acupuncture, and homoeopathy) [10].

We anticipate that the methods recommended in these five guidelines and those proposed by Cochrane reviews will be sufficient to identify all available methods that are relevant to current clinical practice. Eligible interventions that were identified from the five guidelines and those proposed by Cochrane reviews are listed below.

- Oral misoprostol ($\leq 50 \mu\text{g}$)
- Vaginal misoprostol ($\leq 50 \mu\text{g}$)
- Sublingual or buccal misoprostol ($\leq 50 \mu\text{g}$)
- Controlled-release misoprostol vaginal pessary
- Vaginal dinoprostone (tablet or gel)
- Controlled-release dinoprostone vaginal pessary
- Amniotomy (alone)
- Oxytocin (alone)
- Nitric oxide donors
- Balloon catheters (Foley urinary catheters, double-balloon devices, and extra-amniotic saline infusion)
- Osmotic cervical dilators
- Oxytocin plus amniotomy
- Balloon catheters plus oxytocin
- Balloon catheters plus misoprostol ($\leq 50 \mu\text{g}$; oral or vaginal routes)
- Comparator interventions (placebo, no treatment, or expectant management)

In the protocol, we did not intend to limit the dosages of vaginal misoprostol. However, during the review process, we discovered several trials that evaluated the effects of high-dose vaginal misoprostol ($> 50 \mu\text{g}$). As this dosage is deemed unsafe in modern obstetrics [13, 45], we excluded these trials from our review.

We did not include membrane sweeping in this review because it is considered to be an adjunctive treatment employed to avoid the use of formal induction of labour methods. A Cochrane review evaluating the effects of membrane sweeping for labour induction was recently published [24].

We had planned to include cervical ripening and induction of labour methods that emerge during the conduct of the review if they are included in revised editions of ACOG, NICE, RCOG, FIGO, or WHO guidelines. During the review process, we only found one updated guideline from the WHO [44], which did not recommend any new methods for cervical ripening or labour induction. Therefore, we evaluated only the labour induction methods that were mentioned earlier for their effects in this review.

Trials were eligible if they administered cervical ripening and induction of labour methods mentioned in the list of eligible interventions at or beyond term labour and compared them with another method listed in the intervention list, placebo, no treatment, or expectant management (waiting for labour to start spontaneously without treatments for inducing labour). We amalgamated placebo, no treatment, and expectant management into a single comparator node.

We combined different dosages of misoprostol, dinoprostone, and oxytocin into the same node. We did not include studies

comparing different dosages of the same interventions (e.g. $25 \mu\text{g}$ oral misoprostol versus $50 \mu\text{g}$ oral misoprostol).

We included multi-arm trials in this review. We assume that any participant who meets the inclusion criteria of the review is equally likely to be randomised to any eligible intervention.

Outcome measures

We included studies that met our eligibility criteria regardless of whether or not they reported on our outcomes of interest. The outcomes of this review were based on the core outcome set for trials on induction of labour [46], and priority outcomes for decision-making proposed by the WHO [11].

Critical outcomes

For woman

- Failure to achieve vaginal delivery within 24 hours (from the start of cervical ripening or labour induction).
- Caesarean section due to any causes.
- Caesarean section due to non-reassuring fetal status.
- Uterine hyperstimulation with changes in the heartbeat of the baby before birth.

For infant

- Perinatal death (defined as the sum of early neonatal death and stillbirth).
- Severe neonatal morbidity as defined by trial authors.

Main outcomes for summary of findings tables

We planned to present all six critical outcomes in the summary of findings tables.

Important outcomes

For woman

- Failure to achieve vaginal delivery with the primary method.
- Maternal not satisfied with the care related to induction of labour (defined as the dissatisfaction of the woman related to induction of labour experience, measured by any measurement before discharge of the woman from the postpartum ward).
- Time from induction of labour to delivery (measured as hours).
- Time from induction of labour to vaginal delivery (measured as hours).
- Blood loss of 500 mL or more.
- Blood loss of 1000 mL or more.
- Mean volume of blood loss (mL).
- Uterine scar dehiscence or rupture (defined as an incomplete uterine scar separation with intact serosa and a full-thickness disruption of the uterine wall, respectively).
- Oxytocin augmentation.
- Maternal death (defined as death during and following childbirth within 42 days).
- Admission to intensive care unit.
- Pain related to induction of labour.

For infant

- Use of mechanical ventilation.

- Admission to the neonatal intensive care unit (NICU).
- Apgar score < 7 at 5 minutes.

Search methods for identification of studies

Electronic searches

We searched the Cochrane Pregnancy and Childbirth Trials Register in collaboration with their Information Specialist (1 February 2023).

The Cochrane Pregnancy and Childbirth Trials Register is a database containing over 34,000 reports of controlled trials in the field of pregnancy and childbirth. It represents the contemporary results of over 30 years of searching, including handsearched journals and conference proceedings, and journals reviewed via a current awareness service. For the detailed search strategies used to populate Cochrane Pregnancy and Childbirth's Trials Register for the Cochrane Central Register of Controlled Trials (CENTRAL), MEDLINE, Embase, and CINAHL (Cumulative Index to Nursing and Allied Health Literature), see [Supplementary material 1](#).

Briefly, the Cochrane Pregnancy and Childbirth's Trials Register was maintained by their Information Specialist and contains trials identified from the following.

- Monthly searches of CENTRAL. CENTRAL contains Cochrane's centralised searches of MEDLINE/PubMed, Embase, CINAHL, ClinicalTrials.gov, and the WHO International Clinical Trials Registry Platform (ICTRP).
- Weekly searches of MEDLINE (Ovid).
- Weekly searches of Embase (Ovid).
- Monthly searches of CINAHL (EBSCO).
- Regular scanning of 30 journals and the proceedings of major conferences.
- Weekly current awareness alerts for a further 44 journals plus monthly BioMed Central email alerts.

Search results are screened, and the full texts of all relevant trial reports identified through the searching activities described above are reviewed. Based on the intervention described, each trial report is assigned a number that corresponds to a specific Cochrane Pregnancy and Childbirth review topic (or topics), and is then added to the Register. The Information Specialist searches the Register for each review using this topic number rather than keywords. This results in a more specific search set that has been fully accounted for in the relevant review sections (Included, Excluded, Awaiting Classification, or Ongoing).

In addition, we searched ClinicalTrials.gov (clinicaltrials.gov/) and the WHO ICTRP (apps.who.int/trialsearch/) for unpublished, planned, and ongoing trial reports using the search methods detailed in [Supplementary material 1](#).

We did not apply any language or date restrictions.

Searching other resources

We reviewed the reference lists of both included and excluded studies from the previous NMA study [10] to identify any additional trials. We also checked the reference lists of all retrieved studies for further relevant studies.

Data collection and analysis

Selection of studies

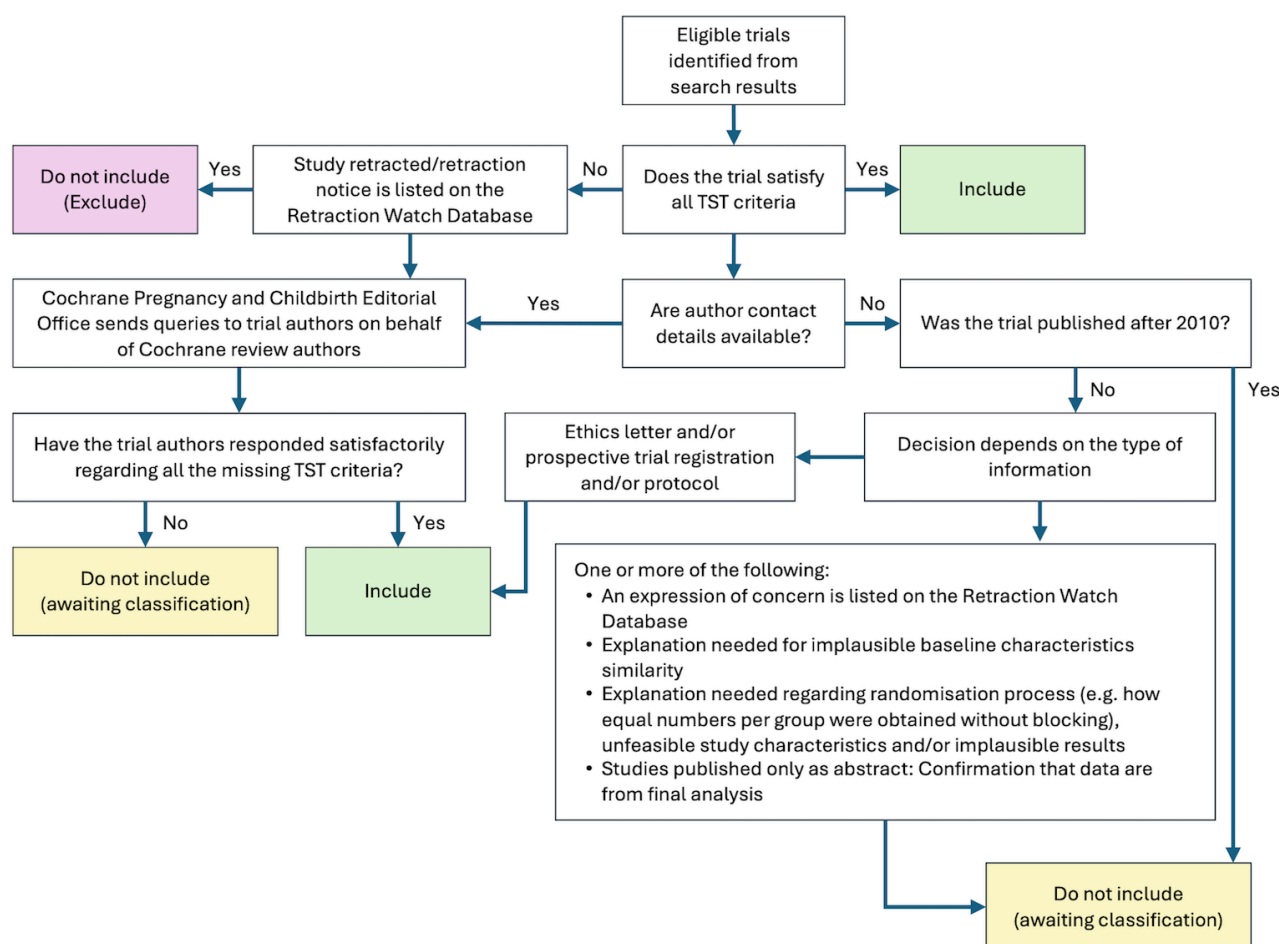
Study selection

We used Mendeley software to merge reference lists [47], after which at least two review authors (of SR, CK, NE, KLS, and KNT) independently screened the titles and abstracts and full articles via Covidence software [48]. If eligibility was unclear, we contacted the trial authors for additional information. Any disagreements were resolved by discussion or by consulting a third review author if required. We created a study flow diagram to illustrate the process of study selection.

Screening eligible studies for trustworthiness

Two review authors (of SR, CK, NE, KLS, and KNT) independently assessed the trustworthiness of all studies that met our review criteria against the Cochrane Pregnancy and Childbirth Trustworthiness Screening tool (CPC-TST) (see more details in our protocol). We performed this screening process to identify potential research integrity issues among trials that met the review criteria and ensure that only trustworthy trials were included in the review. Research integrity issues may include incorrect analyses, falsified methods or data, or manipulated results. As a result, only studies that were deemed trustworthy were included in our analysis.

For trials that were classified as having a 'high risk' for one or more of the CPC-TST criteria, we tried to contact the authors to address any lack of information or concerns. If we were not able to obtain sufficient information, we categorised the trial as 'awaiting classification' and provided detailed descriptions of our concerns ([Supplementary material 4](#)). The process is described in full in [Figure 1](#).

Figure 1. The Cochrane Pregnancy and Childbirth Trustworthiness Screening Tool (CPC-TST) criteria.

Data extraction and management

We used Microsoft Access to create an electronic data extraction form that was piloted first before being implemented [49]. At least two review authors (of SR, CK, NE, KLS, and KNT) independently extracted information from the eligible studies using the agreed-upon form. The data, specifically for numerical data, short text fields such as author name and email address, data from drop-down lists, and Yes/No data, were then imported into Stata software for Mac version 18.0 to check for discrepancies between the two review authors, which was done by one review author (SR) [50]. Any disagreements were resolved by discussion or by consulting a third review author if required. If additional information was needed, we contacted the trial authors of the original reports.

Risk of bias assessment in included studies

Two review authors (of SR, CK, NE, KLS, and KNT) independently assessed the risk of bias in each included study using the Cochrane RoB 1 tool for randomised trials and for cluster-randomised trials [51]. After the risk of bias assessment, review authors recorded their judgements in an electronic data extraction form. The risk of bias judgements were then imported into Stata software for Mac version 18.0 to check for discrepancies between the two review authors, which was done by one review author (SR) [50]. Any disagreements were resolved by discussion or by consulting a third review author if required. We have changed the criteria used for judging the risk

of bias in two areas: 1) blinding of participants and personnel and 2) blinding of outcome assessment. Full details are available in [Supplementary material 9](#).

Measures of treatment effect

Relative treatment effects

Dichotomous data

For studies reporting zero events in one arm, we added a correction factor of 0.5 into all cells of the study results to estimate the effect sizes. We excluded studies with zero events in all arms from the statistical analysis.

We expressed the treatment effects of outcomes as risk ratio (RR) and 95% confidence intervals (CIs). We presented NMA estimates of all dichotomous outcomes in league tables, which show summary RRs and CIs for all pair-wise comparisons. We also used forest plots to present the results from pair-wise, indirect, and network (combining direct and indirect) analyses when comparing methods for inducing labour with vaginal misoprostol ($\leq 50 \mu\text{g}$), when data for this method were available. We used vaginal misoprostol ($\leq 50 \mu\text{g}$) as the common comparator for analyses because it is one of the commonly used methods for labour induction, as recommended by WHO guidelines on induction of labour 2011 and 2013 [3, 52]. In addition, based on the information in this review, we found that vaginal misoprostol ($\leq 50 \mu\text{g}$) was the method that most included

studies used to compare with other methods. If vaginal misoprostol ($\leq 50 \mu\text{g}$) was not available, we used other methods, such as oral misoprostol ($\leq 50 \mu\text{g}$), or comparator interventions (placebo, no treatment, or expectant management), as reference methods.

Continuous data

For time from induction of labour to delivery, time from induction of labour to vaginal delivery, and volume of blood loss outcomes, we used mean difference (MD) if outcomes were measured using the same unit of measurement across trials.

We presented NMA estimates of treatment effects for all continuous outcomes in the league tables, which show the summary MD for all pair-wise comparisons with CIs. We also used forest plots to present the results from pair-wise, indirect, and network (combining direct and indirect) analyses for the comparisons of induction of labour methods versus vaginal misoprostol ($\leq 50 \mu\text{g}$).

Relative treatment ranking

We used the surface under the cumulative ranking curve (SUCRA), a Bayesian statistical method for estimating treatment ranking, to rank the effects of all interventions on all critical outcomes. SUCRA value indicates the cumulative probability of efficacy (or safety) for each intervention when compared to the other available interventions. SUCRA values range from 0.00 to 1.00, where a higher value indicates a more effective treatment [40, 53]. In addition to the SUCRA value, we presented mean ranks, the mean of the distribution of the ranking probabilities, with their 95% CI for each intervention. To estimate the 95% CI of ranking for each intervention in the networks, we determined the lowest rank whose cumulative probability was greater than 2.5% and the highest rank whose cumulative probability was less than 97.5%. After conducting an NMA, we ranked treatments with 10,000 iterations in order to increase the precision of ranking probabilities. To illustrate the distribution of ranking probabilities for each intervention and outcome, rankograms were generated using the *sucra* command in Stata software [54].

Unit of analysis issues

Cluster-randomised trials

We planned to include cluster-randomised trials in the analyses along with individually randomised trials. However, we only identified individually randomised trials for inclusion.

Other unit of analysis issues

Trials with multiple intervention groups

We included multiple intervention trials in the review, and accounted for the correlations between the treatment effects in NMA. We treated multi-arm trials as multiple independent two-arm studies in standard pair-wise meta-analysis.

Trials with multiple pregnancies

We planned to include trials with multiple pregnancies in the review. However, out of the 106 included studies, 105 trials recruited women with a singleton pregnancy, and the remaining trial did not specify this information.

Dealing with missing data

We noted levels of attrition in the included studies. We planned to explore the impact of including studies with high levels of missing data in the overall assessment of treatment effect through sensitivity analysis. However, only two studies had a high level of dropout rate (over 20% in all intervention arms). This was due to approximately 30% of participants in each group of these two studies dropping out after randomisation due to protocol violations. We believed that our analysis results would not be affected by excluding these studies, therefore we did not perform this planned sensitivity analysis. For all outcomes, we carried out analyses on an intention-to-treat basis as far as possible, that is we attempted to include all participants randomised to each group in the analyses, and all participants were analysed in the group to which they had been allocated, regardless of whether or not they had received the allocated intervention. The denominator for each outcome in each trial was the number of pregnant women randomised minus any participants whose outcomes were known to be missing.

We first tried to contact the first or corresponding authors of each trial to obtain missing or incomplete data. For trials reporting their findings in graphs only, we used Plot Digitizer software to extract data from graphs [55]. If trials did not report standard errors, we estimated standard errors from the width of CIs. In cases where an included study reported its treatment effects as median and range or interquartile range, we estimated mean and standard deviation from those values using the methods proposed by Wan and colleagues [56].

Reporting bias assessment

For the pair-wise meta-analysis of critical outcomes, we created funnel plots to investigate reporting biases (such as publication bias) if 10 or more studies were included in the meta-analysis. We assessed funnel plot asymmetry visually. If a visual assessment suggested asymmetry, we planned to perform sensitivity analysis to investigate small-study effects by excluding studies with fewer than 400 participants. For NMA of critical outcomes, we examined small-study effects by plotting comparison-adjusted funnel plots, which we assessed visually. We hypothesised that publication bias may occur because studies with positive findings, which compared active labour induction methods with the common method (vaginal misoprostol ($\leq 50 \mu\text{g}$)), are more likely to be published. Thus, we established vaginal misoprostol ($\leq 50 \mu\text{g}$) as a common comparator in funnel plots and compared it with the other active labour induction methods. We planned to examine small-study effects, but did not do so because we did not identify any issues in our analyses.

Synthesis methods

Because there are significant differences in effectiveness, safety, and methods used for cervical ripening and labour induction among 1) women without a history of caesarean delivery and 2) women with a history of caesarean delivery, we analysed and reported the findings separately for these two populations [1, 4, 13]. Trials that recruited a mixed population where more than 50% of participants had no previous caesarean section were categorised as 'women without a history of caesarean delivery'. Consequently, this review presents separate analyses and findings for women:

- without a history of caesarean delivery and a mix of women with or without a history of caesarean delivery;
- with a history of caesarean delivery

For each outcome, we considered whether an overall summary result was meaningful.

Methods for direct treatment comparisons

We conducted standard pair-wise meta-analyses with a random-effects model for all critical and important outcomes with at least two available trials [57]. If there was clinical heterogeneity or considerable statistical heterogeneity, we planned to synthesise the findings narratively and present the results using forest plots without calculating an overall estimate. We used RevMan software for all analyses [58].

Methods for indirect treatment comparisons

Before conducting an NMA, we initially evaluated feasibility by generating network plots. We used the *network map* command in Stata software to create network plots for all outcomes [59]. For each network plot that was suitable for conducting an NMA, and where the distribution of effect-modifying factors across all comparisons met the transitivity assumption, we performed an NMA.

We performed random-effects network meta-analyses within a frequentist framework by pooling study-specific treatment effects. A meta-regression model that includes covariates was used to determine the relative effects and ranking of all available induction methods. Using this modelling approach, the correlations between effect sizes from multi-arm studies are addressed. We used Stata software for Mac version 18.0 with the *network* suite of commands to analyse all NMAs [59].

To assess the presence of clinical heterogeneity, we used descriptive statistics to present the distributions of study and participant characteristics across all included studies. We examined the presence of clinical heterogeneity by comparing this information between each pair-wise comparison. We considered the following characteristics: gestational age, parity, and cervical conditions.

We examined statistical heterogeneity in each pair-wise meta-analysis by visual inspection of the overlap of CIs in the forest plots. We also quantified heterogeneity levels using the I^2 statistic. We interpreted the I^2 statistic according to the guidelines in the *Cochrane Handbook for Systematic Reviews of Interventions* [60], as follows:

- 0% to 40%: might not be important;
- 30% to 60%: may represent moderate heterogeneity;
- 50% to 90%: may represent substantial heterogeneity;
- 75% to 100%: represents considerable heterogeneity.

For NMA, we assessed statistical heterogeneity in the entire network by considering the magnitude of the between-study variance parameter (τ^2) derived from the NMA models. We compared the magnitude of the heterogeneity variance with empirical distribution for dichotomous outcomes, as proposed by Turner and colleagues [61].

Assessment of transitivity across treatment comparisons

We assessed the transitivity assumption by comparing the distribution of potential effect modifiers across the available direct comparisons in the network. In this context, we expect that the transitivity assumption will hold assuming that: 1) the common treatment used to compare different methods for cervical ripening and labour induction indirectly is similar when it appears in different trials (e.g. oral misoprostol is administered in a similar way in oral misoprostol versus vaginal dinoprostone trials and in oral misoprostol versus oxytocin plus amniotomy trials); 2) all direct comparisons in the network do not differ with respect to the distribution of effect modifiers (e.g. the effect modifiers of oral misoprostol versus vaginal dinoprostone trials and of oral misoprostol versus oxytocin plus amniotomy trials). We considered the following potential effect modifiers.

- Gestational age (term labour/post-term labour/mixed)
- Parity (nulliparous/multiparous/mixed)
- Cervical conditions: as defined by trialists using Bishop score (unfavourable/favourable/mixed)

We evaluated the transitivity assumption by comparing the distribution of these three prespecified potential effect modifiers across the different pair-wise comparisons. If all participants in the NMA had similar prespecified effect modifiers, we accepted the transitivity assumption.

Assessment of statistical inconsistency

We evaluated inconsistency in NMA using local and global approaches to complement the evaluation of transitivity.

Local approaches for evaluation of inconsistency

We applied the loop-specific approach to assess the presence of local inconsistency [62]. This method evaluates the consistency assumption in each closed network loop separately by estimating the absolute difference between direct and indirect comparisons (inconsistency factor: IF). $P < 0.05$ indicates evidence of inconsistency between direct and indirect estimates. Due to the low power of this test to detect inconsistency, we interpreted the findings cautiously and looked for inconsistency by considering the range of values contained in CIs of the incoherence factors and statistical significance [40, 53]. We assumed a common heterogeneity estimate within each loop. We used the *ifplot* command in Stata software for Mac version 18.0 to perform loop-specific testing [54].

Global approaches for evaluation of inconsistency

We used the design-by-treatment interaction model to evaluate the assumption of global consistency in each network [63]. We inferred the presence of inconsistency if the P value from the design-by-treatment model was less than 0.05. We used Stata software for Mac version 18.0 with the *mvmeta* command to fit the model [64].

Investigation of heterogeneity and subgroup analysis

Assumptions when estimating heterogeneity

In standard pair-wise meta-analyses, we estimated a different heterogeneity variance for each pair-wise comparison. In NMA, we assumed a common estimate for the heterogeneity variance across the different comparisons.

Subgroup analysis

We performed subgroup analyses by evaluating the relative effects of the following factors for the six critical outcomes.

- Gestational age (early term and full term (37⁺⁰ to 40⁺⁶ weeks)/late term and post-term (≥ 41⁺⁰ weeks)/mixed)
- Parity (nulliparous/multiparous/mixed)
- Cervical conditions: as defined by trialists using Bishop score (unfavourable/favourable/mixed)
- Setting of induction of labour (inpatient setting/outpatient setting/trials comparing between inpatient and outpatient settings)
- Membrane status (intact membrane/ruptured membrane/mixed)

We first visually compared the network diagrams of each subgroup to determine subgroup differences. Following that, for each subgroup, we performed a pair-wise and network meta-analysis and evaluated their relative treatment effects and ranking. We looked qualitatively for interactions between subgroups in which the direction of effect might be reversed, that is if an intervention was beneficial to one subgroup but harmful to another.

Equity-related assessment

We did not plan to investigate health inequity in this review.

Sensitivity analysis

We performed the following sensitivity analyses for all critical outcomes.

- Risk of bias: we removed trials that were judged to be at overall high or unclear risk of bias. We judged studies as being at overall low risk of bias if they were at low risk of bias for random sequence generation and allocation concealment.
- Mixed population studies: we kept studies that only recruited women without a previous caesarean section.
- Split the comparator interventions node: we split interventions in the comparator interventions node and treated them separately as three nodes (placebo, no treatment, and expectant management).
- Split pharmacological methods according to dosage (low dose/high dose): we split pharmacological methods as follows and treated them separately as six nodes:
 - oral misoprostol (≤ 25 µg oral misoprostol/> 25 µg oral misoprostol);
 - vaginal misoprostol (≤ 25 µg vaginal misoprostol/> 25 µg vaginal misoprostol);
 - sublingual or buccal misoprostol (≤ 25 µg sublingual or buccal misoprostol/> 25 µg sublingual or buccal misoprostol).
- Split the balloon catheters node: we split interventions in the balloon catheters node and treated them separately as two nodes (single balloon catheters and double balloon catheters).
- Randomisation unit: we planned to remove cluster-randomised trials to examine the effect of the randomisation unit. However, we did not identify any cluster-RCTs.
- Abstract-only data: we planned to remove trials presented only as abstracts from the analyses. We identified two studies that were published as conference abstracts. After we contacted the

corresponding authors of these studies, they provided us with the full unpublished manuscripts, and we used the information from these manuscripts in our review. Consequently, we did not perform this planned sensitivity analysis.

Certainty of the evidence assessment

We initially planned to create summary of findings tables presenting the evidence for all critical outcomes, with separate presentations for 1) women without a previous caesarean section and a mix of women with or without previous caesarean section, and 2) women with previous caesarean section in this review. However, based on the suggestions from the reviewers and editors, we decided to present only select summary of findings tables in the main review, with the remaining tables included in the supplementary materials ([Supplementary material 16](#); [Supplementary material 17](#); [Supplementary material 18](#); [Supplementary material 19](#)).

In the main review, we have presented summary of findings tables for four critical outcomes among women without a previous caesarean section and a mix of women with or without previous caesarean section:

- failure to achieve vaginal delivery within 24 hours;
- caesarean section due to non-reassuring fetal status;
- uterine hyperstimulation with changes in the heartbeat of the baby before birth; and
- perinatal death.

As a result, the findings summarised in the Abstract, Plain language summary, Summary of main results, and Implications for practice sections focus primarily on these four outcomes for women without a previous caesarean section and a mix of women with or without previous caesarean section.

Each summary of findings table presents evidence comparing all cervical ripening and induction of labour methods to a reference comparator, vaginal misoprostol (≤ 50 µg), when data for this method were available. We used vaginal misoprostol (≤ 50 µg) as the reference because it is one of the commonly used methods for labour induction, as recommended by WHO guidelines on induction of labour 2011 and 2013 [3, 52]. Furthermore, it was the most commonly used comparator across the included studies.

In some analyses, such as caesarean section due to any cause among women with a previous caesarean section, vaginal misoprostol (≤ 50 µg) was not available in the network and therefore could not be used as the reference method. In such cases, we used alternative comparators – either balloon catheters (for women with a previous caesarean section) or inactive interventions (placebo, no treatment, or expectant management) – depending on data availability. Current evidence suggests that no induction of labour method is superior for women with a previous caesarean section in terms of overall benefits [65]. However, we chose balloon catheters as the reference method in this review for women with a previous caesarean section because they were a common comparator in the included studies and, given their likely lower risk of uterine scar rupture, are likely to be recommended in practice [66].

Each summary of findings table presents key features of the evidence relating to a single outcome, with one table for each of our critical outcomes, in accordance with the GRADE approach. We

used the GRADE approach for network meta-analysis to assess the certainty of effect estimates from NMA [38, 67]. There were three steps for grading evidence using the GRADE approach.

- We assessed the certainty of the direct evidence for each critical outcome based on five domains of the standard GRADE approach: 1) limitations in study design (risk of bias), 2) inconsistency of results, 3) indirectness of evidence, 4) imprecision, and 5) publication bias.
- We assessed the certainty of indirect evidence if either the indirect evidence contributed more to the estimates than direct evidence, or the direct evidence showed a lack of certainty (moderate, low, or very low certainty). To assess the certainty of indirect evidence, we selected the most dominant first-order loop (the closed loop of three interventions where all three are directly compared to each other) and rated the certainty of each direct comparison from that loop by applying the five domains of the standard GRADE approach. We chose the lowest of the two ratings to represent the certainty of indirect evidence. Finally, we examined the differences in study characteristics that may affect the treatment effect on the direct comparisons that form the basis of the indirect estimate.
- To assess the certainty of network evidence, we first selected the rating of the most precise of the indirect and direct estimates, or the highest of these two ratings if both contributed similarly. This rating was then adjusted based on 1) the coherence of effect estimates from direct and indirect estimates and 2) the precision of the network estimate.

At each of the above stages, one review author (SR) appraised the certainty ratings of direct, indirect, and network evidence. The certainty of evidence assessed by SR was then reviewed by other review authors. We rated the certainty of network evidence for each outcome as high, moderate, low, or very low, in accordance with the GRADE approach and as explained below.

- High certainty: we are very confident that the true effect lies close to that of the estimated effect.

- Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimated effect, but there is a possibility that it is substantially different.
- Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimated effect.
- Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimated effect.

Consumer involvement

We organised a discussion forum with a small group of obstetricians and gynaecologists working in clinical settings, universities, and policymaking organisations to discuss and share their ideas on our preliminary findings. The clinicians informed us that the priority outcomes they would like to see for induction of labour are uterine hyperstimulation with changes in the baby's heartbeat and vaginal delivery without a time interval. Although we did not plan to evaluate the effects of induction of labour methods on vaginal delivery without a time interval in our review, we have information on the caesarean section outcome, which is the opposite direction outcome of vaginal delivery. As a result, we decided to set the caesarean section as one of our critical outcomes. In addition, we followed the core outcome set for trials on induction of labour, which was developed with consumer involvement (obstetricians, neonatologists, midwives, and women's representatives), to set the review outcomes.

RESULTS

Description of studies

Results of the search

The search results are summarised in the PRISMA flow diagram (Figure 2).

Figure 2. PRISMA flow diagram.

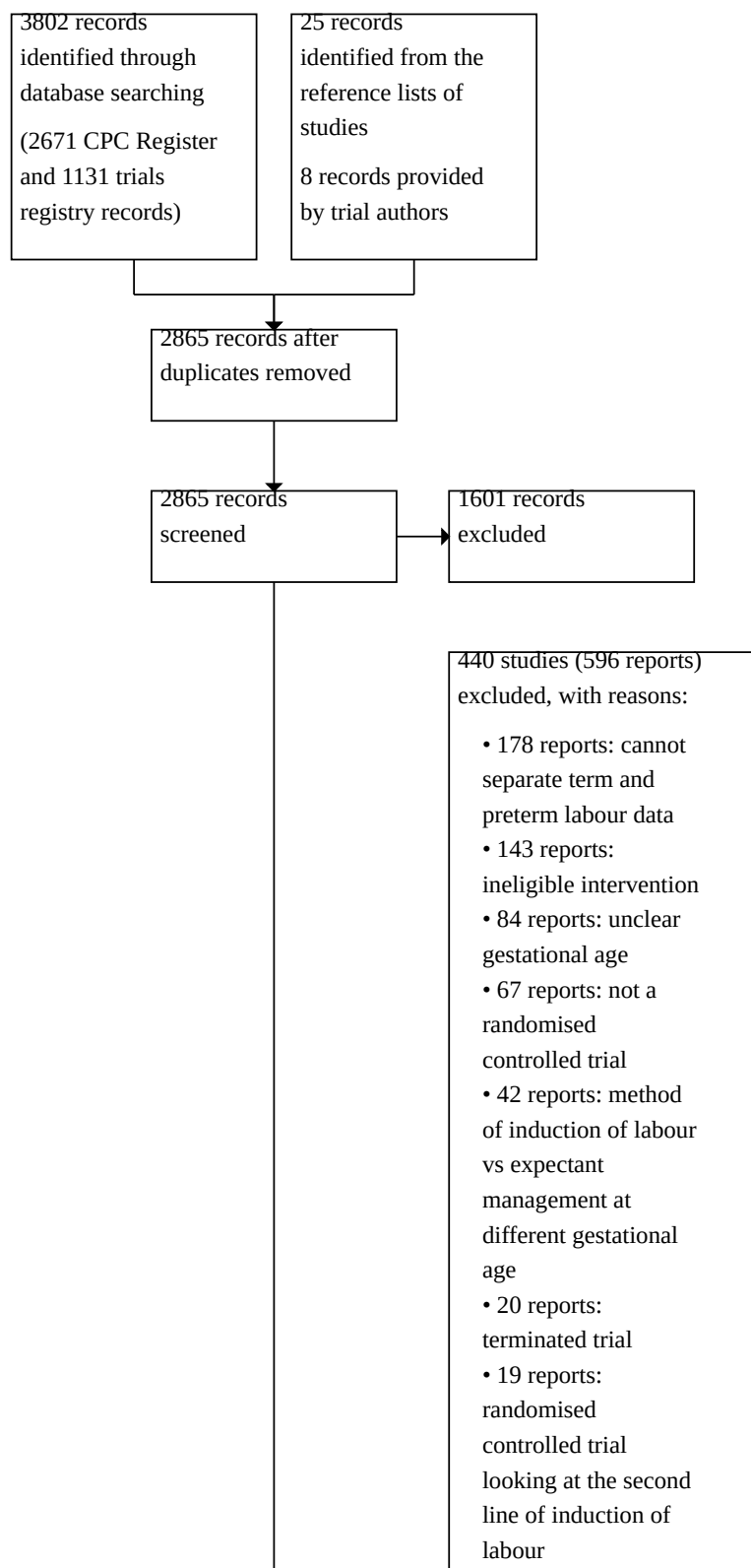


Figure 2. (Continued)

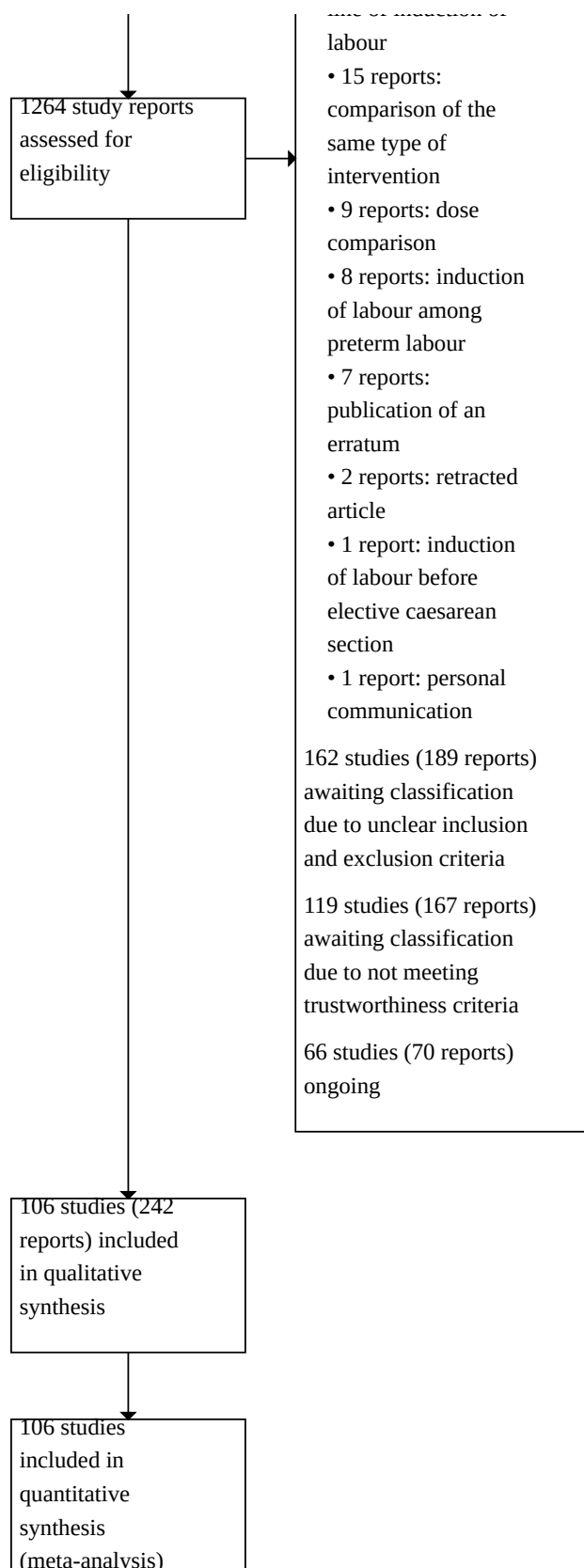


Figure 2. (Continued)

synthesis (meta-analysis)

The search of Cochrane Pregnancy and Childbirth's Trials Register on 1 February 2023 retrieved 2671 records. We also found 1131 records through our searches of the trial registration databases. Additionally, we retrieved 25 records from reference lists of studies and eight records from trial authors. After removing duplicate records, we screened the titles and abstracts of 2865 records, of which 1601 records were excluded. We reviewed the full texts of the 1264 reports that remained. We contacted the trial authors of 188 studies (214 reports) to obtain additional information or clarification about their eligibility criteria, interventions, or methods. We attempted to contact the study authors twice, with the second attempt made two weeks after the first. Of these 188 studies, we were unable to obtain the requested data from trial authors for 162 studies (189 reports) and thus listed these studies as awaiting classification (see [Supplementary material 4](#)). Sixty-six studies (70 reports) are ongoing (see [Supplementary material 5](#)). We excluded 440 studies (596 reports) from this review, with reasons provided in the PRISMA flow diagram ([Figure 2](#)).

A total of 225 studies (409 reports) satisfied our review criteria, excluding ongoing studies and studies awaiting classification. Sixty-nine studies satisfied the Cochrane Pregnancy and Childbirth Trustworthiness criteria. For the remaining 156 studies that did not meet the trustworthiness criteria, we tried contacting the first or corresponding authors. We made two attempts to contact the study authors, with the second attempt made two weeks after the first. We could not find contact details for 23 studies (32 reports) published before 2011, and therefore listed them as awaiting classification. We received satisfactory responses for 37 studies, resulting in 96 studies (135 reports) being reclassified as awaiting classification. In total, 119 studies (167 reports) were reclassified as awaiting classification at this stage (see [Supplementary material 4](#)). [Supplementary material 12](#) presents details for each item of the Cochrane Pregnancy and Childbirth Trustworthiness Screening Tool for the 119 studies awaiting classification.

The reasons that 96 studies did not meet the trustworthiness criteria are listed below.

- After our request, we found that nine studies published after 2010 lacked prospective trial registration and did not provide evidence of an approved protocol.
- Six studies were retrospectively registered with a clinical trial database after they were conducted.
- Eight studies did not fully explain their randomisation process, so that we could not confirm how equal numbers of participants were randomised for each arm of the trial.
- One study required clarification regarding the gestational age of the participants. The study recruited women who were more than 37 weeks pregnant. However, the gestational age means of the two study groups were both around 38 weeks, with standard deviations of approximately 1.3 weeks. These findings appeared to be untrustworthy.

- One study did not report the study period, preventing a determination of whether participant recruitment was feasible.
- The remaining 71 studies raised multiple concerns when assessed with the trustworthiness checklist. We grouped the studies based on the number of concerns, including two, three, four, and five:
 - 41 studies had two areas of concern;
 - 22 studies had three areas of concern;
 - 6 studies had four areas of concern;
 - 2 studies had five areas of concern.

We finally included 106 RCTs (242 reports) in the review.

Included studies

This review included 106 studies involving a total of 30,348 women (see [Supplementary material 2](#)). Three trials were unpublished manuscripts. We received these unpublished manuscripts from the contact author of trial registration (Matega [68, 69]) and trials published as conference abstracts (Bebbington [70, 71, 72]; Lemyre [73, 74]).

The 'Overview of synthesis and included studies' table ([Table 1](#)) summarises the key study characteristics of the included studies.

Characteristics of studies

The included studies were published between 1988 and 2022. Thirty-seven trials (34.9%) were conducted from 2010 onwards. All trials were individually randomised controlled trials. Most of the trials were two-arm trials (98/106, 92.4%). There were six three-arm trials and two four-arm trials. As we had planned in our protocol, we combined multi-arm trials that included arms with the same intervention. The median size of the trials was 160 women (interquartile range 107 to 222).

The included trials were conducted across 35 countries, ranging from low- to high-income countries. Two trials were conducted in multiple countries. Ninety-nine trials were reported in English (93.4%). We used Google Translate [75] to translate the remaining trials into English: French (n = 1), Korean (n = 1), Arabic (n = 2), Portuguese (n = 2), and Spanish (n = 1).

Fifty trials (47.2%) reported that the authors declared no conflicts of interest, whereas five studies (4.7%) disclosed conflicts of interest among the authors. Ten studies (9.4%) reported receiving support from commercial companies. Many studies did not report the source of funding (59/106, 55.7%).

Characteristics of participants

Almost all studies (97/106, 91.5%) recruited women without previous caesarean section, while two trials (1.9%) recruited women with previous caesarean section, and three trials (2.8%) recruited both women with or without previous caesarean section.

In the mixed-population studies, the percentage of women with a prior caesarean section was small: 3.5% (177/5041) (Hannah 1996 [76, 77, 78, 79, 80, 81, 82, 83, 84]), 5.0% (5/100) (O'Brien 1995 [85, 86]), and 7.7% (Gupta 2022 [87, 88, 89, 90, 91]). Four trials did not report this information. However, as more than half of the participants in these four studies were nulliparous, this meant that more than half of the participants had no previous caesarean section, therefore these studies still met our inclusion criteria. We classified these studies as trials that recruited both women with or without previous caesarean section.

Most of the included studies recruited women at or beyond 37⁺⁰ weeks of gestation (91/106, 85.8%). Four studies recruited only early-term labour and full-term labour (gestational age 37⁺⁰ to 40⁺⁶ weeks), and 11 studies recruited only late-term labour and post-term labour (gestational age $\geq$ 41⁺⁰ weeks).

One hundred and five trials included women with a singleton pregnancy. One trial did not specify this information. Eleven trials (10.4%) included only nulliparous women; five trials (4.6%) included only multiparous women; 89 trials (84.0%) included both nulliparous and multiparous women; and one trial did not specify the parity of the women included in the trials.

Out of the 106 trials, 87 (82.1%) evaluated women with an unfavourable cervix; two (1.9%) evaluated women with a favourable cervix; and seven (6.6%) evaluated both. Ten trials did not specify cervical conditions. Most of the included studies used the Bishop score to determine cervical status. However, the cut-off point for the Bishop score varied across studies; several studies determined women as having an unfavourable cervix when the Bishop score was less than 8 ($n = 5$) or less than 9 ($n = 4$).

According to membrane status, 63 trials (59.4%) recruited women with intact membranes; 13 trials (12.3%) recruited women with ruptured membranes; and 17 trials (16.0%) recruited both. Thirteen trials did not specify membrane status.

Women were induced labour in a hospital setting in 90 trials (84.9%) and in an outpatient setting in 13 trials (12.3%). Three trials compared the effects of labour induction methods between outpatient and hospital settings.

Treatment formats

The 106 trials (222 trial arms) used the following methods, either as intervention or comparison.

- Vaginal misoprostol ($\leq 50 \mu\text{g}$), 52 trial arms (23.4%)
 - $\leq 25 \mu\text{g}$, 30 trial arms
 - $> 25 \mu\text{g}$, 22 trial arms
- Oral misoprostol ($\leq 50 \mu\text{g}$), 26 trial arms (11.7%)
 - $\leq 25 \mu\text{g}$, 10 trial arms
 - $> 25 \mu\text{g}$, 16 trial arms
- Sublingual or buccal misoprostol ($\leq 50 \mu\text{g}$), 13 trial arms (5.9%)
 - $\leq 25 \mu\text{g}$, seven trial arms
 - $> 25 \mu\text{g}$, six trial arms

- Controlled-release misoprostol vaginal pessary, one trial arm (0.5%)
- Vaginal dinoprostone (tablet or gel), 28 trial arms (12.6%)
 - $< 3 \text{ mg}$, 18 trial arms
 - 3 mg , 10 trial arms
- Controlled-release dinoprostone vaginal pessary, 19 trial arms (8.6%)
- Oxytocin (alone), 11 trial arms (5.0%)
- Nitric oxide donors, nine trial arms (4.1%)
- Balloon catheters, 24 trial arms (10.8%)
 - Single-balloon catheter, 20 trial arms
 - Double-balloon catheter, four trial arms
- Osmotic cervical dilators, three trial arms (1.4%)
- Oxytocin plus amniotomy, one trial arm (0.5%)
- Balloon catheters plus oxytocin, five trial arms (2.3%)
- Balloon catheters plus misoprostol ($\leq 50 \mu\text{g}$), four trial arms (1.8%)
 - $\leq 25 \mu\text{g}$ vaginal misoprostol, three trial arms
 - $\leq 25 \mu\text{g}$ oral misoprostol, one trial arm
- Comparator interventions, 26 trial arms (11.7%)
 - Placebo, 19 trial arms
 - No treatment, one trial arm
 - Expectant management, six trial arms

We did not identify any included trials that used amniotomy (alone). [Table 1](#) presents the list of intervention comparisons. [Supplementary material 13](#) presents information regarding the dosage of pharmacological methods.

Outcomes

[Supplementary material 14](#) presents the outcomes reported by each included study. The studies included in the analysis of perinatal death, blood loss of 500 mL or more, and blood loss of 1000 mL or more had varying definitions and time frames for measuring these outcomes. Some of these definitions were not consistent with our predefined criteria, and some studies had unclear time frames when measuring the outcomes. However, we still included all of them in our analyses. Information about how the outcomes were defined in the studies is provided in [Supplementary material 15](#).

Excluded studies

We excluded 440 studies (596 reports) (see [Supplementary material 3](#)). One of the most common reasons for exclusion was the inclusion of both premature and term labour without reporting results for each separately. Additionally, some trials evaluated interventions that were not listed in our criteria of interest, and some did not provide a clear definition of term labour or gestational age.

Risk of bias in included studies

We have presented summaries of the methodological quality of the included studies for each of the domains assessed across all studies ([Figure 3](#)) and for each included study ([Figure 4](#)).

Figure 3. Risk of bias graph: review authors' judgements about each risk of bias item presented as percentages across all included studies.

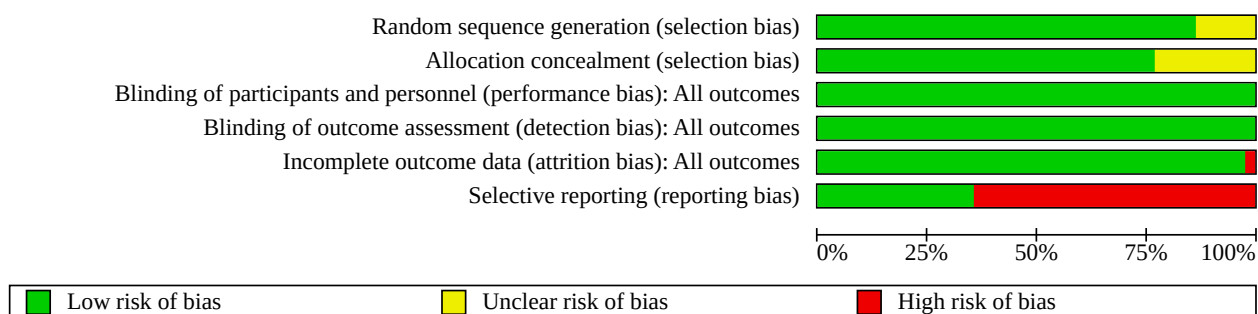


Figure 4. Risk of bias summary: review authors' judgements about each risk of bias item for each included study.

	Random sequence generation (selection bias)	Allocation concealment (selection bias)	Blinding of participants and personnel (performance bias): All outcomes	Blinding of outcome assessment (detection bias): All outcomes	Incomplete outcome data (attrition bias): All outcomes	Selective reporting (reporting bias)
Aalami-Harandi 2013	+	?	+	+	+	+
Aduloju 2016	+	+	+	+	+	-
Al-Ibraheemi 2018	+	+	+	+	+	-
Amarasena 2021	+	+	+	+	+	+
Anjali 2022	+	+	+	+	+	+
Anzeljc 2022	+	+	+	+	+	-
Ayaz 2009	?	+	+	+	+	+
Barda 2018	+	+	+	+	+	-
Bartusevicius 2006	+	+	+	+	+	+
Bebbington	+	+	+	+	+	-
Beigi 2010	+	?	+	+	+	-
Bennett 1998	+	+	+	+	+	-
Bhide 2020	+	?	+	+	+	-
Boddu 2017	?	+	+	+	+	+
Bullarbo 2007	+	+	+	+	+	-
Butt 1999	+	+	+	+	+	-
Calder 2008	+	?	+	+	+	+

Figure 4. (Continued)

Calder 2008	+	?	+	+	+	+
Caliskan 2005	+	+	+	+	+	+
Chaudhuri 2011	+	+	+	+	+	-
Cheung 2006	+	+	+	+	+	-
Dadashaliha 2021	+	+	+	+	+	-
da Graca Krupa 2005	+	+	+	+	+	-
Dallenbach 2003	+	+	+	+	+	+
De Bonrosto Torralba 2019	+	?	+	+	+	-
de Moraes 2005	+	+	+	+	+	-
Devillard 2022	+	+	+	+	+	+
Diguisto 2021	+	+	+	+	+	-
Dodd 2006	+	+	+	+	+	+
Dogra 2019	+	+	+	+	+	-
El-Shawarby 2006	?	+	+	+	-	-
Feitosa 2006	+	?	+	+	+	+
Fuenmavor-Beltran 2017	+	+	+	+	+	-
Gattas 2020	+	+	+	+	+	+
Gavara 2022	+	+	+	+	+	+
Ghanaie 2013	?	?	+	+	+	-
Goonewardene 2014	?	+	+	+	+	-
Grace 2021	+	+	+	+	+	+
Grant 1992	+	+	+	+	+	-
Gupta 2022	+	+	+	+	+	+
Haas 2019	+	+	+	+	+	+
Habib 2008	+	+	+	+	+	-
Hannah 1996	+	+	+	+	+	-
Henry 2013	+	+	+	+	+	+
Hosli 2008	+	+	+	+	+	-
Incerpi 2001	+	?	+	+	+	-
Itoh 2021	+	?	+	+	+	+
Jahromi 2016	?	?	+	+	+	-
Jozwiak 2011	+	+	+	+	+	-
Jozwiak 2013	+	+	+	+	+	-
Khamaiseh 2012	+	?	+	+	+	-
Khoury 2001	+	?	+	+	+	+
Kidanto 2007	+	+	+	+	+	-

Figure 4. (Continued)

Levy 2007	+	+	+	+	+	+
Kidanto 2007	+	+	+	+	+	-
Kim 2005	+	?	+	+	+	-
Kuper 2018	+	+	+	+	+	-
Kwon 2001	+	+	+	+	+	-
Lemyre	?	+	+	+	+	-
Levine 2016	+	+	+	+	+	+
Levy 2007	?	+	+	+	+	-
Lo 2006	+	+	+	+	+	+
Mahmood 1995	+	+	+	+	+	-
Matega	?	+	+	+	+	-
McKenna 2004	+	?	+	+	+	-
Moraes Filho 2010	+	+	+	+	+	+
Mundle 2017	+	+	+	+	+	+
Nassar 2007	+	+	+	+	+	+
Nopdonrattakoon 2003	?	+	+	+	+	+
Nwali 2021	+	+	+	+	+	-
O'Brien 1995	+	?	+	+	+	-
Oboro 2005	+	+	+	+	+	-
Oliveira 2010	+	+	+	+	+	-
Osman 2006	+	+	+	+	+	-
Ozkan 2009	+	+	+	+	+	+
Paisarntantiwong 2005	?	+	+	+	+	-
Pandis 2001	+	+	+	+	-	+
Papanikolaou 2004	+	?	+	+	+	-
PonMalar 2017	+	+	+	+	+	-
Prager 2008	+	+	+	+	+	-
Rahman 2013	+	+	+	+	+	+
Rameez 2007	?	+	+	+	+	-
Rayburn 1988	+	?	+	+	+	-
Rayburn 1992	+	+	+	+	+	-
Rozenberg 2004	+	+	+	+	+	+
Saad 2019	+	+	+	+	+	-
Sarreau 2020	+	+	+	+	+	-
Sawai 1991	?	+	+	+	+	-
Sawai 1994	+	?	+	+	+	-
Sawai 2014	+	+	+	+	+	-

Figure 4. (Continued)

Sawai 1994	+	?	+	+	+	+
Schmitz 2014	+	+	+	+	+	+
Shechter-Maor 2015	+	?	+	+	+	+
Shetty 2001	+	+	+	+	+	+
Shetty 2002	+	+	+	+	+	+
Sifakis 2007	+	?	+	+	+	+
Soliman 2013	+	+	+	+	+	+
Somirathne 2017	+	+	+	+	+	+
Souza 2013	+	?	+	+	+	+
Sulaiman 2023	+	+	+	+	+	+
Surbek 1997	+	?	+	+	+	+
Tabasi 2007	+	?	+	+	+	+
Tan 2010	+	+	+	+	+	+
Ten Eikelder 2016	+	+	+	+	+	+
Triglia 2010	+	?	+	+	+	+
Unthanan 2022	+	+	+	+	+	+
Vidanagamage 2011	?	+	+	+	+	+
Wallstrom 2019	+	+	+	+	+	+
Witter 1996	+	+	+	+	+	+
Young 2020	+	+	+	+	+	+
Zhang 2015	+	+	+	+	+	+

The majority of included studies were at low risk of bias for randomisation processes. We reached a consensus that blinding might be difficult in trials evaluating methods of inducing labour. However, the lack of blinding might not affect our critical and important outcomes. We therefore assessed all studies as at low risk of performance bias and detection bias. We assessed only two trials as at high risk of attrition bias (El-Shawarby 2006 [92]; Pandis 2001 [93]). This was because around 30% of participants in each group dropped out after randomisation due to protocol violations. We assessed the majority of trials as at high risk of reporting bias because they failed to report our critical outcomes.

Synthesis of results

We have presented our results separately for 1) women without previous caesarean section and a mix of women with or without previous caesarean section (women without caesarean section), and 2) women with previous caesarean section. Results are presented primarily for NMA. All results of the direct comparisons included in the NMA are summarised in [Supplementary material 6](#).

Summary of findings tables for each critical outcome are listed below.

- Women without previous caesarean section and a mix of women with or without previous caesarean section
 - Failure to achieve vaginal delivery within 24 hours ([Summary of findings 1](#))
 - Caesarean section due to any causes ([Supplementary material 16](#))
 - Caesarean section due to non-reassuring fetal status ([Summary of findings 2](#))
 - Uterine hyperstimulation with changes in the heartbeat of the baby before birth ([Summary of findings 3](#))
 - Perinatal death ([Summary of findings 4](#))
 - Severe neonatal morbidity ([Supplementary material 17](#))
- Women with previous caesarean section
 - Caesarean section due to any causes ([Supplementary material 18](#))
 - Caesarean section due to non-reassuring fetal status ([Supplementary material 19](#))

We had to exclude some studies from our analyses of adverse maternal and neonatal outcomes, such as perinatal death, uterine rupture, maternal death, NICU admission, and an Apgar score < 7 at 5 minutes. This was because no events were observed in any of the study arms. In addition, we encountered various formats for

reporting continuous outcomes, including mean with interquartile range, mean with min-max, and only mean or median. We could not convert the results to a suitable format for analysis when the studies reported their findings using these methods. As a result, we excluded these studies from our analyses of continuous outcomes.

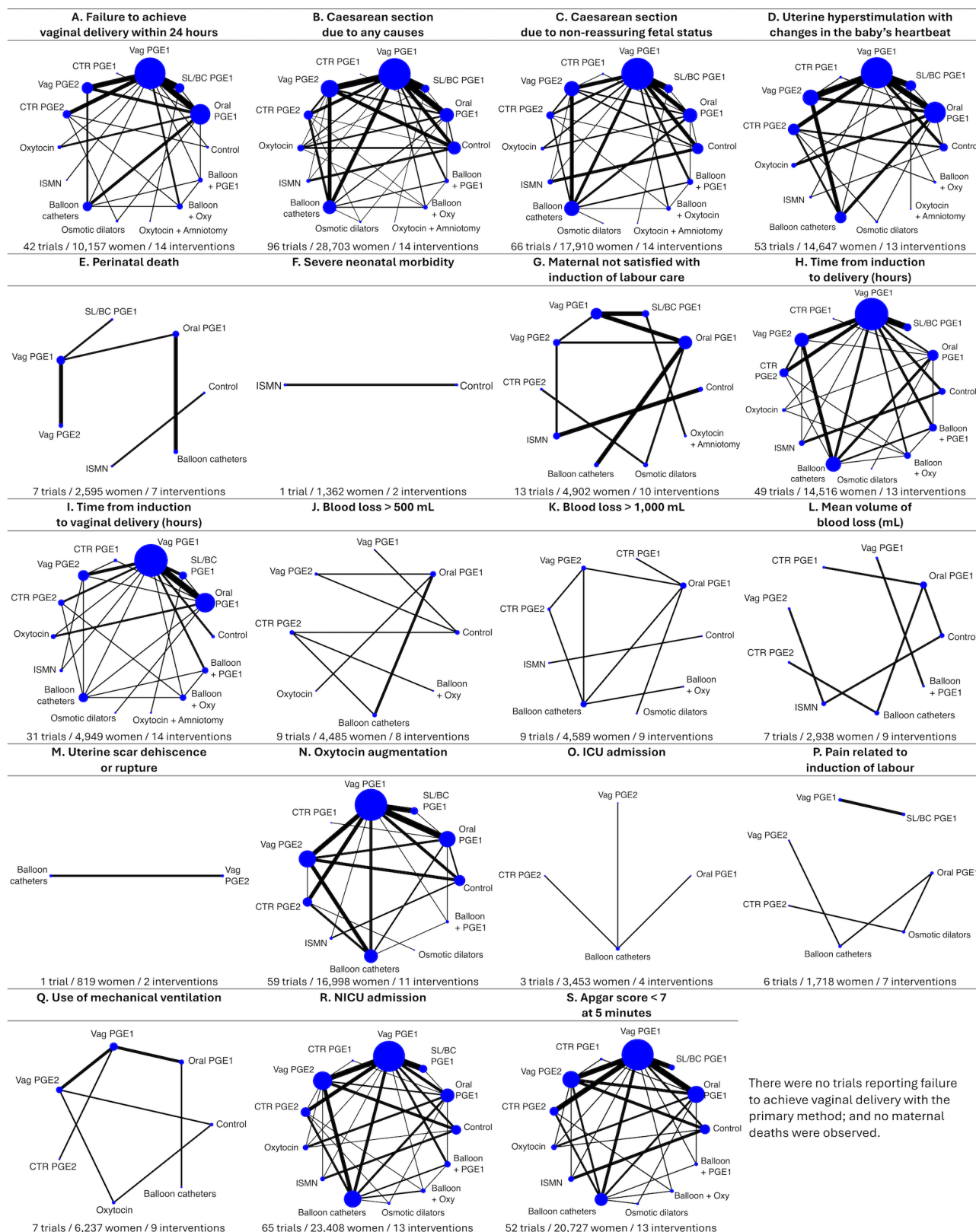
1. Women without previous caesarean section and a mix of women with or without previous caesarean section

Figure 5 shows the network diagrams for all available critical and important outcomes. These diagrams visually represent the relationships between different interventions in the included studies for each outcome. Each node (circle) represents an intervention, with its size proportional to the number of trials

comparing this intervention to any other in the network. Each edge (line) connecting two interventions represents a direct comparison, with its thickness proportional to the number of trials making each direct comparison. Figure 6 displays forest plots showing direct, indirect, and network evidence for the six critical outcomes. Figure 7 presents forest plots of network evidence for all important outcomes, along with direct evidence for outcomes that could not be analysed using NMA due to insufficient data. Full details of analysis results for all available outcomes, which include a network diagram, forest plot presenting the direct, indirect, and network evidence, league table, and rankogram, are provided in [Supplementary material 20](#).

Figure 5. Summary of network diagrams across the critical and important outcomes for women without previous caesarean section and a mix of women with or without previous caesarean section. Note: Network diagram visually represents the relationships between different interventions in the included studies for each outcome. Each node (circle) represents an intervention, with its size proportional to the number of trials comparing this intervention

to any other in the network. Each edge (line) connecting two interventions represents a direct comparison, with its thickness proportional to the number of trials making each direct comparison.



There were no trials reporting failure to achieve vaginal delivery with the primary method; and no maternal deaths were observed.

Abbreviations: Control, Placebo or No treatment or Expectant management; Oral PGE1, Oral misoprostol ($\leq 50 \mu\text{g}$); SL/BC PGE1, Sublingual or buccal misoprostol ($\leq 50 \mu\text{g}$); Vag PGE1, Vaginal misoprostol ($\leq 50 \mu\text{g}$); CTR PGE1, Controlled-release misoprostol vaginal pessary; Vag PGE2, Vaginal

Figure 5. (Continued)

/ trials / 6,237 women / 9 interventions	65 trials / 23,408 women / 13 interventions	52 trials / 20,727 women / 13 interventions
Abbreviations: Control, Placebo or No treatment or Expectant management; Oral PGE1, Oral misoprostol ($\leq 50 \mu\text{g}$); SL/BC PGE1, Sublingual or buccal misoprostol ($\leq 50 \mu\text{g}$); Vag PGE1, Vaginal misoprostol ($\leq 50 \mu\text{g}$); CTR PGE1, Controlled-release misoprostol vaginal pessary; Vag PGE2, Vaginal dinoprostone (tablet or gel); CTR PGE2, Controlled-release dinoprostone vaginal pessary; Oxytocin, Oxytocin (alone); ISMN, Nitric oxide donors (ISMN/ISDN/nitroglycerin/sodium nitroprusside); Balloon catheters, Foley single balloon catheter or Double balloon devices (Atad/Cook balloon or other type); Oxytocin + Amniotomy, Oxytocin plus amniotomy; Balloon + Oxy, Balloon catheters plus oxytocin; and Balloon + PGE1, Balloon catheters plus misoprostol ($\leq 50 \mu\text{g}$; oral or vaginal routes).		

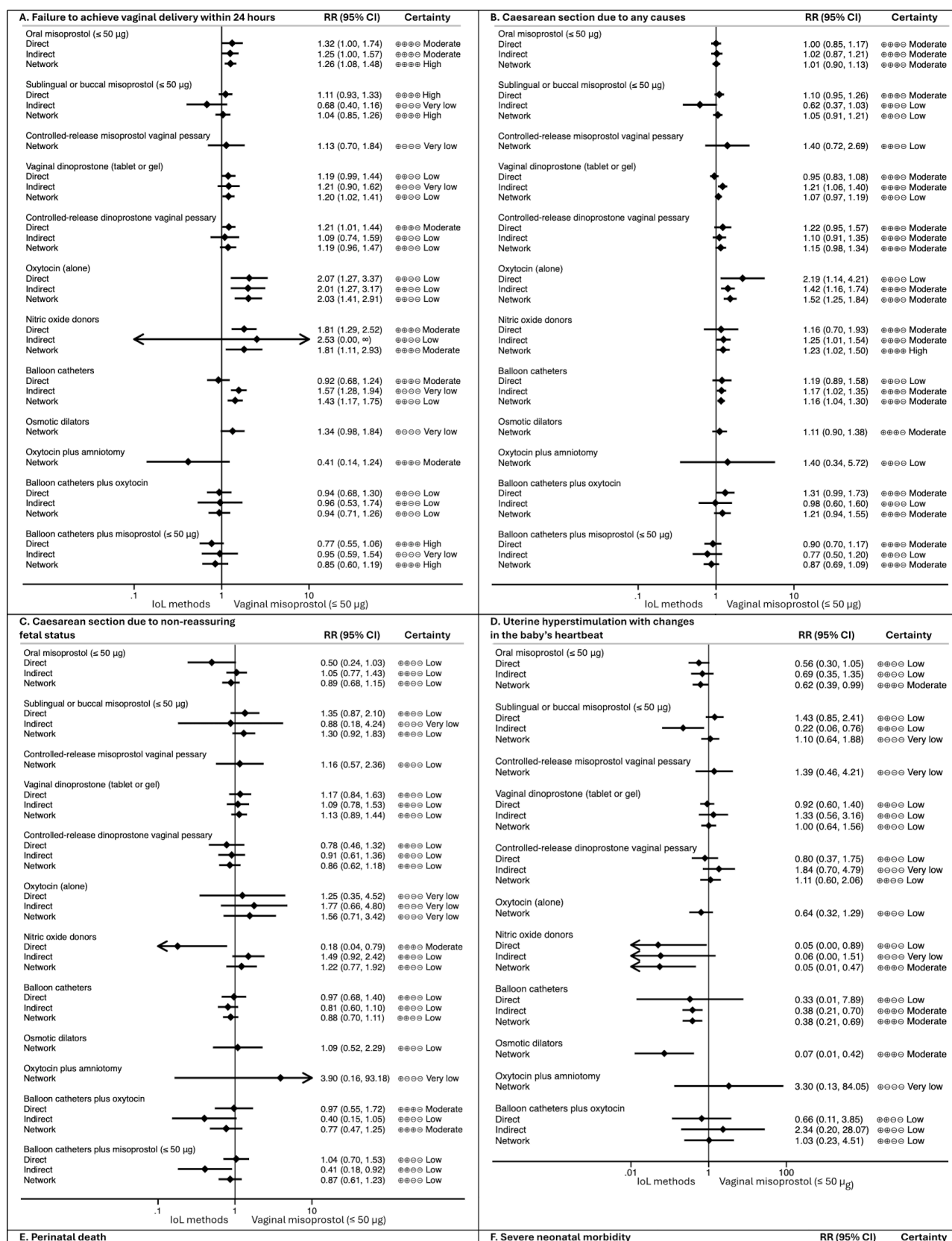
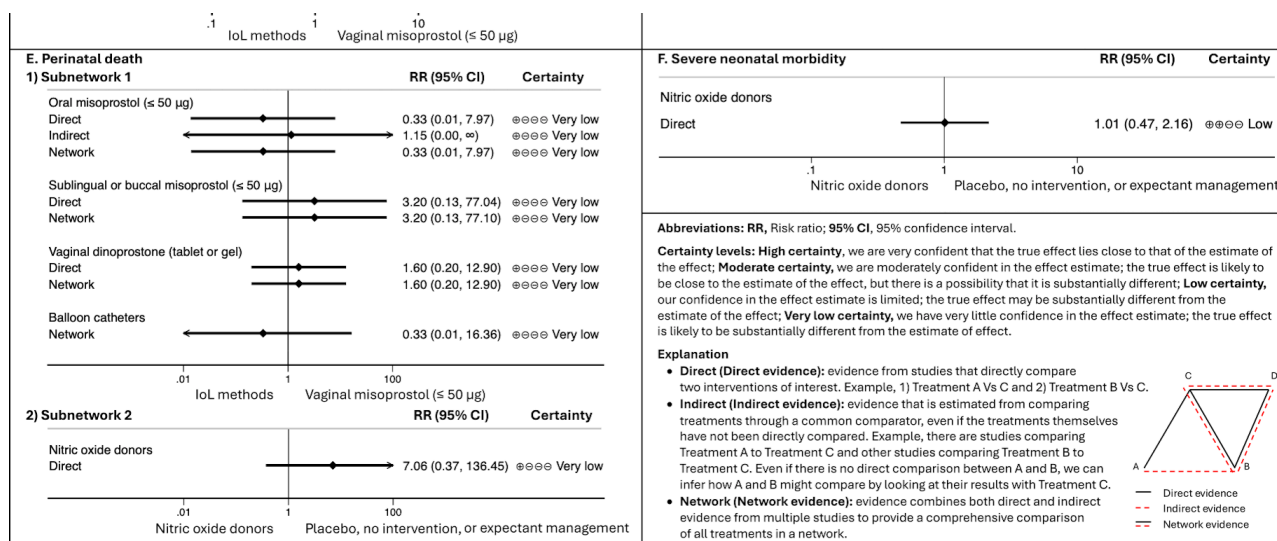
Figure 6. Summary of forest plots across the critical outcomes for women without previous caesarean section and a mix of women with or without previous caesarean section.

Figure 6. (Continued)



A. Maternal not satisfied with IoL care (Network evidence) RR (95% CI)

Intervention	RR (95% CI)
Oral misoprostol (≤ 50 µg)	0.90 (0.58, 1.40)
Sublingual or buccal misoprostol (≤ 50 µg)	0.26 (0.10, 0.67)
Vaginal dinoprostone (tablet or gel)	1.12 (0.64, 1.97)
Controlled-release dinoprostone vaginal pessary	0.47 (0.10, 2.13)
Nitric oxide donors	0.53 (0.22, 1.23)
Balloon catheters	1.07 (0.47, 2.41)
Osmotic dilators	0.75 (0.20, 2.82)
Oxytocin plus amniotomy	1.31 (0.23, 7.53)

B. Time from IoL to delivery (hours) (Network evidence) MD (95% CI)

Intervention	MD (95% CI)
Oral misoprostol (≤ 50 µg)	3.89 (1.41, 6.37)
Sublingual or buccal misoprostol (≤ 50 µg)	0.03 (-2.25, 2.30)
Vaginal dinoprostone (tablet or gel)	2.13 (-4.26, 8.52)
Controlled-release misoprostol vaginal pessary	1.27 (-0.77, 3.30)
Vaginal dinoprostone (tablet or gel)	1.27 (-0.77, 3.30)
Controlled-release dinoprostone vaginal pessary	2.15 (-0.16, 4.46)
Oxytocin (alone)	3.34 (-0.37, 7.06)
Nitric oxide donors	12.09 (7.85, 16.34)
Balloon catheters	4.05 (1.88, 6.23)
Osmotic dilators	3.19 (-3.24, 9.62)
Balloon catheters plus oxytocin	0.11 (-2.96, 3.18)
Balloon catheters plus misoprostol (≤ 50 µg)	-1.57 (-4.34, 1.20)

C. Time from IoL to vaginal delivery (hours) (Network evidence) MD (95% CI)

Intervention	MD (95% CI)
Oral misoprostol (≤ 50 µg)	4.41 (2.42, 6.39)
Sublingual or buccal misoprostol (≤ 50 µg)	-1.23 (-3.94, 1.49)
Controlled-release misoprostol vaginal pessary	1.75 (-2.55, 6.05)
Vaginal dinoprostone (tablet or gel)	1.51 (-1.23, 4.26)
Controlled-release dinoprostone vaginal pessary	3.34 (0.02, 6.67)
Oxytocin (alone)	2.97 (-0.34, 6.28)
Nitric oxide donors	11.61 (7.55, 15.67)
Balloon catheters	1.49 (-1.45, 4.42)
Osmotic dilators	3.51 (-1.95, 8.96)
Oxytocin plus amniotomy	-5.66 (-12.20, 0.88)
Balloon catheters plus oxytocin	-2.48 (-6.38, 1.42)
Balloon catheters plus misoprostol (≤ 50 µg)	-3.94 (-7.35, -0.53)

D. Blood loss > 500 mL (Network evidence) RR (95% CI)

Intervention	RR (95% CI)
Oral misoprostol (≤ 50 µg)	0.54 (0.02, 17.37)
Vaginal dinoprostone (tablet or gel)	0.37 (0.01, 11.80)
Controlled-release dinoprostone vaginal pessary	0.63 (0.02, 20.51)
Oxytocin (alone)	0.18 (0.00, 8.10)
Balloon catheters	0.54 (0.02, 17.56)
Balloon catheters plus oxytocin	0.53 (0.01, 20.27)

E. Blood loss > 1,000 mL

1) Subnetwork 1 (Network evidence) RR (95% CI)

Intervention	RR (95% CI)
Controlled-release misoprostol vaginal pessary	1.28 (0.70, 2.31)
Vaginal dinoprostone (tablet or gel)	1.30 (0.86, 1.97)
Controlled-release dinoprostone vaginal pessary	1.02 (0.43, 2.39)
Balloon catheters	1.01 (0.77, 1.33)
Osmotic dilators	0.34 (0.04, 3.19)
Balloon catheters plus oxytocin	3.08 (0.13, 75.17)

2) Subnetwork 2 (Direct evidence) RR (95% CI)

Intervention	RR (95% CI)
Nitric oxide donors	1.17 (0.57, 2.40)

F. Mean volume of blood loss (mL)

1) Subnetwork 1 (Network evidence) MD (95% CI)

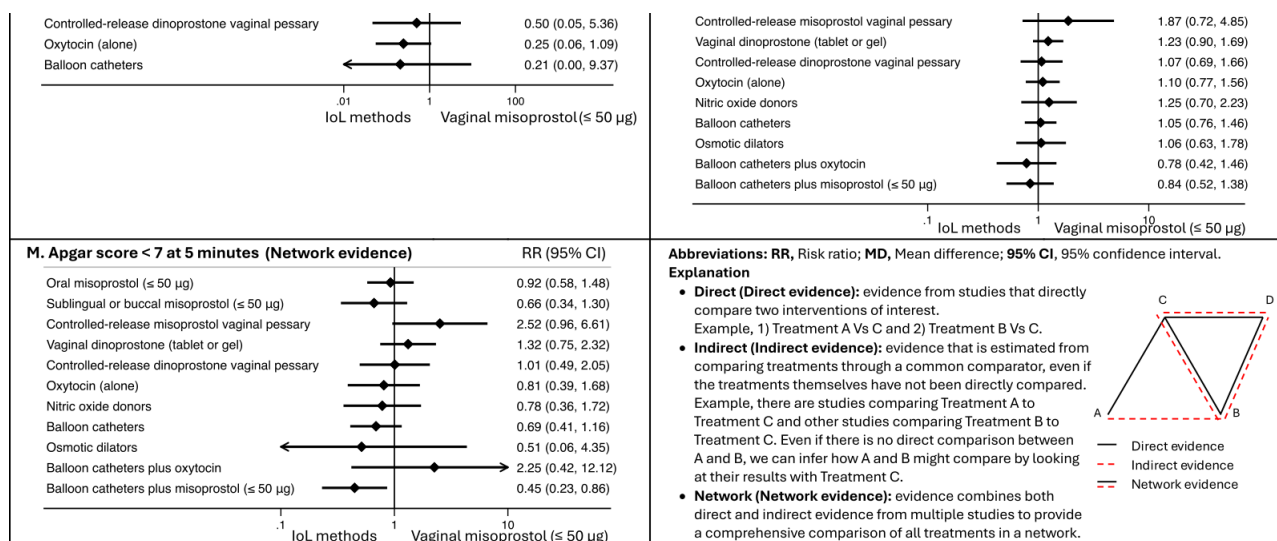
Intervention	MD (95% CI)
Controlled-release misoprostol vaginal pessary	120.90 (-31.79, 273.59)
Vaginal dinoprostone (tablet or gel)	35.22 (-128.85, 199.29)
Controlled-release dinoprostone vaginal pessary	-258.32 (-638.96, 122.32)
Nitric oxide donors	41.89 (-117.62, 201.40)
Balloon catheters	0.00 (-20.39, 20.39)

2) Subnetwork 2 (Direct evidence) MD (95% CI)

Intervention	MD (95% CI)
Balloon catheters plus misoprostol	4.00 (-104.17, 112.17)

G. Uterine scar dehiscence

Figure 7. (Continued)



The league tables present the findings of both direct and network evidence for all available methods regarding: 1) failure to achieve vaginal delivery within 24 hours and uterine hyperstimulation with changes in the heartbeat of the baby before birth (Figure 8), and 2) caesarean section due to any causes and caesarean section due to non-reassuring fetal status (Figure 9). Each cell in the league table displays the RR and its 95% CI for the comparison between two

treatments. Bold numbers indicate statistically significant results. The intersection of a row and a column shows the effect estimate for that specific comparison. Comparisons should be interpreted with the treatment on the left side as the reference for the treatment on the right side. League tables for the other outcomes can be seen in [Supplementary material 20](#).

Figure 8. League table presenting network and direct evidence for failure to achieve vaginal delivery within 24 hours and uterine hyperstimulation with changes in the heartbeat of the baby before birth outcomes for women without previous caesarean section and a mix of women with or without previous caesarean section. Notes: The league table presents the findings of both direct and network evidence for all available methods. Each cell in the league table displays the risk ratio (RR) and its 95% confidence interval (CI) for the comparison between two treatments. The intersection of a row and a column shows the effect estimate for that specific comparison. Bold numbers indicate statistically significant results. Comparisons should be interpreted with the treatment on the left side as the reference for the treatment on the right side. For example: # The network evidence of failure to achieve vaginal delivery within 24 hours: the comparison between Oral PGE1 and Vag PGE1 (reference) shows an RR of 1.26 (95% CI 1.08 to 1.48). # The network evidence of uterine hyperstimulation with changes in the heartbeat of the baby before birth: the comparison between Nitric oxide donors and Oral PGE1 (reference) shows an RR of 0.09 (95% CI 0.01 to 0.80).

Control	3.47 (1.16, 10.35)	2.14 (0.67, 6.81)	3.82 (1.13, 12.87)	4.80 (1.04, 22.29)	3.48 (1.09, 11.14)	3.86 (1.21, 12.38)	2.21 (0.61, 7.96)	0.19 (0.02, 1.67)	1.33 (0.40, 4.37)	0.25 (0.03, 1.92)	11.44 (0.38, 348.44)	3.56 (0.58, 21.96)	-
	1.88 (0.45, 7.87)	5.15 (0.25, 105.29)	-	-	-	8.83 (1.11, 70.12)	-	0.20 (0.01, 4.07)	-	-	-	-	-
0.60 (0.44, 0.82)	Vag PGE1	0.62 (0.39, 0.99)	1.10 (0.64, 1.88)	1.39 (0.46, 4.21)	1.00 (0.64, 1.56)	1.11 (0.60, 2.06)	0.64 (0.32, 1.29)	0.05 (0.01, 0.47)	0.38 (0.21, 0.69)	0.07 (0.01, 0.42)	3.30 (0.13, 84.05)	1.03 (0.23, 4.51)	-
0.75 (0.65, 0.86)		0.56 (0.31, 1.05)	1.43 (0.85, 2.41)	-	0.92 (0.60, 1.40)	0.80 (0.37, 1.75)	-	0.05 (0.00, 0.89)	0.33 (0.01, 7.89)	-	-	0.66 (0.11, 3.85)	-
0.76 (0.54, 1.06)	1.26 (1.08, 1.48)	Oral PGE1	1.78 (0.93, 3.43)	2.24 (0.78, 6.44)	1.62 (1.02, 2.58)	1.80 (0.89, 3.66)	1.03 (0.58, 1.84)	0.09 (0.01, 0.80)	0.62 (0.36, 1.07)	0.12 (0.02, 0.66)	5.34 (0.21, 139.06)	1.66 (0.36, 7.76)	-
-	1.32 (1.00, 1.74)		3.00 (0.13, 71.96)	4.76 (1.41, 16.05)	1.57 (0.92, 2.69)	-	0.74 (0.44, 1.24)	-	0.77 (0.45, 1.30)	0.14 (0.02, 1.15)	-	-	-
0.62 (0.43, 0.90)	1.04 (0.85, 1.26)	0.82 (0.64, 1.05)	SL/BC PGE1	1.26 (0.37, 4.23)	0.91 (0.47, 1.77)	1.01 (0.45, 2.30)	0.58 (0.27, 1.25)	0.05 (0.01, 0.46)	0.35 (0.16, 0.76)	0.07 (0.01, 0.41)	3.00 (0.12, 73.03)	0.93 (0.19, 4.50)	-
-	1.11 (0.93, 1.33)	0.56 (0.37, 0.84)		-	-	-	2.25 (0.72, 7.03)	-	-	-	3.00 (0.13, 70.42)	-	-
0.68 (0.39, 1.20)	1.13 (0.70, 1.84)	0.90 (0.56, 1.42)	1.09 (0.65, 1.84)	CTR PGE1	0.72 (0.25, 2.13)	0.80 (0.24, 2.72)	0.46 (0.14, 1.53)	0.04 (0.00, 0.44)	0.28 (0.09, 0.87)	0.05 (0.01, 0.39)	2.38 (0.08, 72.44)	0.74 (0.12, 4.66)	-
-	-	0.90 (0.66, 1.21)	-		2.53 (0.50, 12.71)	-	-	-	-	-	-	-	-
0.72 (0.51, 1.01)	1.20 (1.02, 1.41)	0.95 (0.80, 1.12)	1.16 (0.90, 1.49)	1.06 (0.65, 1.73)	Vag PGE2	1.11 (0.54, 2.28)	0.64 (0.31, 1.29)	0.05 (0.01, 0.49)	0.38 (0.21, 0.68)	0.07 (0.01, 0.42)	3.29 (0.13, 85.78)	1.02 (0.22, 4.75)	-
-	1.19 (0.99, 1.44)	0.91 (0.81, 1.02)	-	-		-	-	-	0.43 (0.20, 0.96)	-	-	-	-
0.71 (0.52, 0.98)	1.19 (0.96, 1.47)	0.94 (0.73, 1.20)	1.14 (0.85, 1.54)	1.05 (0.62, 1.77)	0.99 (0.76, 1.28)	CTR PGE2	0.57 (0.23, 1.44)	0.05 (0.01, 0.45)	0.34 (0.18, 0.65)	0.06 (0.01, 0.38)	2.96 (0.11, 79.99)	0.92 (0.20, 4.20)	-
0.53 (0.40, 0.70)	1.21 (1.01, 1.44)	-	-	-	-		-	-	0.36 (0.06, 2.01)	0.04 (0.00, 0.74)	-	2.00 (0.19, 21.18)	-
1.22 (0.76, 1.95)	2.03 (1.41, 2.91)	1.60 (1.13, 2.28)	1.96 (1.30, 2.94)	1.79 (1.00, 3.20)	1.69 (1.16, 2.46)	1.71 (1.13, 2.59)	Oxytocin	0.09 (0.01, 0.82)	0.60 (0.27, 1.34)	0.11 (0.02, 0.70)	5.17 (0.19, 137.90)	1.61 (0.32, 8.22)	-
-	2.07 (1.27, 3.37)	1.59 (1.03, 2.44)	-	-	-	-		-	-	-	-	-	-
1.09 (0.61, 1.93)	1.81 (1.11, 2.93)	1.43 (0.86, 2.38)	1.74 (1.03, 2.94)	1.60 (0.80, 3.17)	1.51 (0.90, 2.51)	1.52 (0.90, 2.59)	0.89 (0.49, 1.63)	Nitric oxide donors	6.97 (0.76, 64.00)	1.31 (0.08, 20.92)	60.09 (1.23, 2927.88)	18.71 (1.39, 252.75)	-
-	1.81 (1.29, 2.52)	-	-	-	-	-	-		-	-	-	-	-
0.86 (0.60, 1.23)	1.43 (1.17, 1.75)	1.13 (0.95, 1.35)	1.38 (1.05, 1.82)	1.26 (0.77, 2.07)	1.19 (0.98, 1.45)	1.20 (0.91, 1.60)	0.71 (0.48, 1.04)	0.79 (0.47, 1.34)	Balloon catheters	0.19 (0.03, 1.11)	8.62 (0.32, 230.09)	2.68 (0.57, 12.71)	-
-	0.92 (0.68, 1.24)	1.33 (1.20, 1.48)	-	-	1.12 (0.92, 1.37)	-	-	-		-	-	-	-
0.81 (0.54, 1.21)	1.34 (0.98, 1.84)	1.06 (0.77, 1.45)	1.29 (0.89, 1.87)	1.18 (0.68, 2.07)	1.12 (0.80, 1.56)	1.13 (0.84, 1.51)	0.66 (0.42, 1.05)	0.74 (0.42, 1.32)	0.94 (0.66, 1.33)	Osmotic dilators	45.80 (1.16, 1813.51)	14.26 (1.47, 138.45)	-
-	-	1.07 (0.82, 1.40)	-	-	-	1.12 (1.05, 1.19)	-	-	-		-	-	-
0.25 (0.08, 0.78)	0.41 (0.14, 1.24)	0.33 (0.11, 0.99)	0.40 (0.14, 1.17)	0.37 (0.11, 1.21)	0.35 (0.11, 1.04)	0.35 (0.11, 1.07)	0.20 (0.06, 0.65)	0.23 (0.07, 0.76)	0.29 (0.10, 0.88)	0.31 (0.10, 0.97)	Oxytocin + Amniotomy	0.31 (0.01, 10.93)	-
-	-	-	0.40 (0.14, 1.11)	-	-	-	-	-	-	-		-	-
0.57 (0.37, 0.86)	0.94 (0.71, 1.26)	0.75 (0.54, 1.02)	0.91 (0.64, 1.28)	0.83 (0.48, 1.45)	0.78 (0.57, 1.08)	0.79 (0.56, 1.12)	0.46 (0.29, 0.73)	0.52 (0.30, 0.91)	0.66 (0.48, 0.91)	0.70 (0.47, 1.06)	2.27 (0.73, 7.04)	Balloon + Oxytocin	-
-	0.94 (0.68, 1.30)	-	-	-	-	0.50 (0.26, 0.98)	-	-	0.86 (0.62, 1.20)	-	-		-
0.51 (0.32, 0.80)	0.85 (0.60, 1.19)	0.67 (0.48, 0.94)	0.82 (0.55, 1.21)	0.75 (0.42, 1.33)	0.71 (0.49, 1.01)	0.71 (0.48, 1.06)	0.42 (0.26, 0.68)	0.47 (0.26, 0.85)	0.59 (0.42, 0.84)	0.63 (0.40, 0.99)	2.04 (0.65, 6.43)	0.90 (0.60, 1.34)	Balloon + PGE1
-	0.77 (0.55, 1.06)	0.56 (0.37, 0.85)	-	-	-	-	-	-	0.83 (0.59, 1.17)	-	-	0.97 (0.68, 1.38)	

Network evidence of failure to achieve vaginal delivery within 24 hours

Network evidence of uterine hyperstimulation with changes in the heartbeat of the baby before birth

Direct evidence of failure to achieve vaginal delivery within 24 hours

Direct evidence of uterine hyperstimulation with changes in the heartbeat of the baby before birth

Abbreviations: - (dash), no evidence; **Control**, Placebo or No treatment or Expectant management; **Vag PGE1**, Vaginal misoprostol ($\leq 50 \mu\text{g}$); **Oral PGE1**, Oral misoprostol ($\leq 50 \mu\text{g}$); **SL/BC PGE1**, Sublingual or buccal misoprostol ($\leq 50 \mu\text{g}$); **CTR PGE1**, Controlled-release misoprostol vaginal pessary; **Vag PGE2**, Vaginal dinoprostone (tablet or gel); **CTR PGE2**, Controlled-release dinoprostone vaginal pessary; **Oxytocin**, Oxytocin (alone); **Nitric oxide donors**, ISMN/ISDN/nitroglycerin/sodium nitroprusside; **Balloon catheters**, Foley single balloon catheter or Double balloon devices (Atad/Cook balloon or other type); **Oxytocin + Amniotomy**, Oxytocin plus amniotomy; **Balloon + Oxytocin**, Balloon catheters plus oxytocin; and **Balloon + PGE1**, balloon catheters plus misoprostol ($\leq 50 \mu\text{g}$; oral or vaginal routes).

Figure 9. League table presenting network and direct evidence for caesarean section due to non-reassuring fetal status and caesarean section due to any causes outcomes for women without previous caesarean section and a mix of women with or without previous caesarean section. Notes: The league table presents the findings of both direct and network evidence for all available methods. Each cell in the league table displays the risk ratio (RR) and its 95% confidence interval (CI) for the comparison between two treatments. The intersection of a row and a column shows the effect estimate for that specific comparison. Bold numbers indicate statistically significant results. Comparisons should be interpreted with the treatment on the left side as the reference for the treatment on the right side. For example: # The network evidence of caesarean section due to non-reassuring fetal status: the comparison between Oral PGE1 and Vag PGE1 (reference) shows an RR of 0.89 (95% CI 0.68 to 1.15). # The network evidence of caesarean section due to any causes: the comparison between Nitric oxide donors and Oral PGE1 (reference) shows an RR of 1.22 (95% CI 0.99 to 1.51).

Control	0.78 (0.67,0.90)	0.78 (0.66,0.92)	0.82 (0.67,1.00)	1.08 (0.56,2.11)	0.83 (0.72,0.96)	0.89 (0.73,1.09)	1.18 (0.98,1.42)	0.96 (0.81,1.13)	0.90 (0.76,1.06)	0.86 (0.67,1.11)	1.09 (0.27,4.47)	0.94 (0.71,1.24)	0.68 (0.52,0.88)
	0.78 (0.58,1.05)	0.62 (0.18,2.17)	-	-	0.78 (0.56,1.09)	1.35 (0.59,3.11)	1.13 (0.77,1.67)	1.00 (0.85,1.17)	-	-	-	-	-
Vag PGE1	0.94 (0.63,1.41)	1.01 (0.90,1.13)	1.05 (0.91,1.21)	1.40 (0.72,2.69)	1.07 (0.97,1.19)	1.15 (0.98,1.34)	1.52 (1.25,1.84)	1.23 (1.02,1.50)	1.16 (1.04,1.30)	1.11 (0.90,1.38)	1.40 (0.34,5.72)	1.21 (0.94,1.55)	0.87 (0.69,1.09)
	0.72 (0.45,1.15)	1.00 (0.85,1.17)	1.10 (0.95,1.26)	-	0.95 (0.83,1.08)	1.22 (0.95,1.57)	2.19 (1.14,4.21)	1.16 (0.70,1.93)	1.19 (0.89,1.58)	-	-	1.31 (0.99,1.73)	0.90 (0.70,1.17)
Oral PGE1	0.83 (0.52,1.34)	0.89 (0.68,1.15)	1.04 (0.88,1.25)	1.39 (0.73,2.64)	1.06 (0.94,1.20)	1.14 (0.96,1.35)	1.51 (1.24,1.83)	1.22 (0.99,1.51)	1.15 (1.02,1.29)	1.10 (0.89,1.37)	1.39 (0.34,5.70)	1.20 (0.92,1.56)	0.86 (0.68,1.10)
	0.32 (0.01,7.68)	0.50 (0.24,1.03)	0.53 (0.25,1.14)	1.39 (0.74,2.60)	1.02 (0.81,1.28)	-	1.72 (1.23,2.41)	-	1.20 (1.04,1.37)	0.98 (0.68,1.42)	-	-	0.54 (0.29,1.00)
SL/BC PGE1	1.22 (0.72,2.08)	1.30 (0.92,1.83)	1.47 (0.96,2.24)	1.33 (0.68,2.59)	1.02 (0.86,1.21)	1.09 (0.88,1.34)	1.44 (1.15,1.81)	1.17 (0.92,1.49)	1.10 (0.92,1.32)	1.06 (0.82,1.37)	1.33 (0.33,5.39)	1.15 (0.86,1.52)	0.83 (0.63,1.08)
	-	1.35 (0.87,2.10)	1.00 (0.41,4.72)	-	-	-	2.08 (1.12,3.87)	-	-	-	1.33 (0.33,5.36)	-	-
CTR PGE1	1.09 (0.48,2.45)	1.16 (0.57,2.36)	1.30 (0.65,2.62)	0.89 (0.40,1.95)	0.77 (0.40,1.48)	0.82 (0.42,1.60)	1.09 (0.55,2.14)	0.88 (0.45,1.74)	0.83 (0.43,1.60)	0.80 (0.40,1.57)	1.01 (0.21,4.74)	0.86 (0.43,1.74)	0.62 (0.31,1.24)
	-	-	2.04 (0.64,6.56)	-	-	-	-	-	-	-	-	-	-
Vag PGE2	1.07 (0.68,1.68)	1.13 (0.89,1.44)	1.28 (1.00,1.63)	0.87 (0.58,1.32)	0.98 (0.49,1.94)	1.07 (0.91,1.26)	1.42 (1.17,1.71)	1.15 (0.96,1.39)	1.08 (0.97,1.21)	1.04 (0.83,1.29)	1.31 (0.32,5.36)	1.13 (0.87,1.46)	0.81 (0.64,1.04)
	0.23 (0.04,1.29)	1.17 (0.84,1.63)	1.00 (0.65,1.54)	-	1.23 (0.54,2.85)	0.97 (0.60,1.57)	1.05 (0.83,1.33)	1.07 (0.80,1.43)	1.03 (0.90,1.19)	-	-	-	-
CTR PGE2	0.81 (0.49,1.32)	0.86 (0.62,1.18)	0.97 (0.68,1.37)	0.66 (0.41,1.05)	0.74 (0.35,1.57)	0.76 (0.55,1.05)	1.32 (1.06,1.66)	1.08 (0.85,1.36)	1.01 (0.87,1.18)	0.97 (0.79,1.19)	1.22 (0.30,5.03)	1.05 (0.79,1.40)	0.76 (0.58,0.99)
	2.29 (0.62,8.42)	0.78 (0.46,1.32)	-	-	-	0.73 (0.34,1.54)	-	-	1.00 (0.83,1.20)	1.10 (0.89,1.34)	-	0.71 (0.25,2.06)	-
Oxytocin	1.46 (0.61,3.53)	1.56 (0.71,3.42)	1.75 (0.81,3.80)	1.20 (0.51,2.81)	1.35 (0.48,3.79)	1.37 (0.62,3.04)	1.81 (0.79,4.16)	0.81 (0.64,1.03)	0.76 (0.62,0.94)	0.73 (0.55,0.97)	0.92 (0.22,3.81)	0.79 (0.59,1.08)	0.57 (0.43,0.77)
	-	1.25 (0.35,4.52)	1.98 (0.76,5.17)	-	-	-	-	-	-	-	-	0.83 (0.28,2.51)	-
Nitric oxide donors	1.14 (0.90,1.45)	1.22 (0.77,1.92)	1.37 (0.82,2.30)	0.94 (0.53,1.66)	1.05 (0.45,2.44)	1.07 (0.65,1.77)	1.42 (0.83,2.43)	0.78 (0.32,1.93)	0.94 (0.76,1.15)	0.90 (0.68,1.19)	1.14 (0.28,4.69)	0.98 (0.72,1.33)	0.70 (0.52,0.95)
	1.23 (0.96,1.56)	0.18 (0.04,0.79)	-	-	-	0.08 (0.00,1.36)	-	-	-	-	-	-	-
Balloon catheters	0.83 (0.53,1.30)	0.88 (0.70,1.11)	0.99 (0.80,1.23)	0.68 (0.45,1.02)	0.76 (0.38,1.53)	0.78 (0.64,0.94)	1.03 (0.76,1.38)	0.57 (0.26,1.24)	0.72 (0.44,1.19)	0.96 (0.78,1.18)	1.21 (0.30,4.95)	1.04 (0.80,1.35)	0.75 (0.59,0.95)
	-	0.97 (0.68,1.40)	0.98 (0.74,1.28)	-	-	0.72 (0.57,0.91)	1.08 (0.72,1.60)	-	-	0.78 (0.54,1.14)	-	1.09 (0.74,1.58)	0.86 (0.62,1.20)
Osmotic dilators	1.02 (0.44,2.36)	1.09 (0.52,2.29)	1.23 (0.59,2.56)	0.84 (0.37,1.89)	0.94 (0.35,2.54)	0.96 (0.46,2.00)	1.27 (0.59,2.73)	0.70 (0.24,2.02)	0.89 (0.38,2.12)	1.24 (0.61,2.51)	1.26 (0.31,5.23)	1.09 (0.79,1.49)	0.78 (0.58,1.06)
	-	-	-	-	-	-	-	-	-	1.24 (0.61,2.51)	-	-	-
Oxytocin + Amniotomy	3.67 (0.15,89.85)	3.90 (0.16,93.18)	4.40 (0.18,106.03)	3.00 (0.13,70.28)	3.38 (0.13,87.18)	3.44 (0.14,82.86)	4.55 (0.19,110.34)	2.51 (0.10,65.86)	3.21 (0.13,79.11)	4.44 (0.18,106.79)	3.59 (0.14,93.30)	0.86 (0.21,3.58)	0.62 (0.15,2.57)
	-	-	-	3.00 (0.13,70.42)	-	-	-	-	-	-	-	-	-
Balloon + Oxytocin	0.72 (0.38,1.36)	0.77 (0.47,1.25)	0.87 (0.51,1.46)	0.67 (0.33,1.07)	0.67 (0.29,1.55)	0.68 (0.41,1.13)	0.90 (0.51,1.56)	0.49 (0.20,1.23)	0.63 (0.32,1.23)	0.87 (0.53,1.44)	0.71 (0.30,1.68)	0.20 (0.01,4.88)	0.72 (0.53,0.98)
	-	0.97 (0.55,1.72)	-	-	-	-	0.33 (0.01,7.95)	-	-	0.79 (0.43,1.45)	-	-	0.91 (0.62,1.34)
Balloon + PGE1	0.81 (0.48,1.38)	0.87 (0.61,1.23)	0.98 (0.66,1.45)	0.67 (0.41,1.08)	0.75 (0.35,1.63)	0.76 (0.52,1.13)	1.01 (0.65,1.58)	0.56 (0.24,1.30)	0.71 (0.40,1.26)	0.99 (0.68,1.42)	0.80 (0.36,1.77)	0.22 (0.01,5.40)	1.13 (0.67,1.91)
	-	1.04 (0.70,1.53)	0.33 (0.09,1.20)	-	-	-	-	-	-	0.93 (0.56,1.53)	-	-	1.21 (0.65,2.24)
Legend: Green: Network evidence of caesarean section due to non-reassuring fetal status Yellow: Direct evidence of caesarean section due to non-reassuring fetal status Pink: Network evidence of caesarean section due to any causes Light pink: Direct evidence of caesarean section due to any causes													
Abbreviations: - (dash), no evidence; Control, Placebo or No treatment or Expectant management; Vag PGE1, Vaginal misoprostol (≤ 50 µg); Oral PGE1, Oral misoprostol (≤ 50 µg); SL/BC PGE1, Sublingual or buccal misoprostol (≤ 50 µg); CTR PGE1, Controlled-release misoprostol vaginal pessary; Vag PGE2, Vaginal dinoprostone (tablet or gel); CTR PGE2, Controlled-release dinoprostone vaginal pessary; Oxytocin, Oxytocin (alone); Nitric oxide donors, ISMN/ISDN/nitroglycerin/sodium nitroprusside; Balloon catheters, Foley single balloon catheter or Double balloon devices (Atad/Cook balloon or other type); Oxytocin + Amniotomy, Oxytocin plus amniotomy; Balloon + Oxytocin, Balloon catheters plus oxytocin; and Balloon + PGE1, balloon catheters plus misoprostol (≤ 50 µg; oral or vaginal routes).													

Table 2 presents the ranking of methods based on SUCRA values and mean rank for each outcome. Treatment rankings were not

interpreted for outcomes with limited eligible trials and methods. Supplementary material 20 provides the cumulative probabilities

of each induction of labour method achieving different ranks for preventing each outcome. Treatment hierarchies are presented using the SUCRA, where a higher SUCRA value indicates a higher ranking among all available methods. A SUCRA of 100% represents the most effective methods, while a SUCRA of 0% represents the least effective.

Figure 10 displays spider plots, providing an overview of which induction of labour methods are more effective for each outcome, based on the SUCRA values and certainty of evidence. A larger area on the plot represents a more effective induction of labour method, while a blank area indicates a lack of information.

Figure 10. Spider plots presenting surface under the cumulative ranking curve (SUCRA) values and certainty of evidence of each induction of labour (IoL) method across the critical and important outcomes for women without previous caesarean section and a mix of women with or without previous caesarean section. Note: The spider plots provide an overview of which IoL methods are more effective for each outcome, based on SUCRA values and the certainty of evidence. A larger area on the plot indicates a more effective IoL method, while a blank area indicates a lack of information. SUCRA value (blue line) indicates the cumulative probability of efficacy or safety outcomes for each IoL method compared to the other available IoL methods, ranging from 0% to 100%, where a higher value indicates a more effective method. Certainty of evidence is determined based on the GRADE approach, categorised into four levels: very low (25%), low (50%), moderate (75%), and high certainty (100%). We evaluated the certainty of evidence for only the six critical outcomes: 1) failure to achieve vaginal delivery within 24 hours, 2) caesarean section (CS) due to non-reassuring fetal status, 3) CS due to any causes, 4) uterine hyperstimulation with changes in the heartbeat of baby before birth, 5) perinatal death (PND), and 6) severe neonatal morbidity (SNM). The certainty of network evidence (red line) is based on the results of network evidence comparing each method with vaginal misoprostol (≤ 50 μ g). For nitric oxide donors, the certainty of direct evidence (orange line) is presented instead for SNM and PND outcomes, as the intervention was not available in the network meta-analyses. The certainty of SNM is based on information comparing between nitric oxide donors with placebo, no treatment, or expectant

management, while the certainty of PND is based on comparing nitric oxide donors with vaginal misoprostol (≤ 50 μg).

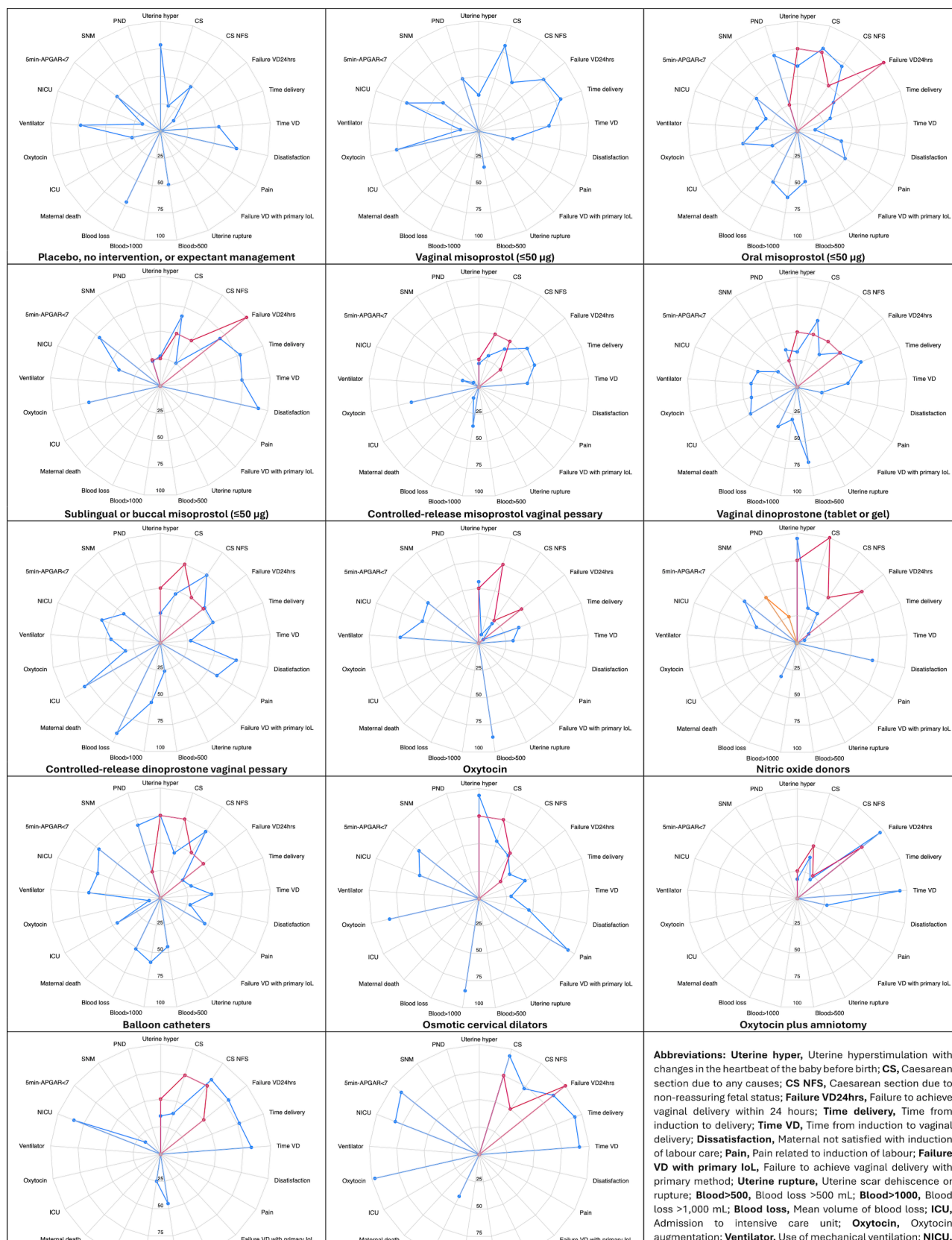
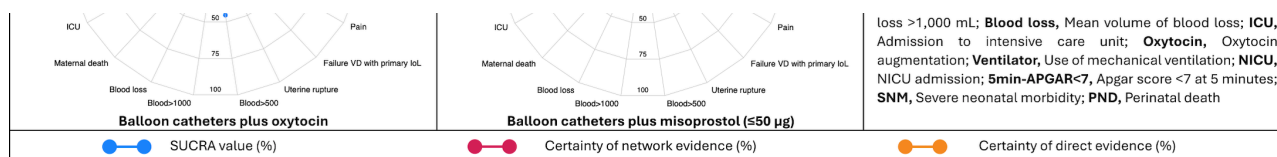


Figure 10. (Continued)



Critical outcomes

1.1 Failure to achieve vaginal delivery within 24 hours

A total of 42 studies, including 10,157 women and 14 interventions, were included in the NMA. The evidence suggests that several interventions likely reduce the risk of this outcome compared with placebo, no treatment, or expectant management. These include vaginal misoprostol ($\leq 50 \mu\text{g}$) (RR 0.60, 95% CI 0.44 to 0.82), sublingual or buccal misoprostol ($\leq 50 \mu\text{g}$) (RR 0.62, 95% CI 0.43 to 0.90), controlled-release dinoprostone vaginal pessary (RR 0.71, 95% CI 0.52 to 0.98), oxytocin plus amniotomy (RR 0.25, 95% CI 0.08 to 0.78), balloon catheters plus oxytocin (RR 0.57, 95% CI 0.37 to 0.86), and balloon catheters plus misoprostol ($\leq 50 \mu\text{g}$) (RR 0.51, 95% CI 0.32 to 0.80) (see [Figure 8](#) or [Figure 1.3](#) in [Supplementary material 20](#)).

There was no clear evidence that any of the induction of labour methods were more effective than vaginal misoprostol ($\leq 50 \mu\text{g}$) in reducing the risk of this outcome. A combination of oxytocin and amniotomy likely results in little to no difference (RR 0.41, 95% CI 0.14 to 1.24; moderate-certainty evidence). Similarly, a combination of balloon catheters and misoprostol ($\leq 50 \mu\text{g}$) (RR 0.85, 95% CI 0.60 to 1.19; high-certainty evidence) and a combination of balloon catheters and oxytocin may result in little to no difference (RR 0.94, 95% CI 0.71 to 1.26; low-certainty evidence) in this outcome. Conversely, some interventions may increase the risk of this outcome compared with vaginal misoprostol ($\leq 50 \mu\text{g}$). These include oral misoprostol ($\leq 50 \mu\text{g}$) (RR 1.26, 95% CI 1.08 to 1.48; high-certainty evidence), vaginal dinoprostone (tablet or gel) (RR 1.20, 95% CI 1.02 to 1.41; low-certainty evidence), oxytocin (alone) (RR 2.03, 95% CI 1.41 to 2.91; low-certainty evidence), nitric oxide donors (RR 1.81, 95% CI 1.11 to 2.93; moderate-certainty evidence), and balloon catheters (RR 1.43, 95% CI 1.17 to 1.75; low-certainty evidence) (see [Summary of findings 1](#) or [Figure 6A](#)).

Based on the evidence from treatment ranking evaluation, despite some methods showing comparable relative treatment effects to vaginal misoprostol ($\leq 50 \mu\text{g}$), oxytocin plus amniotomy (SUCRA 96.2%; mean rank 1.5, 95% CI 1 to 7), balloon catheters plus misoprostol ($\leq 50 \mu\text{g}$) (SUCRA 86.1%; mean rank 2.8, 95% CI 1 to 6), and balloon catheters plus oxytocin (SUCRA 78.8%; mean rank 3.8, 95% CI 1 to 7) were the highest-ranked induction of labour methods for this outcome. The SUCRA of vaginal misoprostol ($\leq 50 \mu\text{g}$) was 75.2% (mean rank 4.2, 95% CI 2 to 6) (see [Table 2](#) or [Figure 1.4](#) in [Supplementary material 20](#)).

The global test of the design-by-treatment interaction model did not show evidence of inconsistency ($P = 0.194$; see bullet point 1.1 in [Supplementary material 21](#)).

1.2 Caesarean section due to any causes

A total of 96 studies, including 28,703 women and 14 interventions, were included in the NMA. We found that vaginal misoprostol ($\leq$

$50 \mu\text{g}$) (RR 0.78, 95% CI 0.67 to 0.90), oral misoprostol ($\leq 50 \mu\text{g}$) (RR 0.78, 95% CI 0.66 to 0.92), vaginal dinoprostone (tablet or gel) (RR 0.83, 95% CI 0.72 to 0.96), and balloon catheters plus misoprostol ($\leq 50 \mu\text{g}$) (RR 0.68, 95% CI 0.52 to 0.88) probably reduce the risk of caesarean section compared with placebo, no intervention, and expectant management (see [Figure 9](#) or [Figure 2.3](#) in [Supplementary material 20](#)).

There was no clear evidence that any of the induction of labour methods were more effective than vaginal misoprostol ($\leq 50 \mu\text{g}$) in reducing the risk of caesarean section. Balloon catheters plus misoprostol ($\leq 50 \mu\text{g}$) (RR 0.87, 95% CI 0.69 to 1.09; moderate-certainty evidence) and oral misoprostol ($\leq 50 \mu\text{g}$) (RR 1.01, 95% CI 0.90 to 1.13; moderate-certainty evidence) probably result in little to no difference in this outcome compared with vaginal misoprostol ($\leq 50 \mu\text{g}$). Interestingly, oxytocin (alone) (RR 1.52, 95% CI 1.25 to 1.84; moderate-certainty evidence), nitric oxide donors (RR 1.23, 95% CI 1.02 to 1.50; high-certainty evidence), and balloon catheters (RR 1.16, 95% CI 1.04 to 1.30; moderate-certainty evidence) likely increase the risk of this outcome compared with vaginal misoprostol ($\leq 50 \mu\text{g}$) (see [Supplementary material 16](#) or [Figure 6B](#)).

According to the evaluation of treatment ranking, despite some methods showing comparable relative treatment effects to vaginal misoprostol ($\leq 50 \mu\text{g}$), balloon catheters plus misoprostol ($\leq 50 \mu\text{g}$) (SUCRA 93.6%; mean rank 1.8, 95% CI 1 to 5), vaginal misoprostol ($\leq 50 \mu\text{g}$) (SUCRA 81.2%; mean rank 3.5, 95% CI 1 to 5), and oral misoprostol ($\leq 50 \mu\text{g}$) (SUCRA 78.7%; mean rank 3.8, 95% CI 1 to 7) were the highest-ranked induction of labour methods for this outcome (see [Table 2](#) or [Figure 2.4](#) in [Supplementary material 20](#)).

The global test of the design-by-treatment interaction model did not show evidence of inconsistency ($P = 0.215$; see bullet point 1.2 in [Supplementary material 21](#)).

1.3 Caesarean section due to non-reassuring fetal status

Sixty-six studies, including 17,910 women and 14 interventions, were included in the NMA. We found that all induction of labour methods may have varying effects on this outcome compared with placebo, no intervention, or expectant management, as well as when compared with vaginal misoprostol ($\leq 50 \mu\text{g}$) (see [Figure 9](#) or [Figure 3.3](#) in [Supplementary material 20](#)).

The NMA findings suggest that a combination of balloon catheters and oxytocin (RR 0.77, 95% CI 0.47 to 1.25; moderate-certainty evidence), controlled-release dinoprostone vaginal pessary (RR 0.86, 0.62 to 1.18; low-certainty evidence), and balloon catheters (RR 0.88, 95% CI 0.70 to 1.11; low-certainty evidence) may result in little to no difference in this outcome when compared with vaginal misoprostol ($\leq 50 \mu\text{g}$) (see [Summary of findings 2](#) or [Figure 6C](#)).

Despite all methods showing comparable relative treatment effects to vaginal misoprostol ($\leq 50 \mu\text{g}$), the evidence from treatment ranking evaluation suggests that balloon catheters plus oxytocin (SUCRA 81.8%; mean rank 3.4, 95% CI 1 to 10), controlled-release dinoprostone vaginal pessary (SUCRA 74.7%; mean rank 4.3, 95% CI 1 to 9), and balloon catheters (SUCRA 73.1%; mean rank 4.5, 95% CI 1 to 8) were the highest-ranked induction of labour methods for this outcome (see [Table 2](#)). The SUCRA of vaginal misoprostol ($\leq 50 \mu\text{g}$) was 53.5% (mean rank 7, 95% CI 3 to 10). However, the ranking was unclear (see [Figure 3.4](#) in [Supplementary material 20](#)).

The global test of the design-by-treatment interaction model did not show evidence of inconsistency ($P = 0.118$; see bullet point 1.3 in [Supplementary material 21](#)).

1.4 Uterine hyperstimulation with changes in the heartbeat of the baby before birth

Fifty-three studies, including 14,647 women and 13 interventions, were included in the NMA. The evidence suggests that vaginal misoprostol ($\leq 50 \mu\text{g}$) (RR 3.47, 95% CI 1.16 to 10.35), sublingual or buccal misoprostol ($\leq 50 \mu\text{g}$) (RR 3.82, 95% CI 1.13 to 12.87), controlled-release misoprostol vaginal pessary (RR 4.80, 95% CI 1.04 to 22.29), vaginal dinoprostone (tablet or gel) (RR 3.48, 95% CI 1.09 to 11.14), and controlled-release dinoprostone vaginal pessary (RR 3.86, 95% CI 1.21 to 12.38) probably increase the risk of this outcome compared with placebo, no intervention, or expectant management (see [Figure 8](#) or [Figure 4.3](#) in [Supplementary material 20](#)).

The evidence from NMA suggests that nitric oxide donors (RR 0.05, 95% CI 0.01 to 0.47; moderate-certainty evidence), osmotic dilators (RR 0.07, 95% CI 0.01 to 0.42; moderate-certainty evidence), balloon catheters (RR 0.38, 95% CI 0.21 to 0.69; moderate-certainty evidence), and oral misoprostol (RR 0.62, 95% CI 0.39 to 0.99; moderate-certainty evidence) probably reduce the risk of this outcome when compared with vaginal misoprostol ($\leq 50 \mu\text{g}$) (see [Summary of findings 3](#) or [Figure 6D](#)).

According to the evaluation of treatment ranking, nitric oxide donors (SUCRA 94.9%; mean rank 1.6, 95% CI 1 to 3), osmotic cervical dilators (SUCRA 93.7%; mean rank 1.8, 95% CI 1 to 3), and placebo, no intervention, or expectant management (SUCRA 78.3%; mean rank 3.6, 95% CI 2 to 7) were the highest-ranked induction of labour methods for this outcome (see [Table 2](#) or [Figure 4.4](#) in [Supplementary material 20](#)).

The global test of the design-by-treatment interaction model did not show evidence of inconsistency ($P = 0.256$; see bullet point 1.4 in [Supplementary material 21](#)).

1.5 Perinatal death

Seven studies, including 2595 infants and seven interventions, were included in the NMA. The network for this outcome was found to be disconnected, resulting in the formation of two separate networks (see [Figure 5E](#)). The subnetwork involved a comparison between nitric oxide donors and placebo, no intervention, or expectant management. Only one trial (1362 women) was available for this subnetwork.

Main network

In the main network, there was no evidence of the effect, but it is still possible that there could be significant effects in either

direction in relation to perinatal death compared with vaginal misoprostol ($\leq 50 \mu\text{g}$) (see [Figure 5.3](#) in [Supplementary material 20](#)), as this outcome was rare (10 perinatal deaths were reported across all trials). The NMA findings suggest that oral misoprostol ($\leq 50 \mu\text{g}$) (RR 0.33, 95% CI 0.01 to 7.97; very low-certainty evidence) and balloon catheters (RR 0.33, 95% CI 0.01 to 16.36; very low-certainty evidence) may have little to no effect on this outcome when compared with vaginal misoprostol ($\leq 50 \mu\text{g}$) (see [Summary of findings 4](#) or [Figure 6E](#)).

Based on the evaluation of treatment ranking, despite all methods showing comparable relative treatment effects to vaginal misoprostol ($\leq 50 \mu\text{g}$), oral misoprostol ($\leq 50 \mu\text{g}$) (SUCRA 72.0%; mean rank 2.1, 95% CI 1 to 4), balloon catheters (SUCRA 69.2%; mean rank 2.2, 95% CI 1 to 4), and vaginal misoprostol ($\leq 50 \mu\text{g}$) (SUCRA 49.8%; mean rank 3, 95% CI 1 to 4) were the highest-ranked induction of labour methods for this outcome (see [Table 2](#)). However, the ranking was unclear (see [Figure 5.4](#) in [Supplementary material 20](#)).

The global test of the design-by-treatment interaction model did not show evidence of inconsistency ($P = 0.658$; see bullet point 1.5 in [Supplementary material 21](#)). However, the consistency of the underlying network could not be conclusively determined due to an insufficient number of direct comparisons.

Subnetwork

For the subnetwork, there was no evidence of a difference in the outcomes between nitric oxide donors and placebo, no intervention, or expectant management (RR 7.06, 95% CI 0.37 to 136.45; very low-certainty evidence) (see [Summary of findings 4](#) or [Figure 6E](#)). However, it is still possible that there could be significant effects in either direction regarding the outcome.

1.6 Severe neonatal morbidity

One study, including 1362 infants and two interventions, was available for this outcome. This study compared nitric oxide donors and placebo, no intervention, or expectant management. We found that nitric oxide donors may have little to no effect on this outcome compared to placebo, no intervention, or expectant management (RR 1.01, 95% CI 0.47 to 2.16; low-certainty evidence) (see [Supplementary material 17](#) and [Figure 6F](#)).

Subgroup analyses

We performed subgroup analyses for our critical outcomes based on five factors: 1) gestational age, 2) parity, 3) cervical conditions, 4) setting of induction, and 5) membrane status, as outlined in [Investigation of heterogeneity and subgroup analysis](#). All results are presented in [Supplementary material 22](#). Overall, the subgroup analyses did not reveal evidence of important subgroup differences for any of the subgroups.

Sensitivity analyses

We performed sensitivity analyses for our critical outcomes by restricting the analyses to five situations: 1) trials with overall low risk of bias, 2) trials that recruited only women without previous caesarean section, 3) splitting the comparator interventions node into three separate nodes, 4) splitting the pharmacological methods (oral misoprostol, vaginal misoprostol, and sublingual or buccal misoprostol according to dosage (low dose/high dose)), and 5) splitting the balloon catheters node into two separate nodes

(single balloon catheters and double balloon catheters), as outlined in [Sensitivity analysis](#). Information on these analyses can be found in [Supplementary material 23](#). The sensitivity analyses indicated that the overall findings remained unchanged by the mentioned criteria.

Assessment of reporting biases

We examined the small-study effects for four critical outcomes that included sufficient studies to perform the analysis. These outcomes included failure to achieve vaginal delivery within 24 hours, caesarean section due to any causes, caesarean section due to non-reassuring fetal status, and uterine hyperstimulation with changes in the heartbeat of the baby before birth. The comparison-adjusted funnel plots of these four outcomes appeared symmetrical, indicating no evidence of small-study effects ([Supplementary material 24](#)).

Important outcomes

1.7 Failure to achieve vaginal delivery with the primary method

There were no trial results available for this outcome.

1.8 Maternal not satisfied with the care related to induction of labour

Thirteen studies, including 4902 women and 10 interventions, were included in the NMA. We found that sublingual or buccal misoprostol ($\leq 50 \mu\text{g}$) (RR 0.26, 95% CI 0.10 to 0.67) probably reduces maternal dissatisfaction when compared with vaginal misoprostol ($\leq 50 \mu\text{g}$) (see [Figure 7A](#)). The evidence also suggests that controlled-release dinoprostone vaginal pessary (RR 0.47, 95% CI 0.10 to 2.13) may result in little to no difference in this outcome when compared with vaginal misoprostol ($\leq 50 \mu\text{g}$).

Despite some methods showing comparable relative treatment effects to vaginal misoprostol ($\leq 50 \mu\text{g}$), the evidence from the treatment-ranking evaluation suggests that sublingual or buccal misoprostol ($\leq 50 \mu\text{g}$) (SUCRA 91.3%; mean rank 1.8, 95% CI 1 to 4), placebo, no intervention, or expectant management (SUCRA 70.8%; mean rank 3.6, 95% CI 1 to 8), and controlled-release dinoprostone vaginal pessary (SUCRA 70.7%; mean rank 3.6, 95% CI 1 to 8) were the highest-ranked induction of labour methods for this outcome. The SUCRA of vaginal misoprostol ($\leq 50 \mu\text{g}$) was 31.7% (mean rank 7.1, 95% CI 4 to 8) (see [Table 2](#) or [Figure 7.4](#) in [Supplementary material 20](#)).

The global test of the design-by-treatment interaction model did not show evidence of inconsistency ($P = 0.077$; see bullet point 1.8 in [Supplementary material 21](#)). However, the consistency of the underlying network could not be conclusively determined due to an insufficient number of direct comparisons.

1.9 Time from induction of labour to delivery

Forty-nine studies, including 14,516 women and 13 interventions, were included in the NMA. The evidence suggests that all induction of labour methods except nitric oxide donors were probably effective in reducing the time from induction to delivery compared with placebo, no intervention, and expectant management (see [Figure 8.3](#) in [Supplementary material 20](#)).

However, we found no clear evidence that any of the induction of labour methods were more effective than vaginal misoprostol ($\leq 50 \mu\text{g}$) in reducing the time from induction of labour to delivery (see [Figure 7B](#)). Evidence from the NMA suggests that a combination

of balloon catheters and misoprostol ($\leq 50 \mu\text{g}$) (MD -1.57 hours, 95% CI -4.34 to 1.20) and sublingual or buccal misoprostol ($\leq 50 \mu\text{g}$) (MD 0.03 hours, 95% CI -2.25 to 2.30) probably result in little to no difference in this outcome when compared with vaginal misoprostol ($\leq 50 \mu\text{g}$).

According to the evaluation of treatment ranking, despite some methods showing comparable relative treatment effects to vaginal misoprostol ($\leq 50 \mu\text{g}$), balloon catheters plus misoprostol ($\leq 50 \mu\text{g}$) (SUCRA 93.2%; mean rank 1.8, 95% CI 1 to 5), vaginal misoprostol ($\leq 50 \mu\text{g}$) (SUCRA 79.8%; mean rank 3.4, 95% CI 1 to 5), and sublingual or buccal misoprostol ($\leq 50 \mu\text{g}$) (SUCRA 77.9%; mean rank 3.7, 95% CI 1 to 7) were the highest-ranked induction of labour methods for this outcome (see [Table 2](#) or [Figure 8.4](#) in [Supplementary material 20](#)).

The global test of the design-by-treatment interaction model did not show evidence of inconsistency ($P = 0.745$; see bullet point 1.9 in [Supplementary material 21](#)).

1.10 Time from induction of labour to vaginal delivery

Thirty-one studies, including 4949 women and 14 interventions, were included in the NMA. The findings suggest that most induction of labour methods may have little to no effect on this outcome compared with placebo, no intervention, or expectant management, except nitric oxide donors. However, nitric oxide donors was the only intervention that might prolong the time from induction to vaginal delivery (MD 10.74 hours, 95% CI 4.45 to 17.04) (see [Figure 9.3](#) in [Supplementary material 20](#)).

Among the interventions compared with vaginal misoprostol ($\leq 50 \mu\text{g}$), only a combination of balloon catheters and misoprostol ($\leq 50 \mu\text{g}$) (MD -3.94 hours, 95% CI -7.35 to -0.53) was likely to significantly reduce the time from induction to vaginal delivery (see [Figure 7C](#)). The NMA results for other combinations were as follows: 1) oxytocin plus amniotomy (MD -5.66 hours, 95% CI -12.20 to 0.88), and 2) balloon catheters and oxytocin (MD -2.48 hours, 95% CI -6.38 to 1.42), both of which showed little to no difference. In addition, the NMA results suggest that oral misoprostol ($\leq 50 \mu\text{g}$) (MD 4.41 hours, 95% CI 2.42 to 6.39), controlled-release dinoprostone vaginal pessary (MD 3.34 hours, 95% CI 0.02 to 6.67), and nitric oxide donors (MD 11.61 hours, 95% CI 7.55 to 15.67) may prolong the time until vaginal delivery compared with vaginal misoprostol ($\leq 50 \mu\text{g}$).

Based on the evaluation of treatment ranking, despite some methods showing comparable relative treatment effects to vaginal misoprostol ($\leq 50 \mu\text{g}$), oxytocin plus amniotomy (SUCRA 92.8%; mean rank 1.8, 95% CI 1 to 6), balloon catheters plus misoprostol ($\leq 50 \mu\text{g}$) (SUCRA 90.3%; mean rank 2.1, 95% CI 1 to 4), and balloon catheters plus oxytocin (SUCRA 81.2%; mean rank 3.3, 95% CI 1 to 7) were the highest-ranked induction of labour methods for this outcome. The SUCRA of vaginal misoprostol ($\leq 50 \mu\text{g}$) was 66.5% (mean rank 5.7, 95% CI 4 to 8) (see [Table 2](#) or [Figure 9.4](#) in [Supplementary material 20](#)).

The global test of the design-by-treatment interaction model did not show evidence of inconsistency ($P = 0.057$; see bullet point 1.10 in [Supplementary material 21](#)).

1.11 Blood loss of 500 mL or more

Nine studies, including 4485 women and eight interventions, were included in the NMA. While there was no conclusive evidence

of an effect, it remains possible that significant effects could exist in either direction for this outcome compared with placebo, no intervention, or expectant management, as well as when compared with vaginal misoprostol ($\leq 50 \mu\text{g}$) (see Figure 10.3 in [Supplementary material 20](#)). The NMA results suggest that oxytocin (alone) (RR 0.18, 95% CI 0.00 to 8.10) and vaginal dinoprostone (tablet or gel) (RR 0.37, 95% CI 0.01 to 11.80) may result in little to no difference in this outcome when compared with vaginal misoprostol ($\leq 50 \mu\text{g}$) (see Figure 7D).

Despite all methods showing comparable relative treatment effects to vaginal misoprostol ($\leq 50 \mu\text{g}$), the evidence from the treatment-ranking evaluation suggests that oxytocin (alone) (SUCRA 86.4%; mean rank 2, 95% CI 1 to 6), vaginal dinoprostone (tablet or gel) (SUCRA 69.2%; mean rank 3.2, 95% CI 1 to 6), and placebo, no treatment, or expectant management (SUCRA 49.1%; mean rank 4.6, 95% CI 1 to 8) were the highest-ranked induction of labour methods for this outcome. The SUCRA of vaginal misoprostol ($\leq 50 \mu\text{g}$) was 33.0% (mean rank 5.7, 95% CI 1 to 8) (see Table 2 or Figure 10.4 in [Supplementary material 20](#)).

The global test of the design-by-treatment interaction model did not show evidence of inconsistency ($P = 0.221$; see bullet point 1.11 in [Supplementary material 21](#)). However, the consistency of the underlying network could not be conclusively determined due to an insufficient number of direct comparisons.

1.12 Blood loss of 1000 mL or more

Nine studies, including 4589 women and nine interventions, were included in the NMA. The network for this outcome was found to be disconnected, forming two separate networks (see Figure 5K). The subnetwork consisted of a single comparison between nitric oxide donors and placebo, no intervention, or expectant management, with only one trial (200 women) contributing to this analysis.

Main network

In the main network, all induction of labour methods may have little to no effect on this outcome compared with oral misoprostol ($\leq 50 \mu\text{g}$) (see Figure 11.3 in [Supplementary material 20](#)). Specifically, osmotic dilator (RR 0.34, 95% CI 0.04 to 3.19) and balloon catheters (RR 1.01, 95% CI 0.77 to 1.33) may result in little to no difference in this outcome when compared with oral misoprostol ($\leq 50 \mu\text{g}$) (see Figure 7E).

According to the evaluation of treatment ranking, despite all methods showing comparable relative treatment effects to oral misoprostol ($\leq 50 \mu\text{g}$), osmotic cervical dilators (SUCRA 84.5%; mean rank 1.9, 95% CI 1 to 6), oral misoprostol ($\leq 50 \mu\text{g}$) (SUCRA 61.0%; mean rank 3.3, 95% CI 1 to 5), and balloon catheters (SUCRA 59.1%; mean rank 3.5, 95% CI 1 to 5) were the highest-ranked induction of labour methods for this outcome (see Table 2). However, the ranking was unclear (see Figure 11.4 in [Supplementary material 20](#)).

The global test of the design-by-treatment interaction model did not show evidence of inconsistency ($P = 0.532$; see bullet point 1.12 in [Supplementary material 21](#)). However, the consistency of the underlying network could not be determined due to an insufficient number of direct comparisons.

Subnetwork

For the subnetwork, the evidence suggests that nitric oxide donors may have little to no effect on this outcome compared with placebo, no intervention, or expectant management (see Figure 7E).

1.13 Mean volume of blood loss (mL)

Seven studies, including 2938 women and nine interventions, were included in the NMA. The network for this outcome was found to be disconnected, resulting in the formation of two separate networks (see Figure 5L). The subnetwork involved a comparison between a combination of balloon catheters and misoprostol ($\leq 50 \mu\text{g}$) and vaginal misoprostol ($\leq 50 \mu\text{g}$). Only one trial (200 women) was available for this subnetwork.

Main network

In the main network, all induction of labour methods may have little to no effect on this outcome compared with placebo, no intervention, or expectant management, as well as compared with oral misoprostol ($\leq 50 \mu\text{g}$) (see Figure 12.3 in [Supplementary material 20](#)). The NMA results suggest that controlled-release dinoprostone vaginal pessary (MD -258.32 mL, 95% CI -638.96 to 122.32) may result in little to no difference in this outcome when compared with oral misoprostol ($\leq 50 \mu\text{g}$) (see Figure 7F).

Despite some methods showing comparable relative treatment effects to oral misoprostol ($\leq 50 \mu\text{g}$), the evidence from the treatment-ranking evaluation suggests that controlled-release dinoprostone vaginal pessary (SUCRA 91.4%; mean rank 1.5, 95% CI 1 to 6), placebo, no intervention, or expectant management (SUCRA 71.4%; mean rank 2.7, 95% CI 1 to 5), and oral misoprostol ($\leq 50 \mu\text{g}$) (SUCRA 51.3%; mean rank 3.9, 95% CI 2 to 6) were the highest-ranked induction of labour methods for this outcome (see Table 2 or Figure 12.4 in [Supplementary material 20](#)).

The global test of the design-by-treatment interaction model did not show evidence of inconsistency ($P = 0.183$; see bullet point 1.13 in [Supplementary material 21](#)). However, the consistency of the underlying network could not be conclusively determined due to an insufficient number of direct comparisons.

Subnetwork

For the subnetwork, there was no evidence of a difference in the outcomes between a combination of balloon catheters and misoprostol ($\leq 50 \mu\text{g}$) and vaginal misoprostol ($\leq 50 \mu\text{g}$) (MD 4.00 mL, 95% CI -104.17 to 112.17) (see Figure 7F).

1.14 Uterine scar dehiscence or rupture

One study, including 819 women and two interventions, was available for this outcome. This study compared vaginal dinoprostone (tablet or gel) and balloon catheters. There was no evidence of a difference in the outcomes between vaginal dinoprostone (tablet or gel) and balloon catheters (RR 5.04, 95% CI 0.24 to 104.59) (see Figure 7G). However, it is still possible that there could be significant effects in either direction regarding the outcome.

1.15 Oxytocin augmentation

Fifty-nine studies, including 16,998 women and 11 interventions, were included in the NMA. The NMA findings suggest that vaginal misoprostol ($\leq 50 \mu\text{g}$) (RR 0.69, 95% CI 0.57 to 0.85), sublingual or buccal misoprostol ($\leq 50 \mu\text{g}$) (RR 0.75, 95% CI 0.56 to 0.98),

and balloon catheters plus misoprostol ($\leq 50 \mu\text{g}$) (RR 0.41, 95% CI 0.25 to 0.66) probably reduce the need of oxytocin for labour augmentation when compared with placebo, no intervention, or expectant management. In contrast, nitric oxide donors was the only method that might increase the need for oxytocin (RR 1.55, 95% CI 1.16 to 2.07) (see Figure 14.3 in [Supplementary material 20](#)).

When compared to vaginal misoprostol ($\leq 50 \mu\text{g}$), the only intervention found likely to be more effective in reducing oxytocin use was a combination of balloon catheters and misoprostol (RR 0.58, 95% CI 0.38 to 0.91) (see [Figure 7H](#)). In addition, osmotic cervical dilators (RR 0.78, 95% CI 0.41 to 1.49) likely result in little to no difference in this outcome when compared with vaginal misoprostol ($\leq 50 \mu\text{g}$). However, oral misoprostol ($\leq 50 \mu\text{g}$) (RR 1.21, 95% CI 1.05 to 1.39), vaginal dinoprostone (tablet or gel) (RR 1.27, 95% CI 1.11 to 1.46), controlled-release dinoprostone vaginal pessary (RR 1.36, 95% CI 1.14 to 1.63), nitric oxide donors (RR 2.23, 95% CI 1.60 to 3.11), and balloon catheters (RR 1.71, 95% CI 1.48 to 1.99) may increase the need of oxytocin for labour augmentation.

Based on the evaluation of treatment ranking, despite some methods showing comparable relative treatment effects to vaginal misoprostol ($\leq 50 \mu\text{g}$), balloon catheters plus misoprostol ($\leq 50 \mu\text{g}$) (SUCRA 97.2%; mean rank 1.3, 95% CI 1 to 2), osmotic cervical dilators (SUCRA 83.2%; mean rank 2.7, 95% CI 1 to 8), and vaginal misoprostol ($\leq 50 \mu\text{g}$) (SUCRA 76.2%; mean rank 3.4, 95% CI 2 to 4) were the highest-ranked induction of labour methods for this outcome (see [Table 2](#) or Figure 14.4 in [Supplementary material 20](#)).

The global test of the design-by-treatment interaction model showed evidence of inconsistency ($P = 0.01$; see bullet point 1.15 in [Supplementary material 21](#)). However, out of 15 close-loops available for this outcome, only two of them – 1) placebo, no treatment, or expectant management versus vaginal misoprostol ($\leq 50 \mu\text{g}$) versus oral misoprostol ($\leq 50 \mu\text{g}$), and 2) placebo, no treatment, or expectant management versus oral misoprostol ($\leq 50 \mu\text{g}$) versus vaginal dinoprostone (tablet or gel) – showed statistically significant evidence for inconsistency as evaluated from the local test (loop-specific approach). We did not plan to evaluate the source of inconsistency for important outcomes. Therefore, the findings of this outcome should be interpreted cautiously.

1.16 Maternal death

No maternal deaths were observed in 17 studies (6147 women) that reported this outcome.

1.17 Admission to intensive care unit

Three studies, including 3453 women and four interventions, were included in the NMA. There were three direct comparisons available for this network: oral misoprostol ($\leq 50 \mu\text{g}$) versus balloon catheters, vaginal dinoprostone (tablet or gel) versus balloon catheters, and controlled-release dinoprostone vaginal pessary versus balloon catheters (see [Figure 5O](#)). The network evidence indicated no significant differences between vaginal dinoprostone (tablet or gel) (RR 0.50, 95% CI 0.02 to 10.93), controlled-release dinoprostone vaginal pessary (RR 0.25, 95% CI 0.02 to 3.97), or balloon catheters (RR 0.50, 95% CI 0.05 to 5.52) when compared with oral misoprostol ($\leq 50 \mu\text{g}$) for this outcome (see [Figure 7I](#)).

The global test of the design-by-treatment interaction model did not show evidence of inconsistency ($P = 0.321$; see bullet point 1.17 in [Supplementary material 21](#)). However, the consistency of the

underlying network could not be conclusively determined due to an insufficient number of direct comparisons.

1.18 Pain related to induction of labour

Six studies, including 1718 women and seven interventions, were included in the NMA. The network for this outcome was found to be disconnected, resulting in the formation of two separate networks (see [Figure 5P](#)). The subnetwork involved a comparison between sublingual or buccal misoprostol ($\leq 50 \mu\text{g}$) and vaginal misoprostol ($\leq 50 \mu\text{g}$). There were two trials (142 women) available for this subnetwork.

Main network

In the main network, osmotic cervical dilators (RR 0.72, 95% CI 0.53 to 0.98) probably reduce pain related to labour induction compared with oral misoprostol ($\leq 50 \mu\text{g}$), while the use of vaginal dinoprostone (tablet or gel) may increase pain (RR 2.35, 95% CI 1.12 to 4.89) (see [Figure 7J](#)).

The global test of the design-by-treatment interaction model did not show evidence of inconsistency ($P = 0.397$; see bullet point 1.18 in [Supplementary material 21](#)). However, the consistency of the underlying network could not be conclusively determined due to an insufficient number of direct comparisons.

Subnetwork

For the subnetwork, the evidence suggests that sublingual or buccal misoprostol ($\leq 50 \mu\text{g}$) (RR 1.10, 95% CI 0.81 to 1.50) may have little to no effect on this outcome compared with vaginal misoprostol ($\leq 50 \mu\text{g}$) (see [Figure 7J](#)).

1.19 Use of mechanical ventilation for infants

Seven studies, including 6237 infants and nine interventions, were included in the NMA. We found that all available induction of labour methods may have varying effects on this outcome compared with placebo, no intervention, or expectant management, except vaginal misoprostol ($\leq 50 \mu\text{g}$), which may increase the risk of this outcome (RR 4.08, 95% CI 1.01 to 16.44) (see Figure 17.3 in [Supplementary material 20](#)). When compared with vaginal misoprostol ($\leq 50 \mu\text{g}$), the NMA findings suggest that oxytocin (alone) (RR 0.25, 95% CI 0.06 to 1.09) and balloon catheters (RR 0.21, 95% CI 0.00 to 9.37) likely result in little to no difference in this outcome (see [Figure 7K](#)).

Based on the evaluation of treatment ranking, despite some methods showing comparable relative treatment effects to vaginal misoprostol ($\leq 50 \mu\text{g}$), placebo, no intervention, or expectant management (SUCRA 72.7%; mean rank 2.6, 95% CI 1 to 4), oxytocin (alone) (SUCRA 71.6%; mean rank 2.7, 95% CI 1 to 5), and balloon catheters (SUCRA 65.1%; mean rank 3.1, 95% CI 1 to 7) were the highest-ranked induction of labour methods for this outcome. However, the ranking was unclear. The SUCRA of vaginal misoprostol ($\leq 50 \mu\text{g}$) was 16.5% (mean rank 6, 95% CI 4 to 7) (see [Table 2](#) or Figure 17.4 in [Supplementary material 20](#)).

The global test of the design-by-treatment interaction model did not show evidence of inconsistency ($P = 0.567$; see bullet point 1.19 in [Supplementary material 21](#)). However, the consistency of the underlying network could not be conclusively determined due to an insufficient number of direct comparisons.

1.20 Admission to the neonatal intensive care unit

Sixty-five studies, including 23,408 infants and 13 interventions, were included in the NMA. The NMA findings suggest that only vaginal misoprostol ($\leq 50 \mu\text{g}$) (RR 0.68, 95% CI 0.48 to 0.97) probably reduces the risk of this outcome compared with placebo, no intervention, or expectant management (see Figure 18.3 in [Supplementary material 20](#)).

However, we found that all induction of labour methods may have little to no effect on this outcome compared with vaginal misoprostol ($\leq 50 \mu\text{g}$). A combination of balloon catheters and oxytocin (RR 0.78, 95% CI 0.42 to 1.46) and a combination of balloon catheters and misoprostol ($\leq 50 \mu\text{g}$) (RR 0.84, 95% CI 0.52 to 1.38) likely result in little to no difference in this outcome when compared with vaginal misoprostol ($\leq 50 \mu\text{g}$) (see [Figure 7L](#)).

Despite all methods showing comparable relative treatment effects to vaginal misoprostol ($\leq 50 \mu\text{g}$), the evidence from the treatment-ranking evaluation suggests that balloon catheters plus oxytocin (SUCRA 84.4%; mean rank 2.9, 95% CI 1 to 10), balloon catheters plus misoprostol ($\leq 50 \mu\text{g}$) (SUCRA 81.9%; mean rank 3.2, 95% CI 1 to 9), and vaginal misoprostol ($\leq 50 \mu\text{g}$) (SUCRA 69.9%; mean rank 4.6, 95% CI 1 to 8) were the highest-ranked induction of labour methods for this outcome (see [Table 2](#) or Figure 18.4 in [Supplementary material 20](#)).

The global test of the design-by-treatment interaction model did not show evidence of inconsistency ($P = 0.496$; see bullet point 1.20 in [Supplementary material 21](#)).

1.21 Apgar score < 7 at 5 minutes

Fifty-two studies, including 20,727 infants and 13 interventions, were included in the NMA. The NMA findings suggest that only controlled-release misoprostol vaginal pessary (RR 2.78, 95% CI 1.10 to 7.07) probably increases the chances of an infant having an Apgar score < 7 at 5 minutes compared with placebo, no intervention, or expectant management (see Figure 19.3 in [Supplementary material 20](#)).

When compared with vaginal misoprostol ($\leq 50 \mu\text{g}$), only a combination of balloon catheters and misoprostol ($\leq 50 \mu\text{g}$) (RR 0.45, 95% CI 0.23 to 0.86) was likely more effective in preventing an infant from having an Apgar score < 7 at 5 minutes. Balloon catheters (RR 0.69, 95% CI 0.41 to 1.16) and sublingual or buccal misoprostol ($\leq 50 \mu\text{g}$) (RR 0.66, 95% CI 0.34 to 1.30) likely result in little to no difference in this outcome when compared with vaginal misoprostol ($\leq 50 \mu\text{g}$) (see [Figure 7M](#)).

Despite some methods showing comparable relative treatment effects to vaginal misoprostol ($\leq 50 \mu\text{g}$), the evidence from the treatment-ranking evaluation suggests that balloon catheters plus misoprostol ($\leq 50 \mu\text{g}$) (SUCRA 90.6%; mean rank 2.1, 95% CI 1 to 5), balloon catheters (SUCRA 71.3%; mean rank 4.4, 95% CI 2 to 8), and sublingual or buccal misoprostol ($\leq 50 \mu\text{g}$) (SUCRA 70.6%; mean rank 4.5, 95% CI 1 to 10) were the highest-ranked induction of labour methods for this outcome. The SUCRA of vaginal misoprostol ($\leq 50 \mu\text{g}$) was 41.2% (mean rank 8.1, 95% CI 4 to 11) (see [Table 2](#) or Figure 19.4 in [Supplementary material 20](#)).

The global test of the design-by-treatment interaction model did not show evidence of inconsistency ($P = 0.829$; see bullet point 1.21 in [Supplementary material 21](#)).

2. Women with previous caesarean section

There were only two direct comparisons available for this network: 1) controlled-release dinoprostone vaginal pessary versus balloon catheters, and 2) oxytocin (alone) versus balloon catheters. Each comparison was based on information from one trial.

[Figure 11](#) and [Figure 12](#) show the network diagrams and the forest plot of available outcomes for this population, respectively. Full details of analysis results for all available outcomes, including a network diagram, forest plot presenting the direct, indirect, and network evidence, league table, and rankogram, are presented in [Supplementary material 25](#). Due to limited data, we decided not to estimate and report the treatment ranking for this population.

Figure 11. Summary of network diagrams across the critical and important outcomes for women with previous caesarean section. Note: Network diagram visually represents the relationships between different interventions in the included studies for each outcome. Each node (circle) represents an intervention, with its size proportional to the number of trials comparing this intervention to any other in the network. Each edge (line) connecting two interventions represents a direct comparison, with its thickness proportional to the number of trials making each direct comparison.

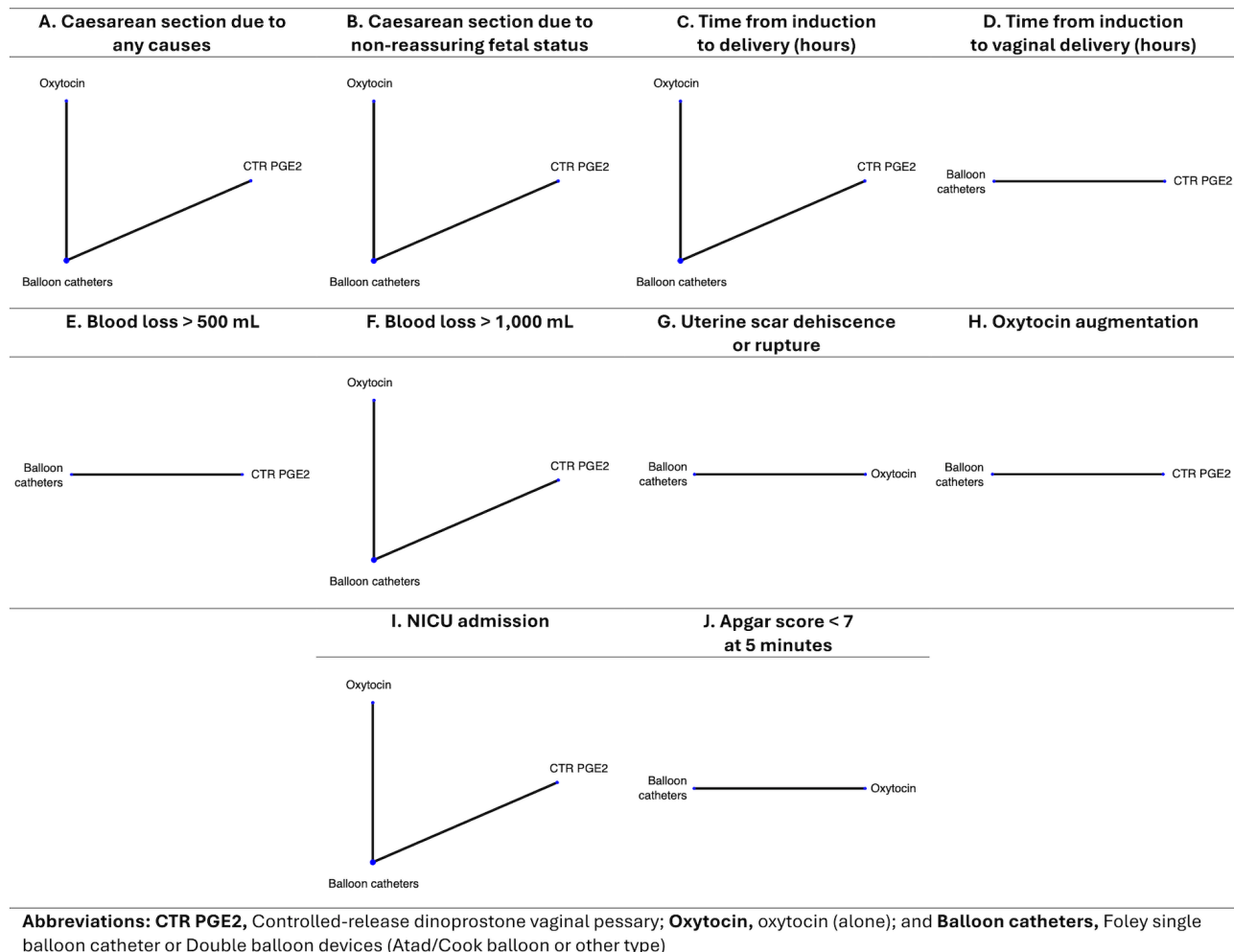
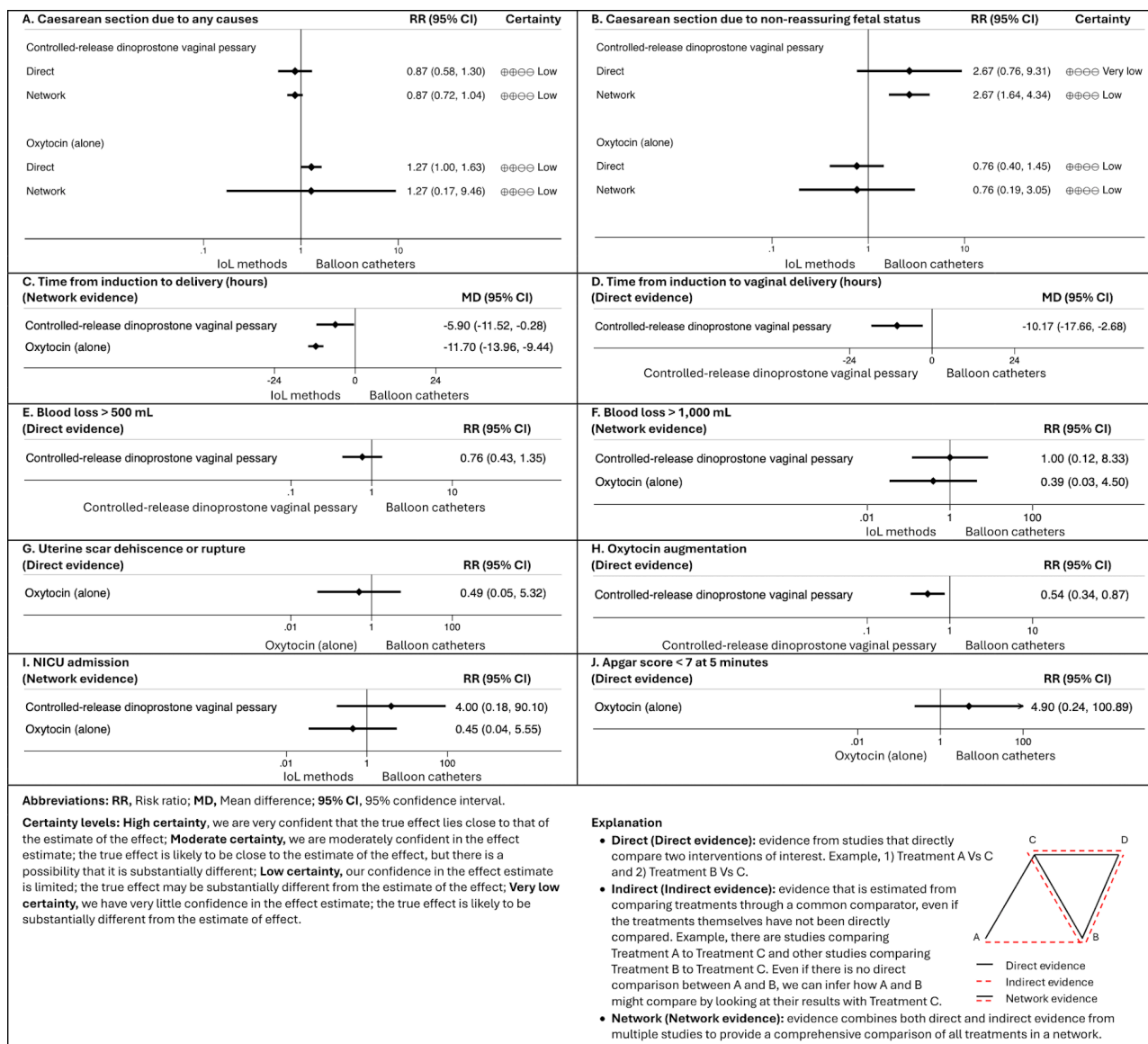


Figure 12. Summary of forest plots across the critical and important outcomes for women with previous caesarean section.

Critical outcomes

2.1 Failure to achieve vaginal delivery within 24 hours

There were no trial results available for this outcome.

2.2 Caesarean section due to any causes

Two studies, including 282 women and three interventions, were included in the NMA. The findings from the NMA indicate that controlled-release dinoprostone vaginal pessary (RR 0.87, 95% CI 0.72 to 1.04; low-certainty evidence) may have little to no effect on this outcome compared to balloon catheters (see [Supplementary material 18](#) or [Figure 12A](#)). There is no evidence of a difference in the outcomes between oxytocin (alone) and balloon catheters (RR 1.27, 95% CI 0.17 to 9.46; low-certainty evidence). However, it is still possible that there could be significant effects in either direction regarding the outcome.

The global test of the design-by-treatment interaction model did not show evidence of inconsistency ($P = 0.812$; see bullet point 2.2 in [Supplementary material 21](#)). However, the consistency of the underlying network could not be conclusively determined due to an insufficient number of direct comparisons.

2.3 Caesarean section due to non-reassuring fetal status

Two studies, including 282 women and three interventions, were included in the NMA. The evidence suggests that controlled-release dinoprostone vaginal pessary (RR 2.67, 95% CI 1.64 to 4.34; low-certainty evidence) may increase the risk of this outcome (see [Supplementary material 19](#) or [Figure 12B](#)).

The global test of the design-by-treatment interaction model did not show evidence of inconsistency ($P = 0.701$; see bullet point 2.3 in [Supplementary material 21](#)). However, the consistency of the underlying network could not be conclusively determined due to an insufficient number of direct comparisons.

2.4 Uterine hyperstimulation with changes in the heartbeat of the baby before birth

There were no trial results available for this outcome.

2.5 Perinatal death

There were no trial results available for this outcome.

2.6 Severe neonatal morbidity

There were no trial results available for this outcome.

Subgroup analysis and sensitivity analysis

We were unable to conduct subgroup and sensitivity analyses due to a lack of variability in the prespecified factors between the two included studies for the critical outcomes.

Assessment of reporting biases

We were unable to perform the analysis due to lack of evidence.

Important outcomes

2.7 Failure to achieve vaginal delivery with the primary method

There were no trial results available for this outcome.

2.8 Maternal not satisfied with the care related to induction of labour

There were no trial results available for this outcome.

2.9 Time from induction of labour to delivery

Two studies, including 282 women and three interventions, were included in the NMA. We found that controlled-release dinoprostone vaginal pessary (MD -5.90 hours, 95% CI -11.52 to -0.28) and oxytocin (alone) (MD -11.70 hours, 95% CI -13.96 to -9.44) may reduce the time from induction of labour to delivery compared to balloon catheters (see [Figure 12C](#)).

The global test of the design-by-treatment interaction model did not show evidence of inconsistency ($P < 0.001$; see bullet point 2.9 in [Supplementary material 21](#)). However, the consistency of the underlying network could not be conclusively determined due to an insufficient number of direct comparisons.

2.10 Time from induction of labour to vaginal delivery

There was only one available comparison for this outcome: controlled-release dinoprostone vaginal pessary versus balloon catheters (one trial, 78 women). The trial reported that controlled-release dinoprostone vaginal pessary (MD -10.17 hours, 95% CI -17.66 to -2.68) may reduce the time from induction of labour to vaginal delivery compared with balloon catheters (see [Figure 12D](#)).

2.11 Blood loss of 500 mL or more

There was only one available comparison for this outcome: controlled-release dinoprostone vaginal pessary versus balloon catheters (one trial, 78 women). The trial reported that controlled-release dinoprostone vaginal pessary (RR 0.76, 95% CI 0.43 to 1.35) may have little to no effect on this outcome compared with balloon catheters (see [Figure 12E](#)).

2.12 Blood loss of 1000 mL or more

Two studies, including 282 women and three interventions, were included in the NMA. There is no evidence of a difference in the outcomes either between controlled-release dinoprostone vaginal

pessary (RR 1.00, 95% CI 0.12 to 8.33) or oxytocin (alone) (RR 0.39, 95% CI 0.03 to 4.50) and balloon catheters (see [Figure 12F](#)). However, it is still possible that there could be significant effects in either direction regarding the outcome.

The global test of the design-by-treatment interaction model did not show evidence of inconsistency ($P = 0.452$; see bullet point 2.12 in [Supplementary material 21](#)). However, the consistency of the underlying network could not be conclusively determined due to an insufficient number of direct comparisons.

2.13 Mean volume of blood loss (mL)

There were no trial results available for this outcome.

2.14 Uterine scar dehiscence or rupture

There was only one comparison for this outcome: oxytocin (alone) versus balloon catheters (one trial, 204 women). There is no evidence of a difference between oxytocin (alone) (RR 0.49, 95% CI 0.05 to 5.32) and balloon catheters (see [Figure 12G](#)). However, it is still possible that there could be significant effects in either direction regarding the outcome.

2.15 Oxytocin augmentation

There was only one comparison for this outcome: controlled-release dinoprostone vaginal pessary versus balloon catheters (one trial, 78 women). The trial reported that controlled-release dinoprostone vaginal pessary (RR 0.54, 95% CI 0.34 to 0.87) may reduce the chance of oxytocin use for augmentation of labour compared with balloon catheters (see [Figure 12H](#)).

2.16 Maternal death

There were no maternal deaths in the one trial (204 women) that reported this outcome.

2.17 Admission to intensive care unit

There were no trial results available for this outcome.

2.18 Pain related to induction of labour

There were no trial results available for this outcome.

2.19 Use of mechanical ventilation for infants

There were no trial results available for this outcome.

2.20 Admission to the neonatal intensive care unit

Two studies, including 282 infants and three interventions, were included in the NMA. There is no evidence of a difference in the outcomes either between controlled-release dinoprostone vaginal pessary (RR 4.00, 95% CI 0.18 to 90.10) or oxytocin (alone) (RR 0.45, 95% CI 0.04 to 5.55) and balloon catheters (see [Figure 12I](#)). However, it is still possible that there could be significant effects in either direction regarding the outcome.

The global test of the design-by-treatment interaction model did not show evidence of inconsistency ($P = 0.530$; see bullet point 2.20 in [Supplementary material 21](#)). However, the consistency of the underlying network could not be conclusively determined due to an insufficient number of direct comparisons.

2.21 Apgar score < 7 at 5 minutes

There was only one comparison for this outcome: oxytocin (alone) versus balloon catheters (one trial, 204 infants). There is no evidence of a difference in the outcome between oxytocin (alone) (RR 4.90, 95% CI 0.24 to 100.89) and balloon catheters (see [Figure 12J](#)). However, it is still possible that there could be significant effects in either direction regarding the outcome.

DISCUSSION

Summary of main results

This NMA evaluated the relative effects of 13 currently recommended methods for inducing labour in women with a live fetus who are at or beyond term labour (gestational age ≥ 37 weeks). It included 106 trials assessing the induction methods in women without a previous caesarean section, as well as a mix of women with and without a previous caesarean section. Only two trials focused specifically on women with a previous caesarean section.

Direct estimates were possible for about half of the comparisons analysed in this review. However, many of these comparisons were based on only a few trials or a small number of participants, or both. When direct comparison data could be pooled, the results generally aligned with the indirect estimates. No significant inconsistency was found in the NMAs, although some outcomes had limited data for assessment.

The following is a summary of the main results for the four critical outcomes among women without caesarean section and a mix of women with or without previous caesarean section, as presented in the summary of findings tables: 1) failure to achieve vaginal delivery within 24 hours, 2) caesarean section due to non-reassuring fetal status, 3) uterine hyperstimulation with changes in the heartbeat of the baby before birth, and 4) perinatal death.

For women without previous caesarean section and a mix of women with or without previous caesarean section, the NMA found no clear evidence that any of the induction of labour methods were more effective than vaginal misoprostol (≤ 50 μ g) in reducing the risk of failure to achieve vaginal delivery within 24 hours or in lowering the rates of caesarean section due to non-reassuring fetal status. Nitric oxide donors, osmotic cervical dilators, balloon catheters, and oral misoprostol (≤ 50 μ g) were probably more effective than vaginal misoprostol (≤ 50 μ g) in reducing the risk of uterine hyperstimulation with fetal heart rate changes. However, evidence remained insufficient regarding perinatal death.

There is insufficient evidence to indicate the safest and most effective labour induction method for women with previous caesarean section.

Limitations of the evidence included in the review

According to our review criteria, this review is applicable for women at or beyond term labour (gestational age $\geq 37^{+0}$ weeks). We included studies that focused on women both with or without previous caesarean section. Of the 106 trials assessed in this review, only two trials assessed women with a previous caesarean section. Among the six critical outcomes of women with previous caesarean section, only caesarean section outcomes were reported: 1) caesarean section due to any causes and 2) caesarean section due to non-reassuring fetal status.

This review assessed the effects of labour induction methods recommended by five international guidelines (ACOG, NICE, RCOG, FIGO, and WHO), as well as the methods suggested for further research by previous Cochrane reviews. We anticipate that these recommended and proposed methods are relevant to current clinical practice. However, it is possible that some methods used in real-world practice may not have been included in this review. Additionally, we did not include studies comparing different dosages of the same pharmacological interventions. It is important to note that limiting the types of interventions evaluated in this NMA may result in excluding relevant indirect evidence if such excluded evidence would satisfy the transitivity assumption.

The number of included studies and participants, or both, for some outcomes were limited, particularly for adverse maternal and neonatal outcomes and women's satisfaction and pain associated with induction. Additionally, not all labour induction methods were reported across the outcomes of interest. Many studies were excluded from analyses of adverse outcomes, such as perinatal death, uterine rupture, maternal death, NICU admission, and Apgar score < 7 at 5 minutes, because no events were observed in the studied groups, potentially leading to underpowered analyses. Furthermore, variations in the reporting formats for continuous outcomes, such as mean with interquartile range, mean with min-max, or mean/median alone, prevented us from converting results into a suitable format for meta-analysis. Consequently, these studies were excluded from our analysis.

We applied the GRADE approach for network meta-analysis to assess the certainty of evidence for the six critical outcomes in this NMA. The overall certainty of evidence ranged from very low to high, despite our only including studies that met the criteria for trustworthiness. We downgraded the certainty of the evidence due to: 1) unclear reporting of the methods used for random sequence generation and allocation concealment among the included studies, 2) heterogeneity of relative effects among direct comparisons, 3) issues with imprecision, and 4) inconsistency between direct and indirect estimates. We identified only one case of inconsistency across all evaluations, and we found no transitivity issues for this evidence.

Limitations of the review processes

The evidence in this review was obtained through a systematic search with no restrictions on publication date or language. The search was conducted by Cochrane Pregnancy and Childbirth's Information Specialist. We also sought out published articles by contacting authors of trial registrations and reviewing references from a related NMA [9]. However, it is possible that some published trials may have been missed.

The findings presented in this review are based on the evidence identified up to 1 February 2023. To assess whether an update was necessary, we conducted a scoping search on 28 February 2025 to identify any new relevant RCTs and performed preliminary analyses by incorporating any newly identified data into the original NMAs. The search covered three databases, Ovid (MEDLINE), CENTRAL, and Embase, for studies published between 1 February 2023 and 28 February 2025. We also searched ClinicalTrials.gov and the WHO ICTRP trial registries during this period, and reviewed the full-text reports of studies classified as ongoing in our previous search. After removing duplicates, we screened the titles and abstracts of 1747 records, then reviewed 70 full-text articles. We found

eight new RCTs (19 reports) that met both our review criteria and the Cochrane Pregnancy and Childbirth Trustworthiness Screening Tool criteria. We then performed preliminary analyses by incorporating the data of these new trials into our NMAs for four critical outcomes, excluding perinatal death and severe neonatal morbidity because no new trials reported these outcomes. Although the inclusion of these new trials slightly changed the findings for the critical outcomes, the overall conclusions of the review remained unchanged. We therefore decided not to incorporate the data from these eight RCTs into the current version of the review. The findings presented here are thus based on evidence searched and reviewed up to 1 February 2023. We plan to update the search and analyses in a future version of the review.

According to the inclusion criteria, this review pertains to women at or beyond term (gestational age ≥ 37 weeks). However, many trials did not clearly define the term pregnancy, leading to the exclusion of a significant number of studies. Clear reporting of this information is crucial for healthcare practitioners and policymakers when making clinical decisions and recommendations.

At least two review authors independently screened, reviewed, extracted data, and assessed risk of bias and the trustworthiness of the studies. We discovered that several studies did not meet the trustworthiness criteria of the Cochrane Pregnancy and Childbirth Trustworthiness Screening tool. As a result, these studies were categorised as awaiting classification. Before categorising these studies as such, we attempted to contact the first and corresponding authors twice to obtain additional information or clarification.

We conducted subgroup and sensitivity analyses for only the six critical outcomes, even though we found no significant heterogeneity among them. However, for oxytocin augmentation, an important outcome of this review, caution is advised when interpreting the results due to statistical inconsistencies. In addition, we did not plan and conduct subgroup analysis according to indications for induction, such as post-term pregnancy, hypertension/pre-eclampsia, or premature rupture of membrane. The indications for induction may have an effect on the benefits and harms of induction of labour [13].

Finally, this review included several changes to our protocol, which are highlighted in [Supplementary material 9](#).

Agreements and disagreements with other studies or reviews

We identified five previous NMAs on methods of induction of labour published between 2015 and 2024 [10, 94, 95, 96, 97]. Four of these NMAs focused on specific types of induction of labour methods or settings. Only one NMA was similar to this review [9, 10], as it evaluated a broad range of induction of labour methods without restriction to specific settings. That NMA reviewed 611 studies covering 31 induction of labour methods and used a placebo as a common comparator. The analysis found that oxytocin plus amniotomy and ≥ 50 μ g vaginal misoprostol were probably more effective than placebo in reducing the risk of failure to achieve vaginal delivery within 24 hours. However, no induction of labour method was found to reduce the risk of uterine hyperstimulation with fetal heart rate changes compared with placebo. In addition, it identified the following three methods as the highest-ranked for reducing the risk of

caesarean section: titrated low-dose misoprostol, sublingual or buccal misoprostol, and < 50 μ g vaginal misoprostol. However, this previous NMA included induction of labour methods no longer in current clinical use, such as acupuncture, corticosteroids, and intravenous prostaglandin. Additionally, it reviewed only one combined method: the use of oxytocin and amniotomy.

Our NMA specifically examined the methods recommended by various internationally accepted guidelines and Cochrane reviews. We only included studies deemed trustworthy according to the Cochrane Pregnancy and Childbirth Trustworthiness Screening Tool. Unlike previous reviews, which reported odds ratios, we reported treatment effects using risk ratios. Additionally, we combined different dosages of the same pharmacological methods into a single node. These methodological differences, along with variations in the types of induction of labour methods evaluated, the number of included studies, and study settings (inpatient versus outpatient), may explain differences in findings compared to previous NMAs. To compare our findings with those of the previous NMA that used a placebo as a common comparator [10], the following information outlines our findings when comparing all available induction of labour methods with placebo, no treatment, and expectant management. According to the three highest-ranked methods identified in our NMA, oxytocin plus amniotomy, balloon catheters plus misoprostol (≤ 50 μ g), and balloon catheters plus oxytocin may be more effective than placebo or no treatment in achieving vaginal delivery within 24 hours. However, none of the methods reduced the risk of uterine hyperstimulation with fetal heart rate changes compared to the control method. Additionally, balloon catheters plus misoprostol (≤ 50 μ g), vaginal misoprostol (≤ 50 μ g), oral misoprostol (≤ 50 μ g), and vaginal dinoprostone (tablet or gel) were likely to reduce the risk of caesarean section due to any causes compared to the control method.

AUTHORS' CONCLUSIONS

Implications for practice

This review found no clear evidence that any of the induction of labour methods were more effective than vaginal misoprostol (≤ 50 μ g) in reducing the risks of failure to achieve vaginal delivery within 24 hours, caesarean section due to non-reassuring fetal status, and perinatal death. The certainty of this evidence ranged from very low to high. Notably, some induction of labour methods, including oxytocin, nitric oxide donors, and balloon catheters, may increase the risk of failure to achieve vaginal delivery within 24 hours compared to vaginal misoprostol (≤ 50 μ g). Additionally, not all induction of labour methods were more effective than placebo, no intervention, or expectant management in reducing the risks of failure to achieve vaginal delivery within 24 hours, caesarean section due to non-reassuring fetal status, and perinatal death.

None of the induction of labour methods reduced the risk of uterine hyperstimulation with fetal heart changes when compared with placebo, no intervention, or expectant management. However, nitric oxide donors, osmotic cervical dilators, balloon catheters, and oral misoprostol (≤ 50 μ g) probably lower this risk compared to vaginal misoprostol (≤ 50 μ g).

An interesting finding is that the combination of balloon catheters and misoprostol (≤ 50 μ g) may reduce the time from induction to vaginal delivery, may decrease the need for oxytocin augmentation, and may lower the risk of infants having an Apgar score < 7 at

5 minutes compared to vaginal misoprostol ($\leq 50 \mu\text{g}$). However, there is insufficient evidence regarding the safety of this method with regard to uterine hyperstimulation with fetal heart changes, perinatal death, severe neonatal morbidity, haemorrhage, and important maternal-reported outcomes such as satisfaction and pain related to induction. It is possible that combined methods may also cause more pain than single methods. It is also important to note that the evidence for combined balloon catheter and misoprostol was based on only four trials, all of which used low-dose misoprostol ($25 \mu\text{g}$) administered either orally or vaginally. Since this network meta-analysis grouped both routes together, it does not clarify whether the oral or vaginal route is preferable. In future updates of this review with more data, we will assess the effects of this combination separately for each route.

The evidence remains insufficient to determine the effects of induction of labour methods on uterine hyperstimulation with fetal heart changes, perinatal death, severe neonatal morbidity, uterine scar dehiscence or rupture, haemorrhage, intensive care admission for women, and maternal satisfaction and pain related to induction. Additionally, not all induction of labour methods were available to assess these outcomes.

The findings of this review are primarily based on studies conducted in hospital settings and among women without a previous caesarean section and a mix of women with or without previous caesarean section. Only two trials specifically assessed the effects of induction of labour methods in women with a previous caesarean section. Healthcare practitioners and policymakers should consider these limitations when using these findings to inform their decisions.

Implications for research

The variability in outcome measures across the trials included in this review hindered comparisons between trials for many outcomes, particularly for undesirable outcomes such as uterine hyperstimulation with fetal heart changes, perinatal death, severe neonatal morbidity, maternal death, uterine scar dehiscence or rupture, haemorrhage, intensive care admission for women, and women's satisfaction and pain associated with induction. Recently, standard sets of outcome measures, known as core outcome sets, for trials assessing labour induction have been proposed to ensure consistent reporting of clinically important outcomes [46]. Future research should adhere to these core outcome sets to enable meaningful comparisons and facilitate data pooling across different trials.

This review adhered to the trustworthiness policy set by Cochrane and the Cochrane Pregnancy and Childbirth group, which requires evaluating and addressing trustworthiness issues in included studies. Many trials did not meet these trustworthiness criteria and were excluded from the analyses. The main issues were unclear methodological descriptions and a lack of prospective trial registration. Future trials should follow the CONSORT guidelines to ensure comprehensive and transparent reporting [98]. Additionally, all trial protocols must be registered prospectively before enrolling participants.

Although the combination of balloon catheters and misoprostol ($\leq 50 \mu\text{g}$) may lead to the shortest time from induction to vaginal delivery and fewer side effects compared to other methods, this conclusion is based on data from only four included trials. One trial

used $25 \mu\text{g}$ oral misoprostol, while the other three trials used $25 \mu\text{g}$ vaginal misoprostol. It is unclear which route of misoprostol is more effective for this combination method. There is a need for well-designed trials to resolve this issue.

There is limited evidence on the efficacy and side effects of labour induction methods for women with a previous caesarean section. Among the over 100 trials included in this review, only two specifically assessed pregnant women with a history of caesarean section, limiting our ability to draw meaningful conclusions. There is a need for future trials to identify the most effective labour induction methods for this group.

SUPPLEMENTARY MATERIALS

Supplementary materials are available with the online version of this article: [10.1002/14651858.CD015234.pub2](https://doi.org/10.1002/14651858.CD015234.pub2).

Supplementary material 1 Search strategies

Supplementary material 2 Characteristics of included studies

Supplementary material 3 Characteristics of excluded studies

Supplementary material 4 Characteristics of studies awaiting classification

Supplementary material 5 Characteristics of ongoing studies

Supplementary material 6 Analyses

Supplementary material 7 Data package

Supplementary material 8 Glossary of terms

Supplementary material 9 Differences between the protocol and this review

Supplementary material 10 Recommended methods of induction of labour

Supplementary material 11 List of interventions recommended by Cochrane Reviews for further research to determine their true effects

Supplementary material 12 Trustworthiness screening summary: review authors' judgements about each of the Cochrane Pregnancy and Childbirth Trustworthiness Screening Tool item for each study awaiting classification

Supplementary material 13 Dosage of pharmacological methods

Supplementary material 14 Reported outcomes of each included studies

Supplementary material 15 Definition of perinatal death, blood loss 500 mL or more, and blood loss 1000 mL or more that each included study defined

Supplementary material 16 Summary of findings: Caesarean section due to any causes (women without previous caesarean section and a mix of women with or without previous caesarean section)

Supplementary material 17 Summary of findings: Severe neonatal morbidity (women without previous caesarean section and a mix of women with or without previous caesarean section)

Supplementary material 18 Summary of findings: Caesarean section due to any causes (women with previous caesarean section)

Supplementary material 19 Summary of findings: Caesarean section due to non-reassuring foetal status (women with previous caesarean section)

Supplementary material 20 Original analysis of critical and important outcomes (network diagram, forest plot, league table, and rankogram) – women without previous caesarean section and a mix of women with or without previous caesarean section

Supplementary material 21 Evaluation of local and global inconsistencies

Supplementary material 22 Subgroup analysis – women without previous caesarean section and a mix of women with or without previous caesarean section

Supplementary material 23 Sensitivity analysis – women without previous caesarean section and a mix of women with or without previous caesarean section

Supplementary material 24 Comparison-adjusted funnel plots – women without previous caesarean section and a mix of women with or without previous caesarean section

Supplementary material 25 Original analysis of critical and important outcomes (network diagram, forest plot, league table, and rankogram) – women with previous caesarean section

ADDITIONAL INFORMATION

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We used a review template provided by the Cochrane Pregnancy and Childbirth Group, along with a focused review format template from Cochrane, for the Methods and Results sections. These resources were used to ensure clarity and consistency in presentation of the review. Consequently, some text in this review may be similar or identical to that in other Cochrane reviews.

Editorial and peer-reviewer contributions

Cochrane Pregnancy and Childbirth supported the authors in the development of this review: intervention.

The following people conducted the editorial process for this article:

- Sign-off Editor (final editorial decision): Toby Lasserson, Acting Editor in Chief, Central Executive Team;
- Managing Editor (selected peer reviewers, provided editorial guidance to authors, edited the article): Leanne Jones, Cochrane Central Editorial Service;
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Contributions of authors

Olufemi T Oladapo (OTO), Ioannis D Gallos (IDG), Arri Coomarasamy (AC), Pisake Lumbiganon (PL), Malcolm J Price (MJP), Chumnun Kietpeerakool (CK), and Siwanon Rattananokchai (SR) conceived the idea for this review.

SR, MJP, IDG, AC, CK, and PL designed the review.

SR, CK, Nuntasiri Eamudomkarn (NE), Kyaw Lwin Show (KLS), and Khaing Nwe Tin (KNT) conducted the review (study selection, trustworthiness assessment, data extraction, and risk of bias assessment).

SR, IDG, AC, CK, NE, and PL provided a clinical perspective.

SR, MJP, AC, and IDG provided statistical advice and input.

SR and MJP drafted the manuscript.

IDG, AC, PL, CK, NE, KLS, KNT, OTO, Doris Chou (DC), Ben Willem J Mol (BWM), and Wentao Li (WL) reviewed the manuscript and provided critical feedback.

MJP is the guarantor for this review.

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Declarations of interest

Siwanon Rattananokchai has declared that he has no conflict of interest.

Ioannis D Gallos was an Editor for Cochrane Pregnancy Childbirth, and reports no involvement in the editorial process for this review.

Chumnan Kietpeerakool has declared that he has no conflict of interest.

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Olufemi T Oladapo was an Editor for Cochrane Pregnancy Childbirth, and reports no involvement in the editorial process for this review. OTO reports involvement in World Health Organization guidelines on induction of labour.

Doris Chou declares that she is a staff member of UNDP-UNFPA-UNICEF-WHO-World Bank Special Programme of Research, Development and Research Training in Human Reproduction (HRP), a co-sponsored programme executed by the World Health Organization (WHO). She was the responsible technical officer for the updated recommendations related to the Induction of labour, published by WHO in 2022.

Ben Willem J Mol was an Editor for Cochrane Pregnancy Childbirth, and reports no involvement in the editorial process for this review. BWM also reports receiving consultancy fees from Norgine and being involved in studies eligible for inclusion in this review (PROBAAT I and PROBAAT II).

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Registration and protocol

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Data, code and other materials

As part of the published Cochrane review, the following are made available for download for users of the Cochrane Library: full search strategies for each database, full citations of each unique report for all studies included, ongoing or awaiting classification, or excluded at the full-text screen, in the final review; study data, including study information, study arms, and study results or test data; consensus risk of bias assessments; consensus trustworthiness assessments; and analysis data, including overall estimates, subgroup estimates, sensitivity estimates, and individual data rows. Appropriate permissions have been obtained for such use. Analyses and data management for pairwise meta-analyses were conducted within Cochrane's authoring tool, Review Manager Web, using the inbuilt computation methods [Supplementary material 7](#). We also performed network meta-analyses using STATA software for Mac version 18.0. The commands to perform each analysis related to network meta-analysis can be reviewed in the [Synthesis methods](#) section. The data extraction form, which the review authors created specifically for this review using Microsoft Access software, is available from the authors upon reasonable request.

History

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ADDITIONAL TABLES

Table 1. Overview of synthesis and included studies table

Study	Comparison	Sample size *	Gesta- tional age 1	Parity ²	Cervical condi- tion ³	Setting ⁴	Mem- brane sta- tus ⁵	Type of pregnan- cy	History of caesarean section
Women without previous caesarean section and a mix of women with or without previous caesarean section									
Aalami-Harandi 2013 (Iran) Reference(s): Aalami-Ha- randi 2013 [99, 100]	< 50 µg oral misoprostol vs. oxytocin (alone)	256	Mixed	Mixed	Un- favourable cervix	In-patient setting	Mixed	Singleton pregnancy	Without histo- ry of caesare- an delivery
Aduloju 2016 (Nigeria) Reference(s): Aduloju 2016 [101, 102]	Single balloon catheter vs. < 50 µg vaginal misopros- tol vs. balloon catheters plus < 50 µg vaginal miso- prostol	210	Mixed	Mixed	Un- favourable cervix	In-patient setting	Intact mem- brane	Singleton pregnancy	Without histo- ry of caesare- an delivery
Al-Ibraheemi 2018 (United States) Reference(s): Al-Ibraheemi 2018 [103, 104, 105]	< 50 µg vaginal misopros- tol vs. balloon catheters plus < 50 µg vaginal miso- prostol	200	Mixed	Mixed	Un- favourable cervix	In-patient setting	Intact mem- brane	Singleton pregnancy	Without histo- ry of caesare- an delivery
Amarasena 2021 (Sri Lanka) Reference(s): Amarasena 2021 [106, 107]	50 µg oral misoprostol vs. single balloon catheter	144	Early term and full term (37 ⁺⁰ to 40 ⁺⁶ weeks)	Mixed	Un- favourable cervix	In-patient setting	Intact mem- brane	Singleton pregnancy	Without histo- ry of caesare- an delivery
Anjali 2022	Balloon catheters plus < 50 µg oral misoprostol vs. < 50 µg oral misoprostol	200	Mixed	Mixed	Un- favourable cervix	In-patient setting	Not re- ported	Singleton pregnancy	Without histo- ry of caesare- an delivery

Table 1. Overview of synthesis and included studies table (Continued)

(India)

Reference(s): Anjali 2022

[\[108, 109\]](#)

Anzeljc 2022 (Slovenia)	Vaginal dinoprostone (tablet or gel) vs. controlled-release dinoprostone vaginal pessary	205	Mixed	Mixed	Un-favourable cervix	In-patient setting	Ruptured membrane	Singleton pregnancy	Without history of caesarean delivery
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Reference(s): Anzeljc 2022

[\[110, 111\]](#)

Ayaz 2009 (Pakistan)	50 µg oral misoprostol vs. 50 µg vaginal misoprostol	88	Mixed	Multi-parous	Un-favourable cervix	In-patient setting	Intact membrane	Singleton pregnancy	Without history of caesarean delivery
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Reference(s): Ayaz 2009

[\[112\]](#)

Barda 2018 (Israel)	Vaginal dinoprostone (tablet or gel) vs. single balloon catheter	300	Mixed	Mixed	Un-favourable cervix	In-patient setting	Intact membrane	Singleton pregnancy	Without history of caesarean delivery
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Reference(s): Barda 2018

[\[113, 114\]](#)

Bartusevicius 2006 (Lithuania)	50 µg sublingual or buccal misoprostol vs. < 50 µg vaginal misoprostol	138	Mixed	Mixed	Un-favourable cervix	In-patient setting	Mixed	Singleton pregnancy	Without history of caesarean delivery
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Reference(s): Bartusevicius 2006

[\[115, 116\]](#)

Bebbington (United States)	50 µg vaginal misoprostol vs. vaginal dinoprostone (tablet or gel)	200	Mixed	Mixed	Un-favourable cervix	In-patient setting	Intact membrane	Singleton pregnancy	Without history of caesarean delivery
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Reference(s): Bebbington

Table 1. Overview of synthesis and included studies table (Continued)

Beigi 2010 (Iran) Reference(s): Beigi 2010 [117]	< 50 µg vaginal misopros- tol vs. < 50 µg sublingual or buccal misoprostol	250	Mixed	Mixed	Un- favourable cervix	In-patient setting	Intact mem- brane	Singleton pregnancy	Without histo- ry of caesare- an delivery
Bennett 1998 (Canada) Reference(s): Bennett 1998 [118, 119]	50 µg vaginal misoprostol vs. 50 µg oral misoprostol	206	Mixed	Mixed	Mixed	In-patient setting	Intact mem- brane	Singleton pregnancy	Without histo- ry of caesare- an delivery
Bhide 2020 (United Kingdom) Reference(s): Bhide 2020 [120, 121, 122, 123]	Controlled-release dino- prostone vaginal pes- sary vs. double balloon catheter	38	Mixed	Mixed	Not stated	Outpa- tient set- ting	Intact mem- brane	Singleton pregnancy	Without histo- ry of caesare- an delivery
Boddu 2017 (India) Reference(s): Boddu 2017 [124, 125]	50 µg oral misoprostol vs. oxytocin (alone)	134	Mixed	Mixed	Not stated	In-patient setting	Ruptured mem- brane	Singleton pregnancy	Without histo- ry of caesare- an delivery
Bullarbo 2007 (Sweden) Reference(s): Bullarbo 2007 [126]	Nitric oxide donors vs. placebo	200	Late term and post- term (≥ 41 ⁺⁰ weeks)	Mixed	Un- favourable cervix	Outpa- tient set- ting	Intact mem- brane	Singleton pregnancy	Not stated
Butt 1999	50 µg oral misoprostol vs. oxytocin (alone)	108	Mixed	Mixed	Mixed	In-patient setting	Ruptured mem- brane	Singleton pregnancy	Without histo- ry of caesare- an delivery

Table 1. Overview of synthesis and included studies table (Continued)
(Canada)

Reference(s): Butt 1999
[\[127\]](#)

Calder 2008	< 50 µg vaginal misopros- tol vs. vaginal dinopros- tone (tablet or gel)	626	Mixed	Mixed	Un- favourable cervix	In-patient setting	Intact mem- brane	Singleton pregnancy	Without histo- ry of caesare- an delivery
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(United Kingdom)

Reference(s): Calder 2008
[\[128, 129, 130, 131, 132\]](#)

Caliskan 2005	50 µg vaginal misopros- tol vs. 50 µg sublingual or buccal misoprostol	160	Mixed	Mixed	Un- favourable cervix	In-patient setting	Mixed	Singleton pregnancy	Without histo- ry of caesare- an delivery
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(Turkey)

Reference(s): Caliskan
2005 [\[133\]](#)

Chaudhuri 2011	< 50 µg vaginal misopros- tol vs. vaginal dinopros- tone (tablet or gel)	207	Mixed	Mixed	Mixed	In-patient setting	Ruptured mem- brane	Singleton pregnancy	Without histo- ry of caesare- an delivery
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(India)

Reference(s): Chaudhuri
2011 [\[134, 135\]](#)

Cheung 2006	Placebo vs. 50 µg oral misoprostol	65	Mixed	Mixed	Not stated	In-patient setting	Ruptured mem- brane	Singleton pregnancy	Without histo- ry of caesare- an delivery
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(China)

Reference(s): Cheung 2006
[\[136\]](#)

da Graca Krupa 2005	< 50 µg vaginal misopros- tol vs. expectant manage- ment	150	Mixed	Mixed	Not stated	In-patient setting	Ruptured mem- brane	Singleton pregnancy	Without histo- ry of caesare- an delivery
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(Brazil)

Table 1. Overview of synthesis and included studies table (Continued)

Reference(s): da Graca
Krupa 2005 [137]

Dadashaliha 2021 (Iran)	50 µg sublingual or buccal misoprostol vs. 50 µg vagi- nal misoprostol	200	Mixed	Mixed	Un- favourable cervix	In-patient setting	Intact mem- brane	Singleton pregnancy	Without histo- ry of caesare- an delivery
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Reference(s): Dadashaliha
2021 [138, 139]

Dallenbach 2003 (Switzerland)	< 50 µg oral misoprostol vs. vaginal dinoprostone (tablet or gel)	200	Mixed	Mixed	Un- favourable cervix	In-patient setting	Mixed	Singleton pregnancy	Without histo- ry of caesare- an delivery
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Reference(s): Dallenbach
2003 [140, 141]

De Bonroostro Torralba 2019 (Spain)	< 50 µg vaginal misopros- tol vs. controlled-release dinoprostone vaginal pes- sary	198	Late term and post- term (≥ 41 ⁺⁰ weeks)	Nulli- parous	Un- favourable cervix	In-patient setting	Not re- ported	Singleton pregnancy	Without histo- ry of caesare- an delivery
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Reference(s): De Bon-
roostro Torralba 2019 [142,
143, 144]

de Moraes 2005 (Brazil)	< 50 µg sublingual or buc- cal misoprostol vs. < 50 µg vaginal misoprostol	120	Mixed	Mixed	Un- favourable cervix	In-patient setting	Mixed	Singleton pregnancy	Without histo- ry of caesare- an delivery
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Reference(s): de Moraes
2005 [145]

Devillard 2022 (France)	Balloon catheters plus oxytocin vs. controlled-re- lease dinoprostone vagi- nal pessary	80	Mixed	Mixed	Un- favourable cervix	In-patient setting	Ruptured mem- brane	Singleton pregnancy	Without histo- ry of caesare- an delivery
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Reference(s): Devillard
2022 [146, 147, 148]

Table 1. Overview of synthesis and included studies table (Continued)

Diguisto 2021 (France)	Double balloon catheter vs. controlled-release dinoprostone vaginal pes- sary	1216	Late term and post- term (≥ 41 ⁺⁰ weeks)	Mixed	Un- favourable cervix	In-patient setting	Intact mem- brane	Singleton pregnancy	Without histo- ry of caesare- an delivery
Reference(s): Diguisto 2021 [149, 150, 151]									
Dodd 2006 (Australia)	< 50 µg oral misoprostol vs. vaginal dinoprostone (tablet or gel)	741	Mixed	Mixed	Un- favourable cervix	In-patient setting	Intact mem- brane	Singleton pregnancy	Without histo- ry of caesare- an delivery
Reference(s): Dodd 2006 [152, 153, 154]									
Dogra 2019 (India)	Oxytocin (alone) vs. ex- pectant management	50	Mixed	Mixed	Favourable cervix	In-patient setting	Not re- ported	Singleton pregnancy	Without histo- ry of caesare- an delivery
Reference(s): Dogra 2019 [155, 156]									
El-Shawarby 2006 (United Kingdom)	Controlled-release dino- prostone vaginal pessary vs. vaginal dinoprostone (tablet or gel)	72	Mixed	Mixed	Un- favourable cervix	In-patient setting	Mixed	Singleton pregnancy	Not stated
Reference(s): El-Shawarby 2006									
Feitosa 2006 (Brazil)	< 50 µg sublingual or buc- cal misoprostol vs. < 50 µg vaginal misoprostol	150	Mixed	Mixed	Un- favourable cervix	In-patient setting	Intact mem- brane	Singleton pregnancy	Without histo- ry of caesare- an delivery
Reference(s): Feitosa 2006 [157, 158]									
Fuenmayor-Beltran 2017	Balloon catheters plus oxytocin vs. oxytocin (alone)	80	Mixed	Mixed	Un- favourable cervix	In-patient setting	Intact mem- brane	Singleton pregnancy	Without histo- ry of caesare- an delivery

Table 1. Overview of synthesis and included studies table (Continued)
(Venezuela)

Reference(s): Fuenma-
vor-Beltran 2017 [159]

Gattas 2020	< 50 µg sublingual or buc- cal misoprostol vs. < 50 µg vaginal misoprostol	198	Mixed	Mixed	Un- favourable cervix	In-patient setting	Mixed	Singleton pregnancy	Without histo- ry of caesare- an delivery
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(Brazil)

Reference(s): Gattas 2020
[160, 161, 162]

Gavara 2022	Osmotic cervical dilators vs. < 50 µg oral misopros- tol	303	Mixed	Mixed	Un- favourable cervix	In-patient setting	Intact mem- brane	Singleton pregnancy	Without histo- ry of caesare- an delivery
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(United States)

Reference(s): Gavara 2022
[163, 164]

Ghanaie 2013	Placebo vs. nitric oxide donors	72	Mixed	Nulli- parous	Un- favourable cervix	Outpa- tient set- ting	Intact mem- brane	Singleton pregnancy	Without histo- ry of caesare- an delivery
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(Iran)

Reference(s): Ghanaie
2013 [165, 166]

Goonewardene 2014	Single balloon catheter vs. < 50 µg oral misoprostol	152	Late term and post- term (≥ 41 ⁺⁰ weeks)	Mixed	Un- favourable cervix	In-patient setting	Not re- ported	Singleton pregnancy	Without histo- ry of caesare- an delivery
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(Sri Lanka)

Reference(s): Goonewar-
dene 2014 [167, 168]

Grace 2021	Double balloon catheter vs. vaginal dinoprostone (tablet or gel)	394	Mixed	Mixed	Un- favourable cervix	In-patient setting	Intact mem- brane	Singleton pregnancy	Without histo- ry of caesare- an delivery
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(Singapore and Malaysia)

Table 1. Overview of synthesis and included studies table (Continued)

Reference(s): Grace 2021 [169, 170]									
Grant 1992 (United Kingdom)	Expectant management vs. oxytocin (alone)	444	Mixed	Nulli- parous	Not stated	In-patient setting	Ruptured mem- brane	Singleton pregnancy	Without histo- ry of caesare- an delivery
Reference(s): Grant 1992 [171]									
Gupta 2022 (United Kingdom)	Osmotic cervical dilators vs. controlled-release dinoprostone vaginal pes- sary	672	Mixed	Mixed	Mixed	In-patient setting	Intact mem- brane	Singleton pregnancy	Mixed
Reference(s): Gupta 2022									
Haas 2019 (United States)	< 50 µg vaginal misopros- tol vs. < 50 µg sublingual or buccal misoprostol	300	Mixed	Mixed	Un- favourable cervix	In-patient setting	Intact mem- brane	Singleton pregnancy	Without histo- ry of caesare- an delivery
Reference(s): Haas 2019 [172, 173, 174, 175]									
Habib 2008 (Egypt)	Nitric oxide donors vs. placebo	102	Mixed	Mixed	Un- favourable cervix	Outpa- tient set- ting	Intact mem- brane	Singleton pregnancy	Without histo- ry of caesare- an delivery
Reference(s): Habib 2008 [176]									
Hannah 1996 (Canada, the United King- dom, Australia, Israel, Sweden, and Denmark)	Oxytocin (alone) vs. ex- pectant management vs. vaginal dinoprostone (tablet or gel) vs. expec- tant management	5041	Mixed	Mixed	Not stated	Compar- ison be- tween out- patient and in-pa- tient set- tings	Ruptured mem- brane	Singleton pregnancy	Mixed
Reference(s): Hannah 1996									

Table 1. Overview of synthesis and included studies table (Continued)

Henry 2013 (Australia) Reference(s): Henry 2013 [177, 178, 179, 180, 181]	Single balloon catheter vs. vaginal dinoprostone (tablet or gel)	101	Mixed	Mixed	Un- favourable cervix	Compar- ison be- tween out- patient and in-pa- tient set- tings	Intact mem- brane	Singleton pregnancy	Without histo- ry of caesare- an delivery
Hosli 2008 (Switzerland) Reference(s): Hosli 2008 [182]	50 µg vaginal misoprostol vs. vaginal dinoprostone (tablet or gel)	107	Mixed	Mixed	Mixed	In-patient setting	Mixed	Singleton pregnancy	Without histo- ry of caesare- an delivery
Incerpi 2001 (United States) Reference(s): Incerpi 2001 [183, 184]	< 50 µg vaginal misopros- tol vs. placebo	120	Mixed	Mixed	Un- favourable cervix	Outpa- tient set- ting	Intact mem- brane	Singleton pregnancy	Without histo- ry of caesare- an delivery
Itoh 2021 (Japan) Reference(s): Itoh 2021 [185, 186]	Controlled-release dino- prostone vaginal pessary vs. placebo	113	Late term and post- term (≥ 41 ⁺⁰ weeks)	Mixed	Un- favourable cervix	In-patient setting	Mixed	Singleton pregnancy	Without histo- ry of caesare- an delivery
Jahromi 2016 (Iran) Reference(s): Jahromi 2016 [187, 188, 189]	< 50 µg sublingual or buc- cal misoprostol vs. < 50 µg vaginal misoprostol	200	Mixed	Nulli- parous	Un- favourable cervix	In-patient setting	Mixed	Singleton pregnancy	Without histo- ry of caesare- an delivery
Jozwiak 2011	Single balloon catheter vs. vaginal dinoprostone (tablet or gel)	819	Mixed	Mixed	Un- favourable cervix	In-patient setting	Intact mem- brane	Singleton pregnancy	Without histo- ry of caesare- an delivery

Table 1. Overview of synthesis and included studies table (Continued)
(Netherlands)

Reference(s): Jozwiak
2011 [190, 191, 192]

Jozwiak 2013 (Netherlands)	Single balloon catheter vs. controlled-release dinoprostone vaginal pessary	226	Mixed	Mixed	Un-favourable cervix	In-patient setting	Intact membrane	Singleton pregnancy	Without history of caesarean delivery
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Reference(s): Jozwiak
2013 [193, 194]

Khamaiseh 2012 (Jordan)	Vaginal dinoprostone (tablet or gel) vs. single balloon catheter	414	Mixed	Mixed	Un-favourable cervix	In-patient setting	Intact membrane	Singleton pregnancy	Without history of caesarean delivery
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Reference(s): Khamaiseh
2012 [195]

Khoury 2001 (United States)	Controlled-release dinoprostone vaginal pessary vs. < 50 µg vaginal misoprostol vs. 50 µg vaginal misoprostol	79	Mixed	Mixed	Un-favourable cervix	In-patient setting	Intact membrane	Singleton pregnancy	Without history of caesarean delivery
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Reference(s): Khoury 2001
[196, 197]

Kidanto 2007 (Tanzania)	< 50 µg vaginal misoprostol vs. oxytocin (alone)	142	Mixed	Not stated	Not stated	In-patient setting	Not reported	Singleton pregnancy	Without history of caesarean delivery
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Reference(s): Kidanto
2007 [198]

Kim 2005 (South Korea)	Controlled-release dinoprostone vaginal pessary vs. vaginal dinoprostone (tablet or gel)	197	Mixed	Mixed	Un-favourable cervix	In-patient setting	Intact membrane	Singleton pregnancy	Without history of caesarean delivery
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Table 1. Overview of synthesis and included studies table (Continued)

Reference(s): Kim 2005

[\[199\]](#)

Kuper 2018 (United States) Reference(s): Kuper 2018 [200, 201]	Single balloon catheter vs. balloon catheters plus oxytocin	129	Mixed	Multi-parous	Not stated	Comparison between out-patient and in-patient settings	Mixed	Singleton pregnancy	Without history of caesarean delivery
Kwon 2001 (Canada) Reference(s): Kwon 2001 [202, 203]	50 µg oral misoprostol vs. 50 µg vaginal misoprostol	160	Mixed	Mixed	Un-favourable cervix	In-patient setting	Intact membrane	Singleton pregnancy	Without history of caesarean delivery
Lemyre (France) Reference(s): Lemyre	Single balloon catheter vs. < 50 µg vaginal misoprostol	62	Mixed	Mixed	Un-favourable cervix	In-patient setting	Intact membrane	Singleton pregnancy	Without history of caesarean delivery
Levine 2016 (United States) Reference(s): Levine 2016 [204, 205, 206]	< 50 µg vaginal misoprostol vs. balloon catheters plus < 50 µg vaginal misoprostol vs. single balloon catheter vs. balloon catheters plus oxytocin	491	Mixed	Mixed	Un-favourable cervix	In-patient setting	Intact membrane	Singleton pregnancy	Without history of caesarean delivery
Levy 2007 (Israel) Reference(s): Levy 2007 [207, 208]	50 µg oral misoprostol vs. placebo	130	Mixed	Mixed	Un-favourable cervix	In-patient setting	Ruptured membrane	Singleton pregnancy	Without history of caesarean delivery
Lo 2006	50 µg sublingual or buccal misoprostol vs. oxytocin plus amniotomy	50	Mixed	Nulliparous	Favourable cervix	In-patient setting	Intact membrane	Singleton pregnancy	Without history of caesarean delivery

Table 1. Overview of synthesis and included studies table (Continued)

(Hong Kong)

Reference(s): Lo 2006

[\[209\]](#)

Mahmood 1995	Vaginal dinoprostone (tablet or gel) vs. expectant management	100	Mixed	Multi- parous	Not stated	In-patient setting	Ruptured mem- brane	Singleton pregnancy	Without histo- ry of caesare- an delivery
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(United Kingdom)

Reference(s): Mahmood

1995 [\[210\]](#)

Matega (Democratic Republic of Congo)	50 µg oral misoprostol vs. 50 µg vaginal misoprostol	177	Mixed	Mixed	Not stated	In-patient setting	Not re- ported	Singleton pregnancy	Without histo- ry of caesare- an delivery
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Reference(s): Matega

McKenna 2004	< 50 µg vaginal misopros- tol vs. placebo	68	Mixed	Mixed	Un- favourable cervix	Outpa- tient set- ting	Intact mem- brane	Singleton pregnancy	Without histo- ry of caesare- an delivery
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(United States)

Reference(s): McKenna

2004 [\[211\]](#)

Moraes Filho 2010	< 50 µg vaginal misopros- tol vs. balloon catheters plus oxytocin	240	Mixed	Mixed	Un- favourable cervix	In-patient setting	Intact mem- brane	Singleton pregnancy	Without histo- ry of caesare- an delivery
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(Brazil)

Reference(s): Moraes Filho

2010 [\[212, 213\]](#)

Mundle 2017	Single balloon catheter vs. < 50 µg oral misoprostol	475	Mixed	Mixed	Un- favourable cervix	In-patient setting	Intact mem- brane	Singleton pregnancy	Without histo- ry of caesare- an delivery
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(India)

Reference(s): Mundle 2017

[\[214, 215, 216, 217, 218,](#)
[219, 220, 221\]](#)

Table 1. Overview of synthesis and included studies table (Continued)

Nassar 2007 (Lebanon) Reference(s): Nassar 2007 [222, 223]	50 µg sublingual or buccal misoprostol vs. 50 µg vagi- nal misoprostol	170	Mixed	Mixed	Un- favourable cervix	In-patient setting	Mixed	Singleton pregnancy	Without histo- ry of caesare- an delivery
Nopdonrattakoon 2003 (Thailand) Reference(s): Nopdonrat- takoon 2003 [224]	50 µg oral misoprostol vs. 50 µg vaginal misoprostol	106	Mixed	Mixed	Un- favourable cervix	In-patient setting	Intact mem- brane	Singleton pregnancy	Without histo- ry of caesare- an delivery
Nwali 2021 (Nigeria) Reference(s): Nwali 2021 [225, 226]	Single balloon catheter vs. 50 µg vaginal misoprostol	135	Mixed	Mixed	Un- favourable cervix	In-patient setting	Intact mem- brane	Singleton pregnancy	Without histo- ry of caesare- an delivery
O'Brien 1995 (United States) Reference(s): O'Brien 1995	Vaginal dinoprostone (tablet or gel) vs. placebo	100	Early term and full term (37 ⁺⁰ to 40 ⁺⁶ weeks)	Mixed	Un- favourable cervix	In-patient setting	Not re- ported	Singleton pregnancy	Mixed
Oboro 2005 (Nigeria) Reference(s): Oboro 2005 [227]	< 50 µg vaginal misopros- tol vs. no treatment	77	Mixed	Mixed	Un- favourable cervix	Outpa- tient set- ting	Intact mem- brane	Singleton pregnancy	Without histo- ry of caesare- an delivery
Oliveria 2010	Single balloon catheter vs. < 50 µg vaginal misopros- tol	160	Mixed	Mixed	Un- favourable cervix	In-patient setting	Intact mem- brane	Singleton pregnancy	Without histo- ry of caesare- an delivery

Table 1. Overview of synthesis and included studies table (Continued)
(Brazil)

Reference(s): Oliveira 2010
[228, 229]

Osman 2006	Nitric oxide donors vs. vaginal dinoprostone (tablet or gel)	398	Mixed	Nulli- parous	Un- favourable cervix	Outpa- tient set- ting	Intact mem- brane	Singleton pregnancy	Without histo- ry of caesare- an delivery
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(United Kingdom)

Reference(s): Osman 2006
[230, 231, 232, 233, 234]

Ozkan 2009	50 µg vaginal misopros- tol vs. controlled-release dinoprostone vaginal pes- sary	112	Mixed	Mixed	Un- favourable cervix	In-patient setting	Intact mem- brane	Singleton pregnancy	Without histo- ry of caesare- an delivery
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(Turkey)

Reference(s): Ozkan 2009
[235]

Paisarntantiwong 2005	50 µg oral misoprostol vs. < 50 µg vaginal misopros- tol	146	Mixed	Mixed	Un- favourable cervix	In-patient setting	Intact mem- brane	Singleton pregnancy	Without histo- ry of caesare- an delivery
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(Thailand)

Reference(s): Paisarntanti-
wong 2005 [236, 237]

Pandis 2001	Vaginal dinoprostone (tablet or gel) vs. 50 µg vaginal misoprostol	435	Mixed	Mixed	Un- favourable cervix	In-patient setting	Intact mem- brane	Singleton pregnancy	Without histo- ry of caesare- an delivery
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(United Kingdom)

Reference(s): Pandis 2001

Papanikolaou 2004 (Greece)	50 µg vaginal misoprostol vs. vaginal dinoprostone (tablet or gel)	163	Mixed	Nulli- parous	Un- favourable cervix	In-patient setting	Intact mem- brane	Singleton pregnancy	Without histo- ry of caesare- an delivery
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Reference(s): Papaniko-
laou 2004 [238]

Table 1. Overview of synthesis and included studies table (Continued)

PonMalar 2017 (India) Reference(s): PonMalar 2017 [239, 240]	< 50 µg vaginal misopros- tol vs. placebo	126	Early term and full term (37 ⁺⁰ to 40 ⁺⁶ weeks)	Mixed	Un- favourable cervix	Outpa- tient set- ting	Intact mem- brane	Singleton pregnancy	Without histo- ry of caesare- an delivery
Prager 2008 (Sweden) Reference(s): Prager 2008 [241, 242]	Vaginal dinoprostone (tablet or gel) vs. < 50 µg vaginal misoprostol vs. single balloon catheter	588	Mixed	Mixed	Un- favourable cervix	In-patient setting	Mixed	Singleton pregnancy	Without histo- ry of caesare- an delivery
Rahman 2013 (India) Reference(s): Rahman 2013 [243, 244]	50 µg oral misoprostol vs. < 50 µg vaginal misopros- tol	220	Mixed	Mixed	Un- favourable cervix	In-patient setting	Mixed	Singleton pregnancy	Without histo- ry of caesare- an delivery
Rameez 2007 (Sri Lanka) Reference(s): Rameez 2007 [245, 246]	Nitric oxide donors vs. placebo	156	Late term and post- term (≥ 41 ⁺⁰ weeks)	Mixed	Un- favourable cervix	In-patient setting	Not re- ported	Singleton pregnancy	Not stated
Rayburn 1988 (United States) Reference(s): Rayburn 1988 [247]	Vaginal dinoprostone (tablet or gel) vs. placebo	118	Late term and post- term (≥ 41 ⁺⁰ weeks)	Mixed	Un- favourable cervix	Outpa- tient set- ting	Not re- ported	Singleton pregnancy	Not stated
Rayburn 1992	Placebo vs. controlled-re- lease dinoprostone vagi- nal pessary	215	Mixed	Mixed	Un- favourable cervix	In-patient setting	Intact mem- brane	Singleton pregnancy	Without histo- ry of caesare- an delivery

Table 1. Overview of synthesis and included studies table (Continued)

(United States)

Reference(s): Rayburn
1992 [248, 249, 250]

Rozenberg 2004 (France)	50 µg vaginal misopros- tol vs. controlled-release dinoprostone vaginal pes- sary	140	Late term and post- term (≥ 41 ⁺⁰ weeks)	Mixed	Un- favourable cervix	In-patient setting	Intact mem- brane	Singleton pregnancy	Without histo- ry of caesare- an delivery
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Reference(s): Rozenberg
2004 [251]

Saad 2019 (United States)	Single balloon catheter vs. osmotic cervical dilators	417	Mixed	Mixed	Mixed	In-patient setting	Intact mem- brane	Singleton pregnancy	Without histo- ry of caesare- an delivery
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Reference(s): Saad 2019
[252, 253, 254, 255]

Sawai 1991 (United States)	Vaginal dinoprostone (tablet or gel) vs. placebo	50	Late term and post- term (≥ 41 ⁺⁰ weeks)	Mixed	Un- favourable cervix	Outpa- tient set- ting	Intact mem- brane	Singleton pregnancy	Without histo- ry of caesare- an delivery
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Reference(s): Sawai 1991
[256, 257]

Sawai 1994 (United States)	Placebo vs. vaginal dino- prostone (tablet or gel)	80	Late term and post- term (≥ 41 ⁺⁰ weeks)	Mixed	Un- favourable cervix	Outpa- tient set- ting	Intact mem- brane	Not stated	Without histo- ry of caesare- an delivery
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Reference(s): Sawai 1994
[258, 259]

Schmitz 2014 (France)	Nitric oxide donors vs. placebo	1362	Late term and post- term (≥ 41 ⁺⁰ weeks)	Nulli- parous	Un- favourable cervix	Outpa- tient set- ting	Intact mem- brane	Singleton pregnancy	Without histo- ry of caesare- an delivery
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Table 1. Overview of synthesis and included studies table (Continued)

Reference(s): Schmitz

2014 [260, 261, 262]

Shechter-Maor 2015 (Israel)	Controlled-release dino- prostone vaginal pes- sary vs. double balloon catheter	52	Mixed	Mixed	Un- favourable cervix	In-patient setting	Intact mem- brane	Singleton pregnancy	Without histo- ry of caesare- an delivery
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Reference(s): Shechter-

Maor 2015 [263, 264, 265]

Shetty 2001 (United Kingdom)	50 µg oral misoprostol vs. 50 µg vaginal misoprostol	245	Mixed	Mixed	Un- favourable cervix	In-patient setting	Not re- ported	Singleton pregnancy	Without histo- ry of caesare- an delivery
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Reference(s): Shetty 2001

[266, 267, 268, 269]

Shetty 2002 (Scotland)	50 µg oral misoprostol vs. 50 µg sublingual or buccal misoprostol	100	Mixed	Mixed	Un- favourable cervix	In-patient setting	Not re- ported	Singleton pregnancy	Without histo- ry of caesare- an delivery
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Reference(s): Shetty 2002

[270, 271]

Sifakis 2007 (Greece)	50 µg vaginal misoprostol vs. vaginal dinoprostone (tablet or gel)	415	Mixed	Mixed	Un- favourable cervix	In-patient setting	Intact mem- brane	Singleton pregnancy	Without histo- ry of caesare- an delivery
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Reference(s): Sifakis 2007

[272]

Soliman 2013 (Egypt)	Nitric oxide donors vs. 50 µg vaginal misoprostol	130	Mixed	Nulli- parous	Un- favourable cervix	In-patient setting	Intact mem- brane	Singleton pregnancy	Without histo- ry of caesare- an delivery
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Reference(s): Soliman

2013 [273]

Table 1. Overview of synthesis and included studies table (Continued)

Somirathne 2017 (Sri Lanka) Reference(s): Somirathne 2017 [274, 275, 276]	Single balloon catheter vs. < 50 µg oral misoprostol	180	Early term and full term (37 ⁺⁰ to 40 ⁺⁶ weeks)	Mixed	Un- favourable cervix	In-patient setting	Not re- ported	Singleton pregnancy	Without histo- ry of caesare- an delivery
Souza 2013 (Brazil) Reference(s): Souza 2013 [277, 278, 279]	< 50 µg oral misoprostol vs. < 50 µg vaginal miso- prostol	200	Mixed	Mixed	Un- favourable cervix	In-patient setting	Mixed	Singleton pregnancy	Without histo- ry of caesare- an delivery
Surbek 1997 (Switzerland) Reference(s): Surbek 1997 [280, 281, 282, 283]	50 µg vaginal misoprostol vs. vaginal dinoprostone (tablet or gel)	100	Mixed	Mixed	Un- favourable cervix	In-patient setting	Mixed	Singleton pregnancy	Without histo- ry of caesare- an delivery
Tabasi 2007 (Iran) Reference(s): Tabasi 2007 [284]	50 µg vaginal misoprostol vs. oxytocin (alone)	110	Mixed	Mixed	Un- favourable cervix	In-patient setting	Intact mem- brane	Singleton pregnancy	Without histo- ry of caesare- an delivery
Tan 2010 (Singapore) Reference(s): Tan 2010 [285, 286]	Vaginal dinoprostone (tablet or gel) vs. 50 µg vaginal misoprostol vs. < 50 µg vaginal misoprostol	169	Mixed	Mixed	Un- favourable cervix	In-patient setting	Not re- ported	Singleton pregnancy	Without histo- ry of caesare- an delivery
Ten Eikelder 2016	50 µg oral misoprostol vs. single balloon catheter	1845	Mixed	Mixed	Un- favourable cervix	In-patient setting	Intact mem- brane	Singleton pregnancy	Without histo- ry of caesare- an delivery

Table 1. Overview of synthesis and included studies table (Continued)
(Netherlands)

Reference(s): Ten Eikelder
2016 [287, 288, 289, 290,
291, 292, 293]

Triglia 2010 (Italy)	Controlled-release dino- prostone vaginal pessary vs. vaginal dinoprostone (tablet or gel)	130	Mixed	Mixed	Un- favourable cervix	In-patient setting	Intact mem- brane	Singleton pregnancy	Without histo- ry of caesare- an delivery
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Reference(s): Triglia 2010
[294, 295]

Unthanan 2022 (Thailand)	< 50 µg sublingual or buc- cal misoprostol vs. oxy- tocin (alone)	170	Mixed	Mixed	Un- favourable cervix	In-patient setting	Ruptured mem- brane	Singleton pregnancy	Without histo- ry of caesare- an delivery
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Reference(s): Unthanan
2022 [296, 297]

Vidanagamage 2011 (Sri Lanka)	Nitric oxide donors vs. nitric oxide donors vs. placebo	154	Mixed	Mixed	Un- favourable cervix	In-patient setting	Intact mem- brane	Singleton pregnancy	Without histo- ry of caesare- an delivery
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Reference(s): Vidanagam-
age 2011 [298, 299]

Wallstrom 2019 (Sweden)	< 50 µg oral misopros- tol vs. controlled-release misoprostol vaginal pes- sary	196	Mixed	Nulli- parous	Un- favourable cervix	In-patient setting	Mixed	Singleton pregnancy	Without histo- ry of caesare- an delivery
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Reference(s): Wallstrom
2019 [300, 301, 302, 303]

Witter 1996 (United States)	Placebo vs. controlled-re- lease dinoprostone vagi- nal pessary	193	Mixed	Mixed	Un- favourable cervix	In-patient setting	Intact mem- brane	Singleton pregnancy	Without histo- ry of caesare- an delivery
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Table 1. Overview of synthesis and included studies table (Continued)

Reference(s): Witter 1996
[304, 305]

Young 2020 (Canada)	50 µg oral misoprostol vs. < 50 µg vaginal misopros- tol vs. vaginal dinopros- tone (tablet or gel)	511	Mixed	Mixed	Mixed	In-patient setting	Ruptured mem- brane	Singleton pregnancy	Without histo- ry of caesare- an delivery
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Reference(s): Young 2020
[306, 307, 308]

Zhang 2015 (China)	< 50 µg vaginal misopros- tol vs. placebo	222	Mixed	Nulli- parous	Un- favourable cervix	In-patient setting	Intact mem- brane	Singleton pregnancy	Without histo- ry of caesare- an delivery
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Reference(s): Zhang 2015
[309, 310]

Women with previous caesarean section

Sarreau 2020 (France)	Foley single balloon catheter vs. oxytocin (alone)	204	Mixed	Multi- parous	Un- favourable cervix	In-patient setting	Intact mem- brane	Singleton pregnancy	With history of caesarean delivery
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Reference(s): Sarreau
2020 [311, 312, 313]

Sulaiman 2022 (Malaysia)	Foley single balloon catheter vs. controlled-re- lease dinoprostone vagi- nal pessary	78	Mixed	Multi- parous	Un- favourable cervix	In-patient setting	Intact mem- brane	Singleton pregnancy	With history of caesarean delivery
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Reference(s): Sulaiman
2023 [314, 315]

Note: * Number of women included in analyses

¹ **Gestational age:** Early term and full term (37⁺⁰ to 40⁺⁶ weeks) / Late term and post-term (≥ 41⁺⁰ weeks) / Mixed

² **Parity:** Nulliparous / Multiparous / Mixed

³ **Cervical condition:** Unfavourable cervix / Favourable cervix / Mixed

Table 1. Overview of synthesis and included studies table (Continued)

⁴ **Setting:** In-patient setting / Outpatient setting / Comparison between outpatient and in-patient settings

⁵ **Membrane status:** Intact membrane / Ruptured membrane / Mixed

Table 2. Treatment ranking across the critical and important outcomes for women without previous caesarean section and a mix of women with or without previous caesarean section*

IoL methods	Failure to achieve vaginal delivery within 24 hours		Caesarean section due to any causes		Caesarean section due to non-reassuring fetal status		Uterine hyperstimulation with changes in the heartbeat of the baby before birth	
	SUCRA (%)	Mean rank (95% CI)	SUCRA (%)	Mean rank (95% CI)	SUCRA (%)	Mean rank (95% CI)	SUCRA (%)	Mean rank (95% CI)
Control	15.2	12 (3 to 14)	24.0	10.9 (7 to 13)	48.9	7.6 (1 to 12)	78.3	3.6 (2 to 7)
Vag PGE1	75.2	4.2 (2 to 6)	81.2	3.5 (1 to 5)	53.5	7 (3 to 10)	32.8	9.1 (6 to 11)
Oral PGE1	41.8	8.6 (6 to 10)	78.7	3.8 (1 to 7)	71.3	4.7 (1 to 9)	59.2	5.9 (4 to 8)
SL/BC PGE1	69.3	5 (2 to 8)	66.6	5.3 (1 to 9)	25.2	10.7 (5 to 13)	27.3	9.7 (6 to 12)
CTR PGE1	56.1	6.7 (2 to 12)	29.6	10.2 (1 to 14)	41.5	8.6 (1 to 13)	21.2	10.5 (5 to 13)
Vag PGE2	49.8	7.5 (5 to 10)	63.2	5.8 (2 to 8)	35.8	9.4 (6 to 12)	32.0	9.2 (6 to 12)
CTR PGE2	51.4	7.3 (4 to 10)	46.7	7.9 (3 to 11)	74.7	4.3 (1 to 9)	27.2	9.7 (6 to 12)
Oxytocin	5.2	13.3	8.1	13	21.7	11.2	56.0	6.3

Table 2. Treatment ranking across the critical and important outcomes for women without previous caesarean section and a mix of women with or without previous caesarean section* *(Continued)*

		(11 to 14)		(11 to 14)		(2 to 14)		(3 to 10)
Nitric oxide donors	13.4	12.3 (6 to 14)	33.2	9.7 (4 to 12)	32.4	9.8 (3 to 13)	94.9	1.6 (1 to 3)
Balloon catheters	25.9	10.6 (8 to 12)	42.9	8.4 (5 to 11)	73.1	4.5 (1 to 8)	75.1	4 (2 to 5)
Osmotic dilators	35.6	9.4 (4 to 12)	54.6	6.9 (2 to 11)	47.1	7.9 (1 to 13)	93.7	1.8 (1 to 3)
Oxytocin + amniotomy	96.2	1.5 (1 to 7)	39.1	8.9 (1 to 14)	20.7	11.3 (1 to 14)	17.6	10.9 (2 to 13)
Balloon + oxytocin	78.8	3.8 (1 to 7)	38.6	9 (2 to 13)	81.8	3.4 (1 to 10)	34.7	8.8 (3 to 13)
Balloon + PGE1	86.1	2.8 (1 to 6)	93.6	1.8 (1 to 5)	72.4	4.6 (1 to 10)	-	-
IoL methods	Perinatal death		Severe neonatal morbidity		Failure to achieve vaginal delivery with the primary method		Maternal not satisfied with care related to induction of labour	
	SUCRA	Mean rank	SUCRA	Mean rank	SUCRA	Mean rank	SUCRA	Mean rank
	(%)	(95% CI)	(%)	(95% CI)	(%)	(95% CI)	(%)	(95% CI)
Control	-	-	-	-	-	-	70.8	3.6 (1 to 8)
Vag PGE1	49.8	3 (1 to 4)	-	-	-	-	31.7	7.1 (4 to 8)
Oral PGE1	72.0	2.1 (1 to 4)	-	-	-	-	40.8	6.3 (4 to 8)

Table 2. Treatment ranking across the critical and important outcomes for women without previous caesarean section and a mix of women with or without previous caesarean section* (Continued)

SL/BC PGE1	23.6	4.1 (1 to 5)	-	-	-	-	91.3	1.8 (1 to 4)
CTR PGE1	-	-	-	-	-	-	-	-
Vag PGE2	35.4	3.6 (1 to 5)	-	-	-	-	23.0	7.9 (4 to 9)
CTR PGE2	-	-	-	-	-	-	70.7	3.6 (1 to 8)
Oxytocin	-	-	-	-	-	-	-	-
Nitric oxide donors	-	-	-	-	-	-	70.0	3.7 (1 to 7)
Balloon catheters	69.2	2.2 (1 to 4)	-	-	-	-	27.8	7.5 (3 to 9)
Osmotic dilators	-	-	-	-	-	-	46.4	5.8 (2 to 9)
Oxytocin + amniotomy	-	-	-	-	-	-	27.6	7.5 (2 to 9)
Balloon + oxytocin	-	-	-	-	-	-	-	-
Balloon + PGE1	-	-	-	-	-	-	-	-
IoL methods	Time from induction to delivery (hours)		Time from induction to vaginal delivery (hours)		Blood loss > 500 mL		Blood loss > 1000 mL	
	SUCRA	Mean rank	SUCRA	Mean rank	SUCRA	Mean rank	SUCRA	Mean rank
	(%)	(95% CI)	(%)	(95% CI)	(%)	(95% CI)	(%)	(95% CI)
Control	1.3	12.8	55.5	7.1	49.1	4.6	-	-

Table 2. Treatment ranking across the critical and important outcomes for women without previous caesarean section and a mix of women with or without previous caesarean section* (Continued)

		(12 to 13)		(2 to 12)		(1 to 8)		
Vag PGE1	79.8	3.4 (1 to 5)	66.5	5.7 (4 to 8)	33.0	5.7 (1 to 8)	-	-
Oral PGE1	31.9	9.2 (7 to 11)	18.0	11.9 (10 to 13)	46.3	4.8 (2 to 6)	61.0	3.3 (1 to 5)
SL/BC PGE1	77.9	3.7 (1 to 7)	75.5	4.3 (2 to 7)	-	-	-	-
CTR PGE1	54.2	6.5 (1 to 11)	40.1	8.2 (3 to 12)	-	-	36.0	4.8 (2 to 7)
Vag PGE2	62.4	5.5 (2 to 8)	35.8	8 (4 to 11)	69.2	3.2 (1 to 6)	29.9	5.2 (2 to 7)
CTR PGE2	51.3	6.8 (3 to 10)	32.0	10.4 (6 to 13)	25.9	6.2 (3 to 8)	54.7	3.7 (1 to 6)
Oxytocin	39.2	8.3 (3 to 11)	34.4	9.9 (5 to 13)	86.4	2 (1 to 6)	-	-
Nitric oxide donors	7.2	12.1 (12 to 13)	0.2	14.0 (14 to 14)	-	-	-	-
Balloon catheters	30.0	9.4 (7 to 11)	46.3	7.9 (4 to 11)	44.7	4.9 (2 to 7)	59.1	3.5 (1 to 5)
Osmotic dilators	44.8	7.6 (1 to 11)	31.5	10.2 (4 to 13)	-	-	84.5	1.9 (1 to 6)
Oxytocin + amniotomy	-	-	92.8	1.8 (1 to 6)	-	-	-	-

Table 2. Treatment ranking across the critical and important outcomes for women without previous caesarean section and a mix of women with or without previous caesarean section* (Continued)

Balloon + oxytocin	76.7	3.8 (1 to 7)	81.2	3.3 (1 to 7)	45.4	4.8 (1 to 8)	24.8	5.5 (1 to 7)
Balloon + PGE1	93.2	1.8 (1 to 5)	90.3	2.1 (1 to 4)	-	-	-	-
IoL methods	Mean volume of blood loss (mL)		Uterine scar dehiscence or rupture		Oxytocin augmentation		Maternal death	
	SUCRA (%)	Mean rank (95% CI)	SUCRA (%)	Mean rank (95% CI)	SUCRA (%)	Mean rank (95% CI)	SUCRA (%)	Mean rank (95% CI)
Control	71.4	2.7 (1 to 5)	-	-	26.5	8.3 (6 to 9)	-	-
Vag PGE1	-	-	-	-	76.2	3.4 (2 to 4)	-	-
Oral PGE1	51.3	3.9 (2 to 6)	-	-	50.8	5.9 (4 to 7)	-	-
SL/BC PGE1	-	-	-	-	66.5	4.4 (2 to 6)	-	-
CTR PGE1	11.2	6.3 (2 to 7)	-	-	62.9	4.7 (2 to 9)	-	-
Vag PGE2	39.9	4.6 (1 to 7)	-	-	42.7	6.7 (5 to 8)	-	-
CTR PGE2	91.4	1.5 (1 to 6)	-	-	32.5	7.7 (5 to 9)	-	-
Oxytocin	-	-	-	-	-	-	-	-

Table 2. Treatment ranking across the critical and important outcomes for women without previous caesarean section and a mix of women with or without previous caesarean section* (Continued)

Nitric oxide donors	33.4	5 (2 to 7)	-	-	0.9	10.9 (10 to 11)	-	-
Balloon catheters	51.3	3.9 (2 to 6)	-	-	10.4	10 (9 to 10)	-	-
Osmotic dilators	-	-	-	-	83.2	2.7 (1 to 8)	-	-
Oxytocin + amniotomy	-	-	-	-	-	-	-	-
Balloon + oxytocin	-	-	-	-	-	-	-	-
Balloon + PGE1	-	-	-	-	97.2	1.3 (1 to 2)	-	-
IoL methods	Admission to intensive care unit		Pain related to induction of labour		Use of mechanical ventilation for infants		Admission to the neonatal intensive care unit	
	SUCRA (%)	Mean rank (95% CI)	SUCRA (%)	Mean rank (95% CI)	SUCRA (%)	Mean rank (95% CI)	SUCRA (%)	Mean rank (95% CI)
Control	-	-	-	-	72.7	2.6 (1 to 4)	17.6	10.9 (6 to 13)
Vag PGE1	-	-	-	-	16.5	6 (4 to 7)	69.9	4.6 (1 to 8)
Oral PGE1	26.3	3.2 (1 to 4)	49.9	3 (2 to 4)	36.9	4.8 (1 to 7)	30.9	9.3 (4 to 12)
SL/BC PGE1	-	-	-	-	-	-	40.3	8.2 (2 to 12)
CTR PGE1	-	-	-	-	-	-	15.9	11.1

Table 2. Treatment ranking across the critical and important outcomes for women without previous caesarean section and a mix of women with or without previous caesarean section* (Continued)

							(2 to 13)	
Vag PGE2	49.0	2.5 (1 to 4)	1.0	5 (4 to 5)	42.1	4.5 (2 to 6)	38.5	8.4 (3 to 11)
CTR PGE2	79.5	1.6 (1 to 3)	59.4	2.6 (1 to 4)	45.0	4.3 (1 to 7)	57.1	6.1 (1 to 11)
Oxytocin	-	-	-	-	71.6	2.7 (1 to 5)	55.0	6.4 (1 to 11)
Nitric oxide donors	-	-	-	-	-	-	39.6	8.2 (1 to 12)
Balloon catheters	45.1	2.6 (1 to 4)	46.6	3.1 (1 to 4)	65.1	3.1 (1 to 7)	61.1	5.7 (2 to 9)
Osmotic dilators	-	-	93.2	1.3 (1 to 2)	-	-	57.8	6.1 (1 to 11)
Oxytocin + amniotomy	-	-	-	-	-	-	-	-
Balloon + oxytocin	-	-	-	-	-	-	84.4	2.9 (1 to 10)
Balloon + PGE1	-	-	-	-	-	-	81.9	3.2 (1 to 9)
IoL methods	Apgar score < 7 at 5 minutes							
	SUCRA	Mean rank						
	(%)	(95% CI)						
Control	50.4	6.9 (2 to 10)						

Table 2. Treatment ranking across the critical and important outcomes for women without previous caesarean section and a mix of women with or without previous caesarean section* *(Continued)*

Vag PGE1	41.2	8.1 (4 to 11)
Oral PGE1	47.7	7.3 (3 to 10)
SL/BC PGE1	70.6	4.5 (1 to 10)
CTR PGE1	6.3	12.2 (9 to 13)
Vag PGE2	22.2	10.3 (7 to 12)
CTR PGE2	42.5	7.9 (2 to 11)
Oxytocin	59.0	5.9 (1 to 10)
Nitric oxide donors	60.9	5.7 (1 to 10)
Balloon catheters	71.3	4.4 (2 to 8)
Osmotic dilators	69.6	4.6 (1 to 12)
Oxytocin + amniotomy	-	-
Balloon + oxytocin	17.6	10.9

Table 2. Treatment ranking across the critical and important outcomes for women without previous caesarean section and a mix of women with or without previous caesarean section* (Continued)

(2 to 13)		
Balloon + PGE1	90.6	2.1
(1 to 5)		

*Ranking indicates the cumulative probability of being the best agent, the second best, the third best, etc. We estimated the Surface underneath the Cumulative RAnking line (SUCRA); the larger the SUCRA, the higher its rank among all available agents.

Abbreviations: **SUCRA**, Surface underneath the Cumulative RAnking; **IoL**, induction of labour; **CI**, confidence interval; **Control**, placebo or no treatment or expectant management; **Oral PGE1**, oral misoprostol ($\leq 50 \mu\text{g}$); **SL/BC PGE1**, sublingual or buccal misoprostol ($\leq 50 \mu\text{g}$); **Vag PGE1**, vaginal misoprostol ($\leq 50 \mu\text{g}$); **CTR PGE1**, controlled-release misoprostol vaginal pessary; **Vag PGE2**, vaginal dinoprostone (tablet or gel); **CTR PGE2**, controlled-release dinoprostone vaginal pessary; **Oxytocin**, oxytocin (alone); **Nitric oxide donors**, ISMN/ISDN/nitroglycerin/sodium nitroprusside; **Balloon catheters**, Foley single balloon catheter; double balloon devices (Atad/Cook balloon or other types); **Oxytocin + amniotomy**, oxytocin plus amniotomy; **Balloon + oxytocin**, balloon catheters plus oxytocin; **Balloon + PGE1**, balloon catheters plus misoprostol ($\leq 50 \mu\text{g}$; oral or vaginal routes).